INTERPRETABLE SPARSE MODELING OF LONGITUDINAL SIGNALS VIA CRITICAL-RANGE RECTIFICATION AND ANYTIME RULE COMPRESSION

by

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ABSTRACT

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Old Dominion University, 2026
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High-dimensional longitudinal data arise in clinical monitoring, industrial control systems, and other sensor-driven domains where outcomes are often governed by threshold-and-lag behavior. Traditional longitudinal workflows frequently depend on expert guessing to nominate candidate variables, lag windows, and threshold hypotheses, followed by repeated hypothesis testing over a limited set of manually specified relationships. While such approaches can be useful in narrow settings, they are often less robust in high-dimensional regimes because important interactions may be missed, multicollinearity can destabilize inference, and the resulting process can be labor-intensive and difficult to scale. This dissertation develops an end-to-end framework for interpretable sparse longitudinal modeling that addresses these limitations through three stages: (i) critical-range rectification, which converts each lagged continuous measurement into a {-1,+1} indicator denoting whether it lies within a data-driven range observed during events; (ii) sparse model fitting using an L1-regularized logistic baseline on the rectified design; and (iii) anytime rule compression ("logic polishing"), which collapses small-magnitude coefficients to produce a compact m-of-K rule model tuned to maximize Youden's J at a chosen operating point. Studies on synthetic data with known ground-truth rules and on real-world datasets, including the HAI industrial control benchmark, show that rectification can improve feature-and-lag attribution, reduce false positives, and decrease end-to-end training time when the underlying event structure is threshold-and-lag

aligned, while cross-domain audits also identify settings where those gains weaken or disappear. A tractable theoretical analysis uses the zero-threshold arcsin relation under joint normality to show how sign binarization can contract pairwise correlations, improve conditioning of the Gram matrix, and increase the likelihood of satisfying the LASSO irrepresentable condition. Beyond the prior papers, this work adds repeated-resample stability and ablation, direct interpretable baselines, cross-domain audits, empirical boundary checks, and strict held-out compression validation. In controlled threshold-mediated settings, the compression phase cleanly recovers compact rule forms from the rectified sparse baseline. In real-world data, where threshold-triggered effects may be mixed with smoother additive structure and measurement uncertainty, the dissertation's compression contribution is the explicit decision policy itself: a policy-controlled frontier that provides an auditable basis for accepting or rejecting simplification.

I dedicate my dissertation to my wife Danielle.

ACKNOWLEDGMENTS

I would like to thank my committee, and especially Drs. Sun and Zubair, without whose guidance this document and the work it represents would not have been possible.

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CHAPTER 1

INTRODUCTION

Longitudinal data contain repeated measurements of the same variables across time and are central to many scientific and operational problems. Examples include symptom progression in clinical cohorts, temporal variation in biological markers, and multi-wave surveys of human decision behavior [1–3]. Across these settings, outcomes are often triggered by threshold-and-lag mechanisms: a variable enters a critical range, then a target event appears after a delay as with Figure 1. This structure is common in sensing and monitoring contexts where decision-makers need not only accurate predictions, but also clear explanations of which variables and lags drove the prediction.

From a computer-science perspective, the primary technical readers for this dissertation are ML engineers working with sensor streams, industrial control systems (ICS) security analysts, and clinical-informatics researchers who need lag-aware models with explicit attribution and auditable deployment logic. The motivating technical setting is therefore not threshold-and-lag behavior in the abstract, but threshold-mediated longitudinal decision support under multicollinearity, where the practical questions are how to redesign representation, how to reuse mature sparse-learning software effectively, and how to convert the result into a model artifact that humans can review and deploy.

Related threshold-and-lag patterns also appear in broader scientific and operational domains, including ecosystem dynamics, epidemic spreading, control systems, and financial markets [4–7]. Those examples help motivate the general problem class, but the manuscript's main technical focus is on interpretable machine-learning workflows for sensing and monitoring settings.

In high-dimensional longitudinal modeling, a standard strategy is to construct a lag-expanded design matrix and apply a sparse estimator. This strategy is attractive because sparse models can retain predictive performance while reducing dimensionality [8, 9]. However, lag expansion also introduces strong dependence among adjacent lags and among correlated channels, creating exactly the regimes where support recovery becomes unstable and model-selection assumptions can fail [10, 11]. Practical outcomes include arbitrary swapping among correlated predictors, sensitivity to tuning choices, and weak reproducibility of selected supports across resamples.

A substantial body of work addresses these issues through penalty design and optimization advances, including elastic net, adaptive lasso, group-based penalties, and ordered lag constraints [12–16]. These methods provide important baselines and often improve behavior relative to plain lasso, but they still operate directly on raw correlated representations and do not provide explicit threshold semantics for human review, which can be a useful set of artifacts by themselves. Correlation-aware variants in domain-specific contexts further reinforce this point: dependence can be mitigated, but stable, interpretable attribution remains difficult when predictors are densely coupled [17–19].

A second challenge is interpretability under high-stakes use. Post hoc explanations of complex models can be useful, but interpretable-model-first approaches are often preferable when transparency, auditability, and operational trust are mandatory [20, 21]. Rule-structured models are especially relevant because they align with human review workflows and can be validated directly by domain experts [22–24]. This dissertation therefore focuses on an inherently interpretable sparse pipeline rather than black-box prediction with after-the-fact explanation.

This approach builds from prior work in this research line: a rectification-first longitudinal feature-selection method, a scoped theoretical analysis linking binarized transformation to improved irrepresentable-condition (IC) behavior, and an anytime rule-compression method that converts sparse models into compact logic-like forms [25–27]. The dissertation unifies these components into one framework, strengthens the supporting analysis, broadens empirical evaluation, and emphasizes public, notebook-backed reproducible implementation.

Terminology note: throughout the dissertation, *rectification* is the general pipeline term for converting continuous lagged measurements into threshold indicators. *Zero-threshold sign binarization* is the theorem-scoped special case of rectification used in RQ2 because it yields a tractable closed-form analysis.

1.1 PROBLEM

This work addresses three tensions that arise jointly in longitudinal feature learning:

- Correlation + high dimension $\rightarrow$ unstable sparse support selection and weak lag attribution.
- Operational use $\rightarrow$ limited compute budgets and constraints on model complexity.
- High-stakes deployment $\rightarrow$ demand for transparent, auditable rules instead of opaque scoring functions.

The central research question is whether representation-level rectification can make sparse longitudinal selection more reliable before model fitting, and whether a principled post-selection compression step can preserve discrimination while improving human interpretability. In contrast to approaches that rely only on penalty modifications, this dissertation investigates a preprocessing-plus-selection-plus-compression pipeline designed to explicitly encode threshold behavior, reduce correlation burden, and expose causal hypotheses at the feature-lag level.

Thesis statement: in threshold-and-lag aligned longitudinal settings, critical-range rectification can improve the reliability of sparse feature recovery by contracting harmful dependence structures in lag-expanded data, and anytime rule compression can convert the resulting sparse models into compact, operationally usable rule forms under explicit performance-tolerance policies [26, 27]. The scope emphasizes inherently interpretable sparse models with explicit attribution; deep learning appears only as contextual baselines where needed.

1.2 APPROACH

The dissertation's approach is representation-first. Rather than asking a sparse learner to disentangle raw continuous trajectories directly, the method first changes the feature space so that each candidate variable-lag pair asks a threshold-oriented question: did the measurement fall inside an event-associated operating range? Sparse fitting then operates on indicators that are closer to the target event logic, and downstream compression converts the resulting sparse support into a rule artifact when deployment requires a simpler object.

The pipeline has three main stages. First, it expands longitudinal measurements over lags and rectifies them into critical-range indicators using thresholds estimated from training data only, to prevent contamination of the test data. In the theorem-scoped special case used for RQ2, this step reduces to zero-threshold sign binarization after standardization; in the empirical pipeline, the thresholds can be shifted one-sided or interval-like ranges learned from event-aligned training observations. Second, an L1-regularized logistic baseline is fitted on the rectified design, so nonzero coefficients identify candidate feature-lag triggers after the representation has already encoded threshold structure [8, 28, 29]. Third, when operational interpretability requires an even simpler artifact, the active sparse coefficients are ordered and compressed into nested m-of-K rule candidates by the anytime logic-polishing procedure [27].

This design separates representation engineering from optimizer engineering. Penalty-level methods change the coefficient-space objective; rectification changes the statistical geometry of the feature space before a standard sparse solver is used. That distinction matters for reuse: the method can be layered onto mature sparse-learning software while targeting the dependence patterns that make raw lag-expanded support recovery unstable. It also keeps the deployment interface explicit. A coefficient vector can be useful for analysis, but a compact rule is often easier for an operator, clinician, or model-governance reviewer to audit.

Figure 2 summarizes the operational playbook for applying the pipeline to a new dataset. The sequence starts from a raw lag-expanded sparse baseline because the practical question is whether rectification improves attribution quality, compactness, or deployment usability relative to a conventional sparse starting point. Rectification is promoted only when train-only pilots show threshold-mediated structure and preserve held-out discrimination while improving support concentration, lag localization, or model compactness. Compression is accepted only when the nominated rule clears the held-out policy described in RQ3.

The workflow is intentionally branchable rather than mandatory. If threshold structure is weak, if performance changes erratically under small threshold adjustments, or if rectification degrades held-out fit without compensating interpretability gains, the appropriate branch is to keep the raw sparse baseline or use a hybrid representation that retains selected continuous terms. If the rectified sparse baseline is already interpretable enough, or if the anytime frontier fails the held-out non-inferiority gate, the appropriate branch is to stop before compression. Table 1 lists the conservative defaults used for the first pass through a new dataset.

The reusable modeling core is exposed through the public `cutlass` package [30], while the

Table 1. Conservative default settings for the first pass through the dissertation pipeline.

Item	Default	Reason
Rectifier choice	Use the domain-aligned one-sided or interval rule when it is already meaningful; otherwise start with a simple train-only range rule.	Keeps the representation aligned with event membership while avoiding arbitrary post hoc threshold search.
Threshold estimation	Estimate all bounds on training data only.	Prevents leakage from validation or test labels into the representation.
Penalty selection	Select penalty strength by cross-validation or from a stable near-plateau region.	Avoids overinterpreting a brittle point optimum.
Raw-versus-rectified comparison	Use the same split and the same operating metric on both paths.	Makes the switch decision attributable to representation rather than to evaluation mismatch.
Compression tolerance	Choose the smallest prefix satisfying $J_{\text{train}}(k) \geq 0.98\, J_{\text{train,base}}$.	Imposes a conservative training-side simplification rule before held-out adoption is considered.
Held-out adoption margins	Require paired-bootstrap lower bounds $\Delta\text{AUC} > -0.02$ and $\Delta J > -0.02$.	Provides an explicit non-inferiority gate before a compressed rule is treated as deployment-eligible.

dissertation repository preserves the study layer: notebooks, paired `scripts/_generate_*.py` generators, cached run summaries under `notebooks/runs_new/`, and manuscript-ready figures under `Figures/` [31]. Appendix C maps those artifacts to the figures, tables, and RQ chapters they support.

1.3 CONTRIBUTIONS

This dissertation makes five integrated contributions.

1. **A unified rectification-first longitudinal pipeline.** Building on prior conference work, the dissertation consolidates critical-range transformation, sparse logistic selection, and downstream rule extraction into a single end-to-end framework [25, 26].

2. **A scoped theoretical bridge to sparse-recovery conditions.** The dissertation extends tractable arguments linking transformed representations to improved sparse-support recovery and situates them against established lasso consistency theory and dependence-aware critiques [10, 11, 26].

3. **An anytime rule-compression stage for interpretable deployment.** Sparse coefficient models are converted into compact m-of-K rule forms that can be audited and deployed under explicit performance-tolerance policies [22, 27].

4. **A dissertation-level empirical validation package.** Beyond the prior papers, the dissertation adds repeated-resample stability and ablation studies, expanded sparse and direct interpretable baselines, cross-domain transfer and protocol audits, empirical boundary-condition stress tests, and held-out policy-based compression validation [32, 33].

5. **A reproducible implementation pathway.** The reusable modeling core is exposed through the public `cutlass` package, while the dissertation repository supplies notebook-backed study regeneration and cached evaluation outputs for the reported studies.

1.4 RESEARCH QUESTIONS

This dissertation is organized around three research questions that move from empirical utility to theoretical credibility and then to deployment usability:

- **RQ1:** Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?
- **RQ2:** Why does rectification work when it works, and how far can that mechanism be generalized beyond the current theoretical assumptions?
- **RQ3:** Can the resulting sparse solution be compressed into compact logical rules without unacceptable loss in operational discrimination?

Together, these questions are intentionally sequenced. There is little value in compressing a model into rules if upstream sparse attribution is unstable, and little scientific value in empirical gains if the mechanism producing those gains is not understood. Accordingly, RQ1 tests whether representation-level rectification improves sparse longitudinal selection in practice; RQ2 explains why those improvements should occur and where the theoretical boundary lies; and RQ3 asks whether the recovered sparse structure can be converted into models that are usable in real decision workflows [10, 11, 20, 21].

1.5 EVALUATION PRIORITIES

Across all three research questions, the dissertation evaluates the proposed framework using a common set of priorities:

- **Predictive discrimination:** metrics such as the area under the receiver operating characteristic (ROC) curve (AUC) and Youden's J-index at operational thresholds, interpreted in dataset context [26, 32, 33].

- **Attribution quality and stability:** support recovery behavior, lag localization fidelity, and stability across folds or resamples.
- **Interpretability complexity:** active-feature count, rule count, and rule length as direct proxies for human audit burden [21, 22].
- **Efficiency:** end-to-end training and inference time, including preprocessing and compression overhead.
- **Practical non-inferiority where needed:** when compressed rules are compared against upstream sparse baselines, the evaluation uses prespecified non-inferiority margins to determine whether performance loss is practically small [34].

1.6 EXECUTIVE MAP

For skimming readers, Table 2 compresses the dissertation's core claim into one front-of-manuscript view: the target problem, the pipeline, the regime where the method is strongest, the main boundary cases, and the one-sentence answer to each research question. Later chapters unpack each row in detail.

1.7 THESIS ORGANIZATION

Following this introduction, the dissertation is organized as follows. The background chapter isolates the conceptual ingredients needed for the rest of the manuscript: lag expansion, sparse

Table 2. One-page executive map of the dissertation for skimming readers.

Item	Executive summary
Target problem	High-dimensional lag-expanded longitudinal modeling under multicollinearity, where users need reliable feature-lag attribution and a deployable interpretable artifact rather than only strong raw prediction.
Pipeline	(1) Rectify lagged measurements into critical-range indicators, (2) fit a sparse L1-logistic baseline on the rectified representation, and (3) compress the active support into policy-controlled m-of-K rule candidates when a simpler deployment artifact is needed.
Target regime	The method is strongest when events are at least partly threshold-mediated and lag-structured, raw sparse fits remain predictive but unstable, and plausible one-sided or interval event windows appear at meaningful lags.
Main boundary cases	Benefits weaken when threshold structure is weak, threshold sensitivity is erratic, post-threshold magnitude still carries important signal, or compression fails held-out non-inferiority gates. In those cases, the correct branch is often to stay raw, use hybrid features, or stop before compression.
RQ1 answer	**Yes, conditionally.** Rectification improves support concentration, lag localization, and compactness in threshold-aligned regimes, but not uniformly across all datasets or untuned default settings.
RQ2 answer	**Yes, in a scoped form.** Zero-threshold sign binarization provides a tractable IC-oriented mechanism for why rectification can help, while one-sided practical rules remain the strongest empirical bridge beyond the exact theorem and interval or negative-correlation cases require more caution.
RQ3 answer	**Yes, but policy-controlled.** Compression works cleanly in controlled threshold-mediated settings; in real-world data it should be accepted only when the nominated candidate clears the held-out non-inferiority gates and rejected otherwise.

support recovery under dependence, and interpretability as a design requirement. The related-work chapter then shifts from concepts to literature comparison by positioning competing grouped, ordered, dependence-aware, and rule-based families relative to the dissertation's pipeline [28, 29, 35].

The subsequent chapters address the three research questions directly, and each now includes the approach details specific to that question: empirical evidence for rectification, theoretical analysis of the mechanism, and anytime rule compression for deployment-ready interpretability. The final chapters summarize limitations, future extensions, and deployment implications for longitudinal decision support. Proof details are discussed at a high level in the main text and provided step-by-step in the appendices.

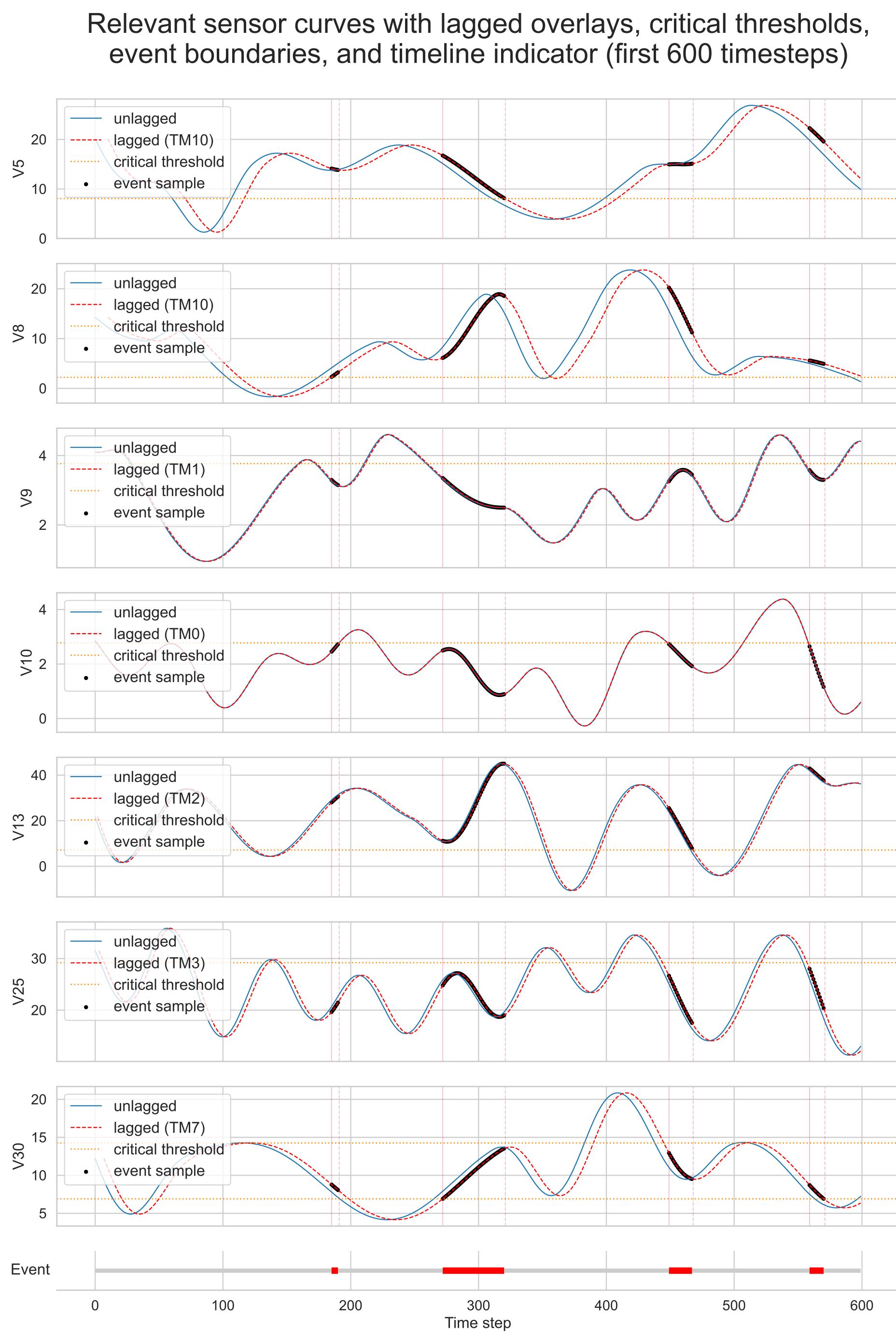


Figure 1. Threshold-and-lag behavior in longitudinal signals showing the first 600 timesteps of the actual synthetic data generated for the experiments discussed later in the document. Time periods shown in red in the timeline along the bottom of the figure show when an event is occurring. The bolded sections of the curve also reflect when an event is occurring on the lagged curve.

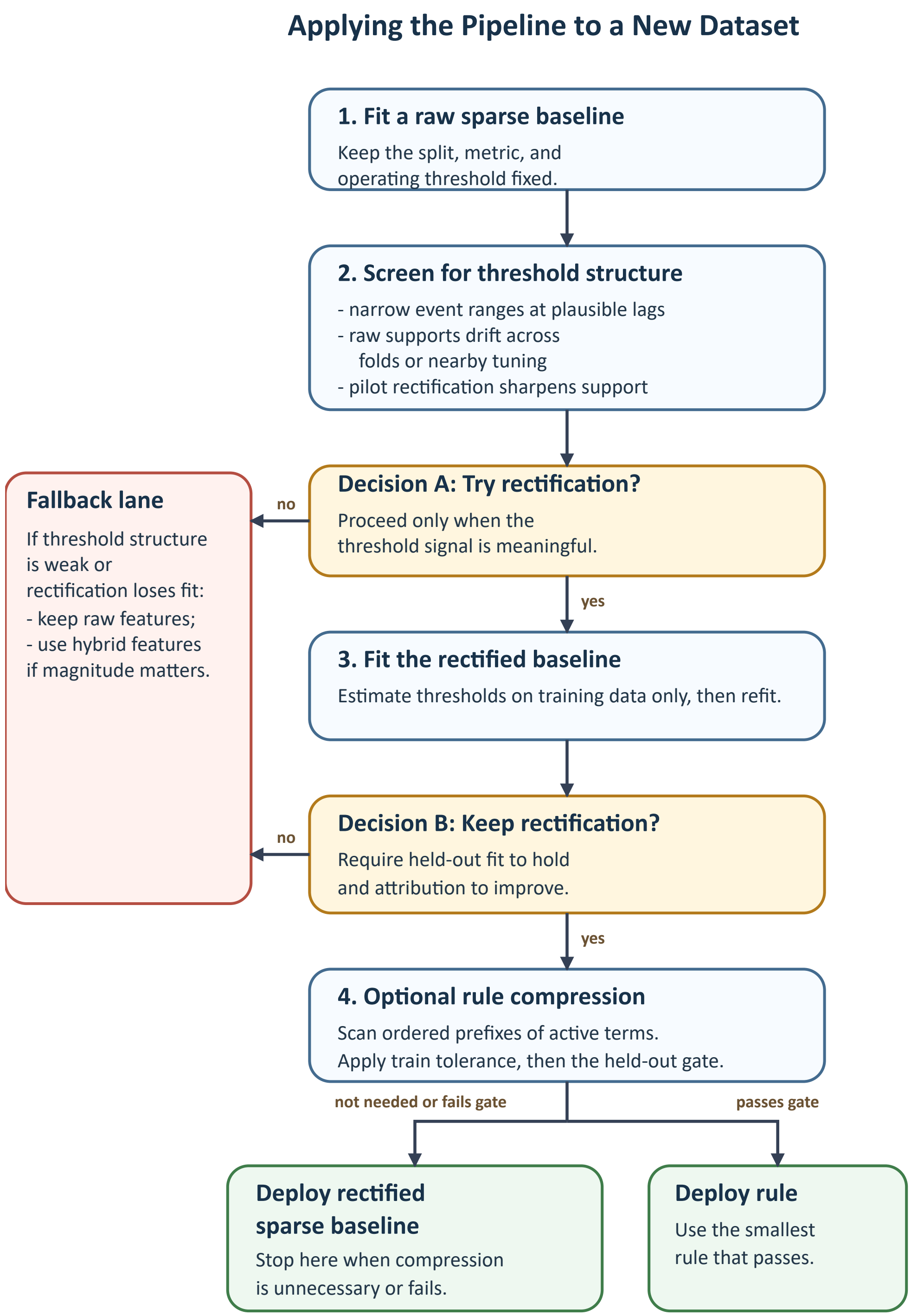


Figure 2. Operational workflow for applying the dissertation's pipeline to a new dataset. The sequence integrates the raw baseline, rectification diagnostic, optional compression stage, and explicit fallback outcomes into one ordered playbook.

CHAPTER 2

BACKGROUND

Sparse modeling with L1 penalties is a standard tool for feature selection in high-dimensional settings because it often yields compact models with useful predictive performance [8, 9]. In longitudinal settings, sparse selection is also used for lag attribution: selected coefficients identify not only which variables matter, but when they matter. This chapter supplies the conceptual background needed for the dissertation: why lag expansion creates difficult dependence, why sparse support recovery can fail under that dependence, and why interpretability is treated here as a design requirement rather than as a post-hoc reporting layer.

Longitudinal applications motivate this framing because repeated-measure data often contain delayed and heterogeneous effects across people, systems, or environments [1–3]. When temporal predictors are expanded across lags, model dimensionality and dependence both increase, so variable selection and interpretability must be considered jointly rather than as separate post-processing steps. Detailed method-family comparison is deferred to Chapter 3; the goal here is to establish the background lenses through which those comparisons will be interpreted.

2.1 L1-REGULARIZED MODELS AND FEATURE SELECTION

The LASSO and L1-regularized logistic regression induce sparsity by shrinking many coefficients to exactly zero, creating an embedded feature-selection mechanism inside model fitting [8]. This property is one reason L1 methods are widely used in longitudinal pipelines: selected nonzero lag terms can provide concise, testable hypotheses about delayed effects.

Modern implementations make these methods computationally practical at scale: coordinate-descent path algorithms with warm starts support efficient generalized linear modeling across many tuning values [28, 29]. This computational maturity is part of the dissertation's design logic: the proposed pipeline reuses stable sparse-learning infrastructure rather than requiring a custom optimizer. Specific extensions and comparator families built on top of this foundation are reviewed comparatively in Chapter 3.

2.2 SUPPORT RECOVERY AND THE IRREPRESENTABLE CONDITION

For feature selection, prediction accuracy alone is insufficient; the key question is whether selected supports match the true active set. Zhao and Yu formalized this issue through the irrepresentable condition (IC), showing that support recovery consistency for lasso depends critically on covariance structure [10]. When off-support predictors are too correlated with active predictors, sparse recovery can fail even when predictive fit appears strong.

This limitation is central in lag-expanded longitudinal designs, where adjacent lags and related channels are frequently collinear. Reviews of lasso under dependence consistently report this selection-versus-prediction tension, with false discoveries and unstable supports common under strong correlation [11]. The dissertation uses this dependence problem as the main background justification for representation-first intervention; concrete dependence-aware alternatives are compared later in Chapter 3.

From a linear-algebra perspective, the issue is closely tied to conditioning and inverse-Gram stability, which govern how sensitive selected supports are to perturbations in data and tuning [36]. For this reason, the dissertation treats representation-level correlation contraction as a first-class objective before sparse fitting, and uses IC-oriented reasoning as the theoretical lens for interpreting support behavior.

2.3 LONGITUDINAL LAG EXPANSION AND STRUCTURED SPARSITY

Lag expansion is the dominant way to represent temporal effects in classical supervised learning, but it can quickly create high-dimensional, redundant design matrices. One general response is to impose temporal or group structure directly in coefficient space, for example through monotone lag assumptions or shared-support assumptions [16, 37].

The conceptual distinction that matters for this dissertation is therefore not between one specific structured penalty and another, but between two intervention points: modify coefficient-space regularization, or modify feature-space representation before fitting. This work takes the second route. Chapter 3 reviews the concrete structured-sparsity families that define the strongest baselines within the first route.

2.4 INTERPRETABILITY AS A DESIGN REQUIREMENT

In high-stakes settings, interpretability is not an afterthought: end-users must understand why a prediction was made and be able to verify it against domain knowledge. Interpretable-model-first arguments emphasize that transparent model classes are often preferable to post hoc explanations layered on top of black boxes [20, 21]. This principle is especially relevant when longitudinal decisions affect operations, safety, or clinical workflows.

Rule-based models provide a natural interface for such settings because they are directly inspectable and can be validated by experts testing each feature and lag individually against specific rules. The dissertation adopts this perspective by treating rule compression as part of model construction, not merely post-hoc explanation. Chapter 3 then situates additive, rule-ensemble, LAD-style, and rule-list families relative to that design requirement.

2.5 BACKGROUND SUMMARY

The background needed for the dissertation is therefore compact. Lag expansion creates many dependent candidate predictors; support recovery under that dependence is structurally harder than prediction alone; and high-stakes deployment places interpretable, audit-ready model forms on equal footing with discrimination. These points motivate the dissertation's representation-first pipeline. The next chapter shifts from these conceptual ingredients to literature comparison and positions the dissertation relative to the major competing method families.

CHAPTER 3

RELATED WORK

This chapter reviews method families most relevant to the problem identified in this work: sparse longitudinal feature selection under strong dependence, with interpretable downstream decision logic. Unlike Chapter 2, which established the conceptual background needed to read the dissertation, this chapter focuses on literature comparison and positioning. The discussion is organized to move from core sparse-learning foundations to structured penalties, dependence-aware alternatives, and rule-based interpretability frameworks. The final sections position this work relative to prior work in the same method lineage.

3.1 SPARSE REGULARIZATION FOUNDATIONS

L1-regularized modeling is the canonical starting point for sparse feature selection. The lasso formulation introduced by Tibshirani established a convex mechanism that performs shrinkage and variable selection simultaneously, making it practical for high-dimensional settings [8]. In classification settings, L1-regularized logistic regression inherits the same sparsity behavior and is widely used when model compactness and feature attribution are important.

Early theoretical and empirical analyses also clarified why L1 penalties are often preferable to purely L2-regularized approaches when many predictors are irrelevant. In sparse-relevance regimes, L1 methods can scale better with growing nuisance dimensionality than rotationally invariant alternatives [9]. This intuition remains central for longitudinal lag-expanded designs, where many candidate lag terms may not be causally relevant.

A second reason this family became dominant is computational maturity. Coordinate-descent path algorithms provide efficient fitting across full regularization paths for generalized linear models and related objectives [28, 29]. These advances made sparse modeling feasible at operational scales and enabled cross-validated tuning in realistic workflows rather than one-off, expensive optimization.

However, plain lasso is not the endpoint of sparse modeling. Elastic net and adaptive lasso were proposed to improve behavior under correlation and to reduce some forms of selection bias [12, 13]. This work treats these methods as important baselines, but not as complete solutions for threshold-and-lag attribution in heavily collinear longitudinal representations.

3.2 SUPPORT RECOVERY UNDER DEPENDENCE

Predictive accuracy and correct feature recovery are distinct goals. Zhao and Yu formalized this distinction by showing that lasso model-selection consistency depends on covariance-structure conditions, particularly the irrepresentable condition (IC) [10]. In practical terms, when inactive predictors are too correlated with active predictors, support recovery may fail even if prediction remains competitive.

This issue is acute in lag-expanded longitudinal data, where neighboring lags and related channels often have strong dependence. Recent reviews emphasize that lasso and many derivatives can produce unstable supports and elevated false positives in dependent designs, especially under common tuning choices optimized for prediction rather than exact support recovery [11].

Dependence-aware variants address parts of this problem. Examples include transformed or weighted formulations for highly correlated predictors and robust rank-based alternatives that reduce sensitivity to link-function misspecification or heavy-tailed behavior [17, 38]. These methods improve robustness in specific regimes, but they generally do not provide a unified threshold semantics and lag-rule representation for operational interpretation.

From a matrix-analysis viewpoint, these selection instabilities reflect conditioning and inverse-Gram sensitivity, which directly affect sparse-support perturbation behavior [36]. This perspective motivates representation-level interventions before sparse fitting, rather than relying only on penalty redesign in the original feature space.

3.3 GROUP, BLOCK, AND ORDERED SPARSITY

Structured penalties were developed to stabilize selection when predictors have known organization. Early simultaneous-selection formulations for multiresponse models and later grouped penalties introduced mechanisms that select predictors at the block level rather than coefficient-by-coefficient [14, 37]. This is relevant in longitudinal settings where features can be grouped by channel, basis expansion, or lag family.

Subsequent work extended grouped sparsity to logistic settings and improved practical optimization. Group lasso for logistic regression and blockwise sparse regression improved feasibility in high-dimensional designs, while unified majorization-descent solvers strengthened practical convergence [15, 35, 39]. These methods remain strong comparators because they encode structural assumptions directly in the objective while retaining convexity in many cases.

Temporal structure has also been added explicitly through ordered penalties. Ordered lasso introduces monotonicity constraints across lag coefficients, reflecting assumptions such as decaying lag influence and yielding interpretable lag profiles when those assumptions hold [16]. This is a particularly relevant baseline for relevant experiments because it targets time-lagged sparsity directly.

Despite these advances, structured penalties still operate in the original correlated representation. They can improve stability and interpretability relative to plain lasso, but they do not inherently convert continuous trajectories into explicit threshold logic. This work positions its rectification step as complementary to these methods: change the representation first, then apply sparse learning.

3.4 CORRELATION-AWARE AND ALTERNATIVE FEATURE SELECTION FAMILIES

Beyond penalized linear models, several approaches attempt to reduce redundancy through direct relevance-redundancy criteria. Katrutsa and Strijov proposed a quadratic-programming feature-selection framework where pairwise similarity and target relevance are balanced in a convex relaxation [18]. This strategy can reduce redundant supports, but it is primarily pairwise in construction and does not natively express lag-threshold rules.

Robust multicollinearity-focused estimators provide another pathway. Methods combining robust regression ideas with conditioning-oriented penalties can improve estimation under contamination and ill-conditioning [40]. These approaches are useful for robustness analysis but typically prioritize fit stability over compact, human-auditable sparse logic.

Filter, wrapper, and embedded selection families remain widely used because they are flexible and model agnostic [41]. Recent hybrid ranking-plus-search pipelines also perform well in domain-specific settings [42]. Even so, these methods can depend on search heuristics, can yield unstable subsets under resampling, and often need extra modeling layers to express temporal threshold rules.

Accordingly, this work includes these methods primarily as conceptual comparators rather than as primary interpretability baselines. Their strengths are pragmatic feature screening and broad applicability; their limitations are weaker guarantees for sparse lag attribution and logic-level interpretability.

3.5 INTERPRETABLE MODELING AND RULE-LEARNING LITERATURE

Interpretability-focused literature increasingly argues that high-stakes deployments should prefer inherently interpretable models over post hoc explanations of black boxes whenever performance is competitive [20, 21]. This framing strongly influences the design choices in this work: interpretability is treated as a primary objective, not a secondary (post-hoc) reporting artifact.

Interpretable additive and rule-ensemble models are especially relevant comparators under this framing. Generalized additive boosting approaches can retain direct per-feature inspection while remaining highly competitive in classification settings [43]. Rule-ensemble methods such as RuleFit occupy a different middle ground: they fit sparse linear weights over conjunction rules extracted from shallow trees, trading a small number of auditable rules against the full complexity of unrestricted tree ensembles [44]. Alongside rule lists, these families are the most direct alternatives when a method claims improvement in operational interpretability rather than only in coefficient sparsity.

Rule-based models are central to this viewpoint. Classical Logical Analysis of Data (LAD) frameworks formalize binarization and logic-pattern extraction, including optimization-based cutpoint selection and pattern construction [23, 24]. These methods establish the value of explicit threshold logic for decision support, but can face combinatorial scaling limits in large, highly correlated feature spaces.

Modern rule-list methods demonstrate that transparent models can be both accurate and principled when optimization is carefully designed. Certifiably optimal rule-list learning shows that exact or bounded-search approaches can produce compact, auditable models with competitive predictive performance on structured tasks [22]. This work does not replicate global rule-list optimization, but it draws from the same interpretability objective and complexity-aware evaluation philosophy.

Relative to this literature, the strategy for this work is to anchor interpretability in sparse model fitting and then compress coefficients into logic-like forms through an "anytime" procedure. This avoids full combinatorial search while still yielding compact rule structures suitable for expert review.

3.6 LONGITUDINAL HIGH-DIMENSIONAL MODELING CONTEXT

Longitudinal methods in broader statistics and machine learning emphasize that repeated measures introduce heterogeneity, temporal dependence, and subgroup structure that can invalidate cross-sectional assumptions. Model-based clustering with regularized mixed-effects formulations demonstrates one advanced path for high-dimensional trajectory modeling [19]. These methods are powerful for discovering latent trajectory classes, but they optimize a different end goal than sparse, explicit lag-rule extraction.

Domain literature further illustrates why interpretable longitudinal modeling matters. Clinical and biological longitudinal studies report heterogeneous temporal pathways and delayed outcome behavior, reinforcing the need for methods that preserve temporal attribution [1, 2]. Similar heterogeneity appears in socio-environmental longitudinal surveys [3]. These contexts motivate models that can be inspected and discussed by domain experts rather than only ranked by black-box metrics.

Benchmark infrastructure also matters for evaluating longitudinal methods. Public anomaly and signal datasets used in this line of inquiry, such as HAI ICS telemetry and ionospheric radar-return data, provide realistic stress tests for lag attribution, sparse recovery, and interpretability tradeoffs [32, 33]. Related engineering contexts with threshold-driven transitions likewise support the relevance of range-based temporal reasoning [45].

3.7 SUMMARY OF GAPS IN PRIOR WORK

The reviewed literature reveals four persistent gaps that motivate this dissertation.

- Sparse penalties are computationally mature and often predictive, but support consistency remains fragile under strong lag-induced dependence [10, 11].
- Structured and dependence-aware penalties improve behavior in specific regimes, yet usually remain tied to raw correlated representations and do not produce threshold-native logic outputs [14, 16, 17].
- Rule-learning frameworks deliver transparency, but direct combinatorial optimization can be difficult to scale in high-dimensional longitudinal spaces [22, 23].
- Prior work in this dissertation line introduced the key pieces, but an end-to-end, reproducible, and extensively benchmarked synthesis is still needed [25–27].

The remainder of the dissertation addresses these gaps through unified methodology, theory-informed analysis, and comparative empirical evaluation.

Table 3. Cross-family positioning of the dissertation pipeline relative to the main competing method families discussed in this chapter.

Method family	Intervention point	Representation change?	Typical output form	Deployment-policy support	Dissertation's intended advantage
L1 / elastic-net / adaptive lasso	Coefficient regularization	No	Sparse weighted coefficients	Indirect; usually analyst-defined	Changes representation before fitting rather than relying only on penalty tuning under the original dependence structure.
Grouped / ordered sparsity	Structured coefficient regularization	No	Grouped or ordered sparse coefficients	Indirect; usually analyst-defined	Targets threshold logic and lag semantics through feature redesign rather than only through coefficient constraints.
Dependence-aware / QP / robust selectors	Redundancy control or robust objective	Usually no	Selected subsets or weighted coefficients	Indirect; usually analyst-defined	Couples redundancy reduction to explicit threshold semantics and a downstream interpretable rule path.
Interpretable additive / rule-ensemble models (EBM / RuleFit)	Model-class redesign	Mixed	Additive terms or sparse rule-weight mixtures	Partial; model-specific	Produces a shared rectified sparse baseline and, when justified, a smaller policy-screened rule artifact rather than leaving rule complexity mostly model-specific.
Direct rule-list / LAD-style methods	Combinatorial rule search	Yes	Explicit rule lists or logical patterns	Often direct, but search-heavy	Reuses mature sparse solvers and restricts logic compression to the active support rather than searching the full lag-expanded space from scratch.
This dissertation	Representation-first pipeline with sparse front end and post-fit compression	Yes	Rectified sparse baseline plus optional policy-controlled m-of-K rule	Explicit held-out policy	Balances support reliability, interpretability, solver reuse, and auditable deployment decisions in one longitudinal pipeline.

CHAPTER 4

RECTIFICATION IMPROVES SPARSE LONGITUDINAL SELECTION (RQ1)

RQ1 asks whether critical-range rectification can produce a stable sparse baseline with reliable feature and lag attribution in longitudinal settings. This chapter answers that question by showing where and why rectification improves sparse longitudinal selection.

This chapter develops the rectification-first lasso logic framework and extends it to later theory-backed case studies [25, 26]. Across studies, the central empirical pattern is consistent: when the data-generating process has threshold-and-lag structure, rectification tends to improve support concentration, lag localization, and sparse-model usability under multicollinearity pressure.

The organization is as follows: establish the sparse-selection problem, develop the critical-range logic engine from simple examples, formalize the fitted pipeline, evaluate it on simulated longitudinal data, and then test real-world applications. Additionally, this chapter will show: repeated-resample stability, stronger comparator packages, direct interpretable baselines, and cross-domain protocol audits.

4.1 BACKGROUND AND RQ1 SCOPE

The sparse learner is intentionally ordinary: an L1-regularized logistic model is fit on the rectified design and compared against a matched raw lag-expanded sparse baseline [8, 28, 29]. This keeps the empirical question focused. RQ1 is not asking whether a custom optimizer can be built for every longitudinal domain; it asks whether changing the representation makes a standard sparse learner recover more stable, lag-localized, and compact support under multicollinearity pressure.

Nonzero coefficients in the rectified fit are therefore interpreted as candidate feature-lag triggers, subject to the stability, attribution, and held-out discrimination checks reported below.

The diagnostic for using rectification is empirical. Rectification is a strong candidate when event windows repeatedly occupy relatively narrow one-sided or interval ranges at plausible lags, raw sparse supports drift across folds or nearby tuning values, the threshold has domain meaning, and a train-only pilot preserves held-out discrimination while improving support concentration, lag localization, or compactness. When no stable event ranges appear, when small threshold shifts produce large performance swings, or when the transformed model loses held-out performance without interpretability gains, RQ1 treats the raw or a hybrid representation as the correct branch rather than forcing rectification.

4.2 CRITICAL-RANGE LOGIC ENGINE

The following method is introduced through a simple logical setting before the full longitudinal pipeline is formalized. The point of this section is to show why a critical-range representation can make the relevant sparse support easier for lasso to isolate.

Consider a *toy example* showing how continuous features can obscure true relationships. Let $\mathbf{X}$ be a 10-feature matrix with columns $\mathrm{A}, \ldots, \mathrm{J}$. Suppose only A and D determine the outcome y. Fitting LASSO on raw continuous data can yield spurious coefficients (Figure 3, top). After *rectifying* features into $+1/-1$ indicators based on a range, irrelevant columns disappear, and A and D stand out (Figure 3, bottom).

Rectification effectively flattens continuous variations that can inflate feature correlations in this toy setting. Intuitively, columns with moderate overlap in continuous space can appear overly correlated. In the simplest sign-binarized setting, mapping to $\{\pm 1\}$ can reduce harmful linear de-

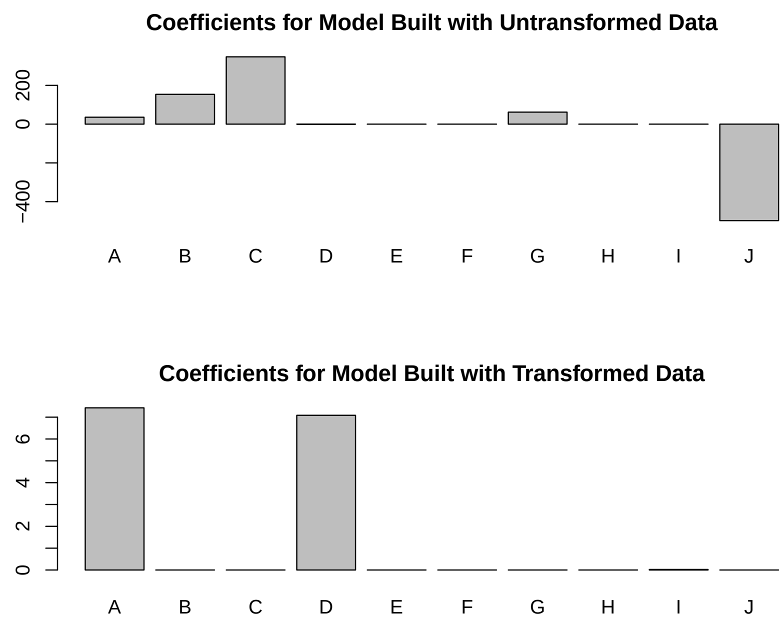


Figure 3. Coefficient vectors for LASSO on the same data before (top) and after (bottom) rectification. The non-zero feature coefficients indicate the selected features. These particular coefficient sets were calculated using the "glmnet" package from the R programming language, but similar results occur using LASSO implementations in other ecosystems.

pendence unless columns are identical or scalar multiples, while preserving coarse event structure. The RQ2 chapter later provides a tractable theoretical case for that mechanism under explicit assumptions; it is intended as an explanatory bridge rather than as a claim that every empirical benefit in the dissertation must arise only through that exact regime.

4.2.1 Toy Example Data

The toy example data demonstrates the behavior of a simple set of instrument readings. A matrix with features 'A' through 'J' shows that even with only 10 features, the coefficients become muddled and tend to gravitate toward spurious correlations rather than the true support. Columns A and D were designed to produce the result vector, while the others were largely random, with top elements intentionally adjusted to ensure they are **not** optimal for producing the result vector.

$$
\mathbf{X} = \begin{array}{c}
\begin{array}{cccccccccc} \mathrm{A} & \mathrm{B} & \mathrm{C} & \mathrm{D} & \mathrm{E} & \mathrm{F} & \mathrm{G} & \mathrm{H} & \mathrm{I} & \mathrm{J} \end{array} \\
\begin{bmatrix}
0.49 & 0.48 & 0.42 & 0.13 & 0.45 & 0.48 & 0.42 & 0.48 & 0.43 & 0.46 \\
0.12 & 0.52 & 0.46 & 0.47 & 0.42 & 0.47 & 0.49 & 0.47 & 0.51 & 0.48 \\
0.23 & 0.11 & 0.38 & 0.57 & 0.99 & 0.51 & 0.41 & 0.71 & 0.29 & 0.56 \\
\mathbf{0.48} & 0.40 & 0.59 & \mathbf{0.46} & 0.41 & 0.43 & 0.40 & 0.41 & 0.60 & 0.52 \\
0.83 & 0.45 & 0.48 & 0.54 & 0.44 & 0.44 & 0.23 & 0.41 & 0.46 & 0.49 \\
\mathbf{0.55} & 0.52 & 0.41 & \mathbf{0.58} & 0.40 & 0.60 & 0.51 & 0.42 & 0.42 & 0.45 \\
0.53 & 0.02 & 0.55 & 0.16 & 0.38 & 0.06 & 0.50 & 0.55 & 0.71 & 0.51 \\
\mathbf{0.54} & 0.60 & 0.56 & \mathbf{0.47} & 0.49 & 0.52 & 0.52 & 0.56 & 0.54 & 0.58 \\
0.33 & 0.22 & 0.59 & 0.49 & 0.47 & 0.49 & 0.59 & 0.47 & 0.91 & 0.57 \\
0.34 & 0.49 & 0.66 & 0.51 & 0.48 & 0.79 & 0.84 & 0.53 & 0.73 & 0.67
\end{bmatrix}
\end{array} \tag{1}
$$

4.2.2 Input Data Transformation

After transformation, the matrix is encoded using $\{+1, -1\}$. In this toy example, values between 0.4 and 0.6 (inclusive) are mapped to $+1$, and all other values are mapped to -1. For present purposes, the key point is that an event occurs when both features A and D lie simultaneously in the interval $[0.4, 0.6]$.

$$\mathbf{X} = \begin{array}{c} \begin{matrix} \mathrm{A} & \mathrm{B} & \mathrm{C} & \mathrm{D} & \mathrm{E} & \mathrm{F} & \mathrm{G} & \mathrm{H} & \mathrm{I} & \mathrm{J} \end{matrix} \\ \begin{bmatrix} 1 & 1 & 1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 \\ \mathbf{1} & 1 & 1 & \mathbf{1} & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & 1 & 1 & -1 & 1 & 1 & 1 \\ \mathbf{1} & 1 & 1 & \mathbf{1} & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & -1 & 1 \\ \mathbf{1} & 1 & 1 & \mathbf{1} & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & 1 \\ -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 & -1 \end{bmatrix} \end{array} \tag{2}$$

The corresponding Boolean response vector is:

$$y^T = \begin{bmatrix} F & F & F & \mathbf{T} & F & \mathbf{T} & F & \mathbf{T} & F & F \end{bmatrix} \tag{3}$$

It can be shown that any minimal logical AND combination of these features that produces the response vector shown must contain the 1st and 4th columns, since the first non-event row is excluded only by column D and the second non-event row is excluded only by column A.

Intuition

The next two figures provide a compact intuition for the regime targeted by the method. Consider a binary event derived from a continuous feature measured over time. Events occur when the feature enters a specific range, shown as the shaded area in Figure 4. This pattern is common in threshold-driven physical systems, where entering an operating region matters more than preserving every continuous fluctuation. As additional observations are collected, the event-associated range can be estimated more precisely. If only one feature is relevant, its rectified indicator aligns closely with the response vector, whereas irrelevant features either span too much of the observed range or fail to separate event times from non-event times.

Figure 4 contrasts a relevant and an irrelevant measurement. The orange curve in the lower panel cannot explain the event pattern on its own because its event-time values overlap too broadly with non-event values. Applied across features, the same logic helps distinguish event-aligned variables from variables that merely fluctuate alongside them.

The **critical range** (CR) of a feature is defined here as the event-associated value interval estimated from the observations. In the simplest version, it is bounded by the minimum and maximum values observed during event times. A useful critical range is one that helps include event rows while excluding non-event rows, either alone or in combination with other features.

4.2.3 Extending the Intuition to Multiple Triggers

The same logic extends naturally to settings in which an event occurs only when several features simultaneously lie within their own critical ranges. In that regime, LASSO becomes useful because it can identify a sparse conjunction of candidate feature indicators rather than relying on

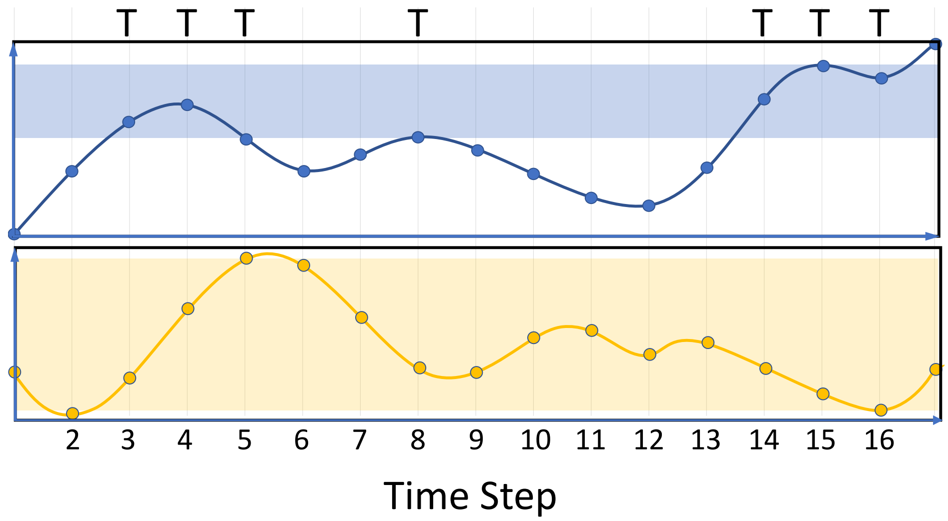


Figure 4. Measurement of two hypothetical features over sixteen time steps. Events are demarcated by the "T" over the curve, with the shaded area spanning the highest and lowest values at which the feature is measured during an event. Hypothetical ranges of interest are shown for an event. If the procedure for finding the critical range is performed on the orange curve (the critical range is shown by the orange shaded region on the bottom curve), it encompasses nearly the entire range of all measured values; this indicates that the feature represented by the orange curve is not relevant to event occurrence.

one feature alone.

For illustration, each feature's CR is estimated from its event-time observations. Figure 5 shows a two-feature case in which events occur only when both features lie in their respective critical ranges.

The observer sees only the response vector and the feature measurements across 17 time steps. In Figure 5, an event is possible when the top curve lies in the upper shaded region and the bottom curve lies in the lower shaded region. When one or both curves fall outside their respective ranges, the event is not supported. Time steps 10 and 13 define the lower shaded region's boundaries for the bottom curve; analogous event-time observations define the range for the top curve.

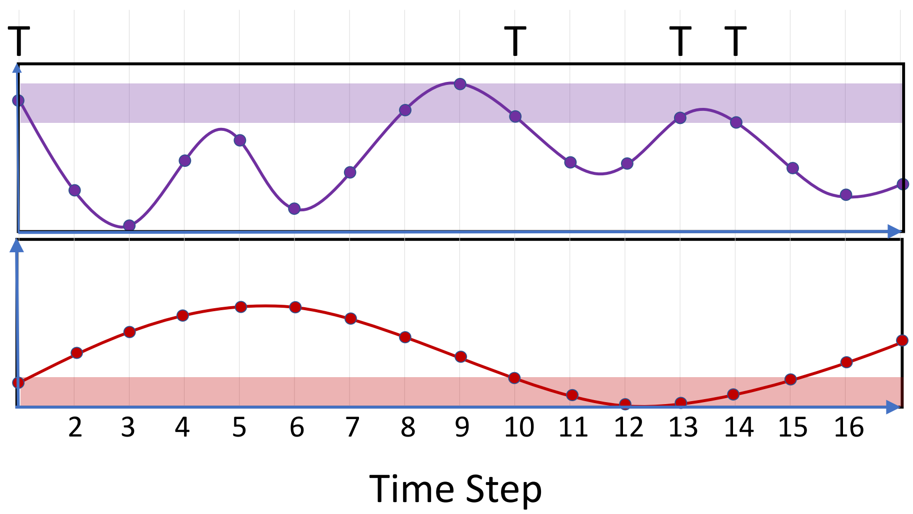


Figure 5. Hypothetical event occurrence in a more complex scenario. The occurrence of four events, each labeled by the "T" symbol, is fully described by the two feature curves shown. For an event to occur, the top curve must be within the upper shaded region at the same time that the bottom curve is within the lower shaded region.

4.2.4 Logical OR Complications

There is also the possibility of a harder case which must be acknowledged: several independent trigger sets can each produce the same event, creating an apparent logical OR relationship [25]. If the rectifier estimates a single event-associated range from all positive rows, the apparent range can become wider than the true range for any one trigger because events caused by another feature still contribute to the observed positive set. One possible remedy is to create contracted versions of candidate critical ranges and let lasso select among those versions as additional columns. In this dissertation, that idea appears as a scope condition rather than as the central RQ1 benchmark: rectification is most defensible when the learned range semantics align with the event mechanism.

4.2.5 Functional Fitting Procedure

The dissertation pipeline tests two stages: lag-expanded critical-range rectification followed by sparse baseline fitting. Starting from longitudinal measurements, the method creates a supervised design matrix in which each column corresponds to a specific variable-lag pair. For each feature column, the rectifier estimates an event-associated range on training data only and converts the raw measurement into an indicator denoting whether that lagged measurement lies inside or outside the learned range. This produces a threshold-aligned representation before any sparse optimization is performed. Algorithm 1 formalizes the process in pseudocode.

With the critical-range examples and functional fitting procedure in place, the chapter now turns from hand-constructed examples to formal evidence. The next section defines the RQ1 evaluation framework, after which the repeated-resample synthetic studies test whether the same attribution-sharpening effect holds under planted ground truth rather than only in an illustrative toy case.

Algorithm 1 Critical-Range Rectification and Sparse Baseline Fitting

Require: $\mathbf{X}, y, \lambda$

Ensure: $\mathbf{A}, \hat{y}, r_j\,[b,c]$

1: Let $r_j\,[b,c]$ be the critical range for feature vector $\mathbf{X_j}$, where b is the minimum and c is maximum defining that range.

2: **procedure** BINARIZATION($\mathbf{X}, y$)

3: Find Z , which is the subset of $\mathbf{X}$ for which y is one of the outcomes (which will be defined as TRUE), $Z \subseteq \mathbf{X}$, where $Z = \{\mathbf{X}|y = \mathbf{TRUE}\}$

4: Compute $r_{j=1..d} = \left[min(Z_j), max(Z_j)\right]$.

5: Compute $\mathbf{A}$, the matrix of transformed feature columns, such that:

$$a_{ij} = \begin{cases} +1 \text{ when } x_{ij} \geq r_j[b] \text{ and } x_{ij} \leq r_j[c] \\ \\ -1 \text{ when } x_{ij} < r_j[b] \text{ or } x_{ij} > r_j[c] \end{cases}$$

6: **end procedure**

7: Finally, compute the LASSO solution on the transformed data:

$\hat{y} = LASSO(\mathbf{A}, y, \lambda)$

4.3 EXPERIMENTAL DESIGN AND EVALUATION FRAMEWORK

The dissertation evaluates RQ1 using three complementary evidence streams:

- **Controlled synthetic studies** with known feature-lag ground truth, where attribution quality can be inspected directly.
- **Comparative baseline studies** against methods designed to address collinearity or redundancy through alternative mechanisms.
- **Real-world case studies** that test whether observed gains survive outside synthetic conditions.

The evaluation criteria are practical rather than purely theoretical: discriminatory performance (for example AUC and Youden's J), sparsity and support clarity, lag-level attribution fidelity, and end-to-end computational cost. This aligns with known limitations of penalty-only sparse methods under dependence, where prediction may remain strong while support recovery becomes unstable [10, 11]. The repeated-resample stability and stage-wise ablation package is summarized in Table 5 and Figures 7–10. The expanded baseline and cross-domain comparison package is summarized in Table 6 and Figures 18–22.

4.4 SIMULATED DATA EXPERIMENTS WITH KNOWN GROUND TRUTH

Synthetic longitudinal datasets are constructed so that events occur when a small subset of variables simultaneously enters critical ranges at specific lags. This creates a hard but controlled setting where lag expansion induces severe multicollinearity by design, closely mirroring the failure modes discussed in lasso consistency literature [10, 11]. In this regime, representation-level rectification is applied before sparse fitting and compared with untransformed baselines.

The synthetic results in Table 4 show a consistent pattern: rectified models improve discrimination and often reduce runtime relative to their raw counterparts. More importantly for RQ1, support concentration is visibly improved at the coefficient level when transformed inputs are used, which directly affects lag-attribution reliability. Note that Adaptive L1 gives the strongest exact-lag recovery in this synthetic panel, but this result should be interpreted as exploratory because the formal theoretical analysis is stated for the standard L1 estimator. The comparison is nevertheless useful: the raw adaptive-L1 fit still fails to recover the generating lags, whereas the rectified adaptive-L1 fit improves substantially. This suggests that rectification and adaptive weighting may be complementary, but a full theory for weighted/adaptive penalties is left for future work.

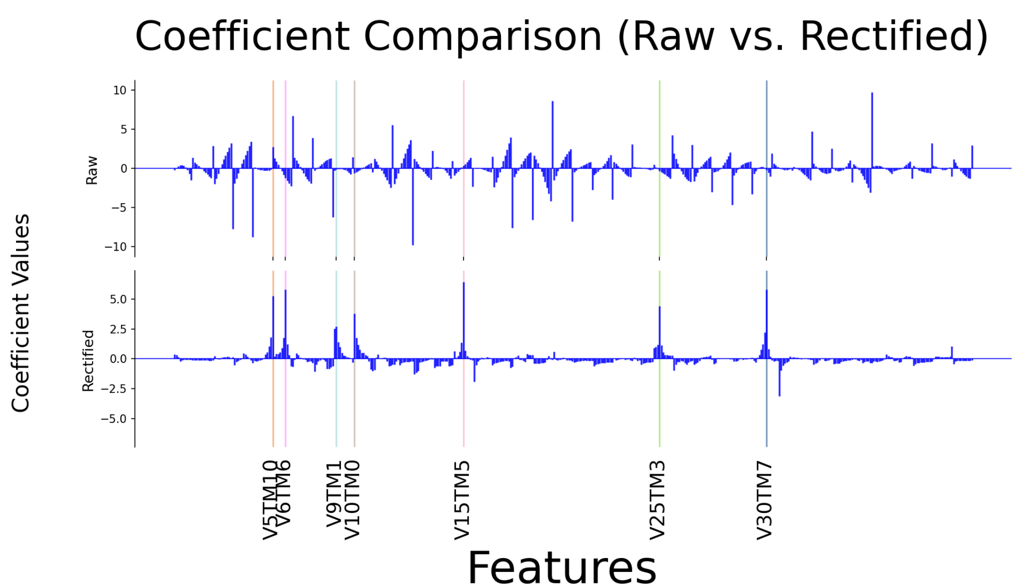


Figure 6. Coefficient profiles on synthetic data: transformed vs untransformed (ground-truth lag locations highlighted).

Figure 6 provides qualitative support for the same conclusion: rectification yields nonzero coefficients that align more closely with true causal lags, while untransformed fitting produces noisier support spread. This behavior is consistent with the idea that binarized critical-range mapping is expected to reduce harmful correlation effects before sparse optimization [25, 26].

4.4.1 Repeated-Resample Stability and Stage-Wise Ablation

To test whether the synthetic gains persist beyond a single seeded draw, this work adds a repeated-resample ablation on the Case 1 generator with fixed planted support and independent random seeds. Each resample is evaluated at three pipeline stages: (i) raw lag-expanded L1-logistic on the continuous design, (ii) rectified L1-logistic after training-only critical-range estimation, and (iii) a compact m-of-K rule derived from the rectified model by coefficient-ranked prefix compression, where m is the number of relevant features and K is the magnitude of the coefficients. This design separates the contribution of the representation change from the contribution of the final logic-compression step.

Table 5 and Figure 7 show that rectification changes the result qualitatively. The gain is not marginal. Relative to raw sparse fitting, the rectified model improves mean test AUC from 0.986 to 0.998, improves Youden's J from 0.882 to 0.971, improves exact-lag F1 from 0.048 to 0.619, and reduces the average number of nonzero coefficients from 340.7 to 190.0. The compression stage then preserves essentially all of the discrimination gain while further increasing exact-lag F1 to 0.759 and reducing the deployed support to an average of six terms. This is exactly the practical pattern sought in RQ1: better attribution with no meaningful loss of operational performance.

The stability figures make the same point from two different angles. Figure 8 shows that pairwise exact-support overlap rises from 0.056 for the raw baseline to 0.452 after rectification,

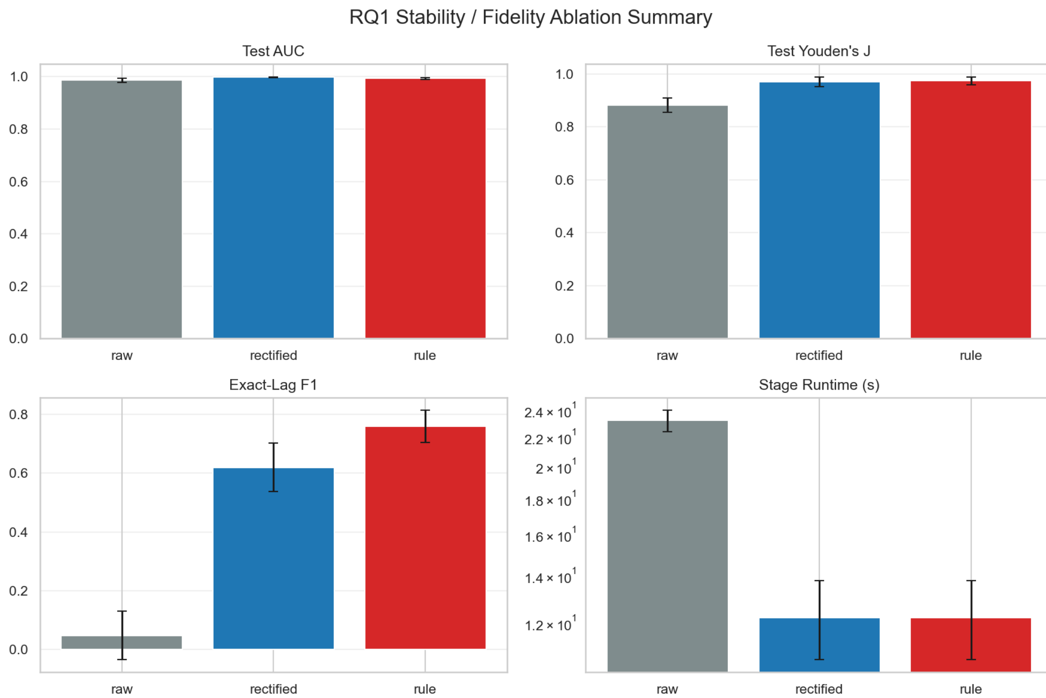


Figure 7. Repeated-resample ablation summary for the synthetic Case 1 study. Rectification improves discrimination and exact-lag recovery over raw sparse fitting, while the compressed rule retains nearly the same discrimination with stronger lag fidelity.

while the compressed rule reaches 0.600. Figure 9 shows why this matters: raw fitting spreads selection mass broadly across the lag-expanded design, whereas rectified models repeatedly return to the same planted variable-lag neighborhoods. The support recovered after compression is even more concentrated because low-weight spillover terms are removed instead of being retained for marginal predictive gain.

Figure 10 shows that the representation change does not impose a prohibitive computational tax. As the lag-expanded feature count grows from 440 to 840, the rectification step remains small (0.015 to 0.034 seconds), while the rectified sparse fit remains materially faster than the raw fit

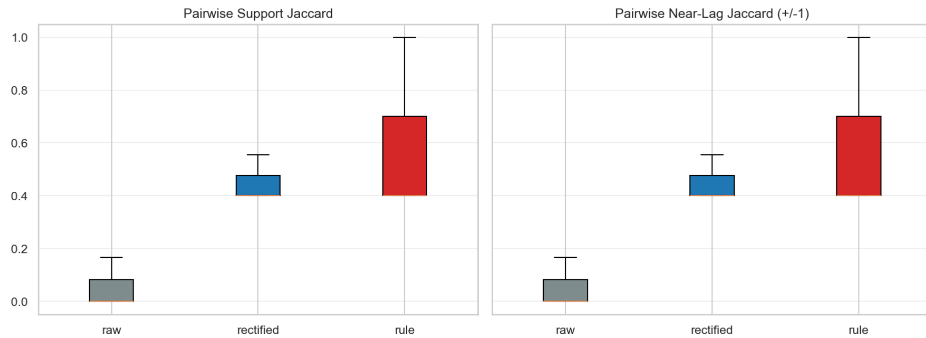


Figure 8. Pairwise exact-support and near-lag support Jaccard distributions across repeated resamples. Rectification produces a much more stable selected support than raw sparse fitting, and the compressed rule concentrates support most aggressively.

(1.19 vs. 2.24 seconds at 440 features, and 2.30 vs. 4.35 seconds at 840 features). This matters for RQ1 because a stable baseline must remain practical under repeated resampling, ablation, model selection, and deployment refresh cycles.

4.5 COMPETING FEATURE-SELECTION BASELINES

RQ1 comparisons also include methods intended to address redundancy or grouped dependence through different mechanisms. Group and block-structured sparse families remain important comparators because they are explicitly designed for correlated predictor settings [14–16, 35, 39]. Across these studies, representation-level rectification combined with standard lasso provides clearer lag attribution in threshold-triggered synthetic settings.

A focused comparator is the quadratic-programming (QP) selector of Katrutsa and Strijov, which optimizes a relevance-redundancy objective using pairwise similarity structure [18]. The method is conceptually aligned with the goal of reducing redundancy, but it emphasizes pairwise

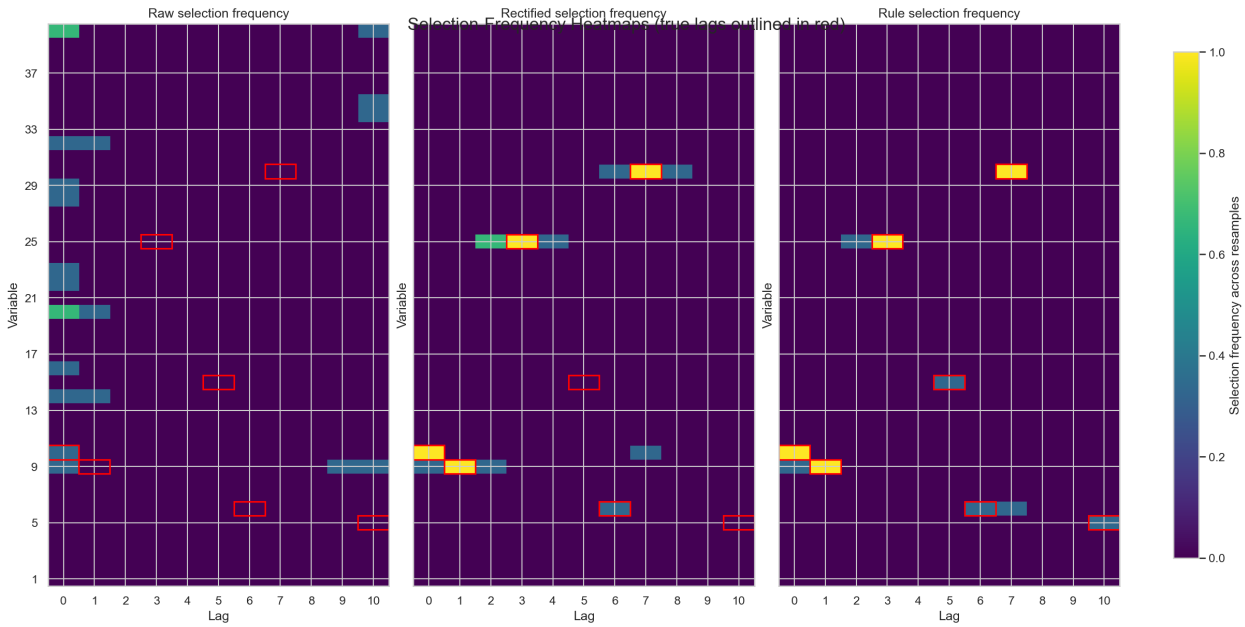


Figure 9. Selection-frequency heatmaps across repeated resamples. Red boxes mark the planted variable-lag pairs. Raw fitting diffuses support over unrelated variables and lags, whereas rectified and compressed models concentrate mass near the planted structure.

criteria and does not directly encode multi-lag conjunctive trigger logic.

In earlier comparisons using the original Katrutsa–Strijov reference code in R, the quadratic-programming approach often identified related variables but missed the correct lag localization and exhibited substantially higher computational cost, with runtime differences on the order of roughly 100–200x relative to rectification plus lasso in that software configuration. The consolidated benchmark reported later in this chapter uses a Python reimplementation of the same selector so that all methods can be compared on a more level software footing. Under that normalized setting, the runtime gap narrows substantially, but the attribution result does not change: the quadratic-programming route may identify related variables while still missing the correct lag

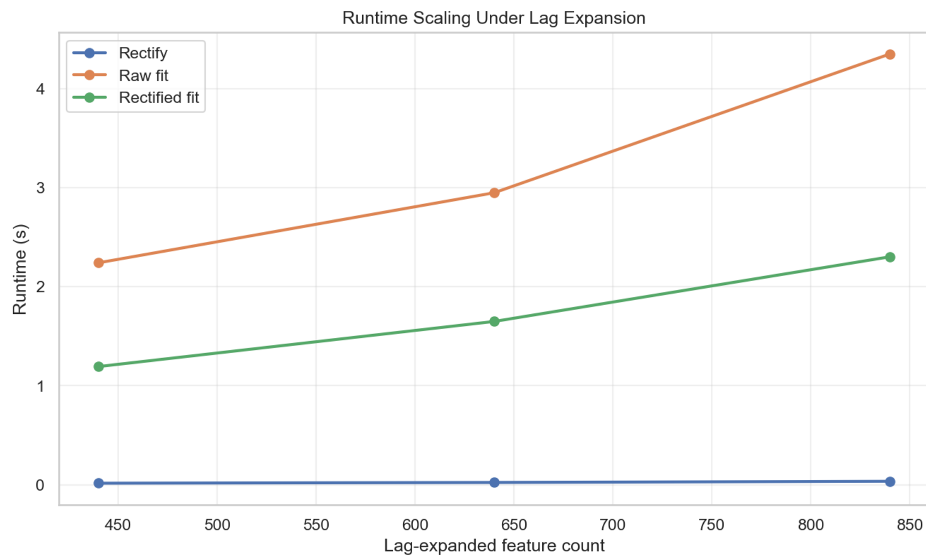


Figure 10. Runtime scaling as the lag-expanded feature count grows. The rectification step remains inexpensive, and the rectified sparse fit remains materially faster than the raw fit across the tested lag windows.

localization. For RQ1, this matters because a "stable sparse baseline" is not only a statistical target; it must also remain practical to run repeatedly across folds, ablations, and deployment refresh cycles.

4.5.1 Expanded Correlated-Feature Baseline Package

To broaden the comparator set beyond a single redundancy-oriented method, the dissertation adds a consolidated synthetic benchmark covering raw lasso, elastic net, adaptive lasso, group lasso, ordered-prefix selection, and the Katrutsa–Strijov quadratic-programming selector, alongside the rectified lasso baseline. All methods are evaluated on the same lag-expanded synthetic case used to stress attribution under multicollinearity. For comparability, the Katrutsa–Strijov se-

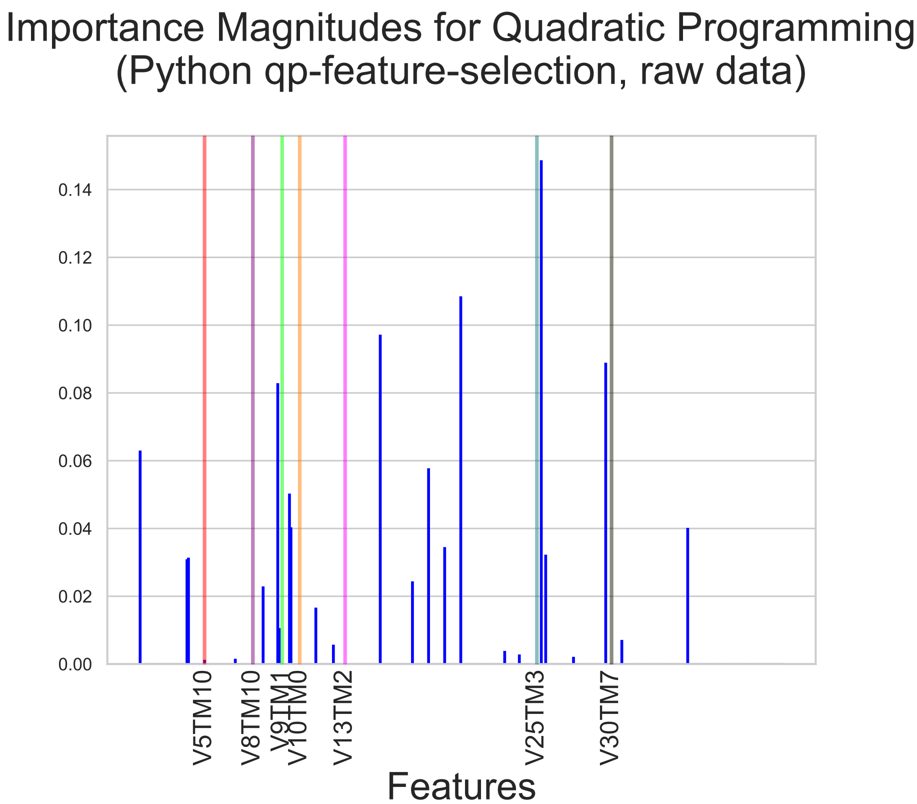


Figure 11. Quadratic-programming importance magnitudes on the synthetic dataset.

lector is run here through a Python reimplementation rather than through the original R reference code used in earlier one-off timing checks. This table uses a separate normalized single-run benchmark on one fixed 70/30 split of Case 1 ($rseed = 1234$), so its values should not be compared row-for-row with the repeated-resample mean $\pm$ standard deviation summaries in Table 5.

Table 6 sharpens the baseline story. Raw lasso, elastic net, and group lasso all remain reasonably predictive, but none of them recover the planted lags exactly. Their support stays concentrated on correlated surrogates in the original continuous representation rather than on the true trigger lags. For Group Lasso, this is despite the groups being explicitly defined by which baseline variable was represented, with all lags for that baseline variable belonging to that group. Adaptive lasso, ordered-prefix selection, and QP reduce model size more aggressively, but they do so by pruning within the same raw representation and therefore sacrifice discrimination without solving the attribution problem. The rectified baseline is the only method in this expanded package that simultaneously attains the best AUC, the best Youden's J, and nonzero exact-lag recovery. For RQ1, this matters more than a marginal runtime win because the question is not merely whether a sparse model predicts well; it is whether the selected support remains interpretable at the feature-lag level.

4.5.2 Direct Interpretable-Model Baseline Package

Because RQ1 is partly an interpretability claim rather than only a sparse-selection claim, a second synthetic benchmark was added against three inherently interpretable model families: an additive Explainable Boosting Machine (EBM), a RuleFit rule ensemble, and a greedy rule-list baseline. All methods were evaluated on the same Case 1 generator across the same three repeated resamples used for the stability study. The fit summary reports held-out discrimination, execution time, and structural complexity, while the selection-frequency heatmap records the top-K exact variable-lag locations ($K = 7$) so lag-recovery behavior can be compared on the same visual scale. That top-K view is a benchmark comparison device, not a direct measure of model utility when the true number of relevant variables is unknown. In this synthetic package, K is known only because the planted support is known. This distinction matters because EBM, RuleFit, and the greedy rule list do not natively tell the analyst how many active locations should be retained for interpretation, whereas the CUTLASS logic-compression step returns a rule size k as part of the fitted output.

The direct-comparator package sharpens the interpretability comparison. EBM achieves the strongest mean discrimination and, in the top-K pane of the dual-view heatmap, repeatedly returns

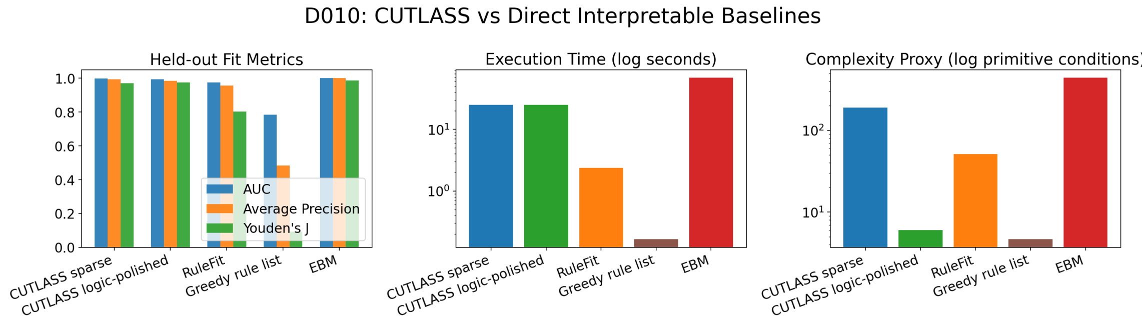


Figure 12. Direct interpretable-model summary on synthetic Case 1. EBM gives the strongest raw fit, but the CUTLASS logic-polished model achieves competitive discrimination with far lower complexity than EBM and materially stronger fidelity than RuleFit or the greedy rule list.

to the planted lag neighborhoods. However, that top-K display should be read as a controlled comparison device rather than as a direct statement of what a practitioner would know to inspect in advance. The full-support pane shows why the distinction matters: for EBM and the other non-CUTLASS baselines, the analyst must still decide how many active locations to keep, and EBM in particular remains dense across the lag-expanded design. CUTLASS logic contributes an additional piece of information by selecting a deployable rule size k natively through its compression policy.

Even with that advantage, EBM in the purely additive configuration used here keeps all 440 exact-lag additive terms active and requires roughly 33 seconds, about 2.6$\times$ the runtime of the CUTLASS pipeline. Runtime comparisons here are package timings rather than implementation-neutral complexity measurements: the current CUTLASS prototype is implemented in Python and NumPy, whereas EBM relies on a compiled native backend. An implementation-neutral comparison would almost certainly be significantly worse for EBM. The issue in this benchmark is therefore not combinatorial rule growth, but dense additive growth: as the lag-expanded feature set increases, the model can retain many nonzero term functions whose combined chatter obscures attribution. If pairwise or higher-order interactions were enabled, this complexity would grow more aggressively, becoming at least quadratic and eventually combinatorial in the number of lagged inputs.

At the other extreme, the greedy rule list is extremely fast and small, yet its discrimination collapses and its heatmap almost never lands on the exact planted lags. RuleFit is a more serious compact baseline: it remains fast at about 1.5 seconds and predictive enough to be plausible in practice. Even so, it trails materially on Youden's J, averages about 29 unique lagged features, and its support heatmap misses several planted lags across seeds.

The CUTLASS direct-comparator result is therefore not that it dominates every interpretable family on raw fit. EBM wins that comparison. The relevant finding is instead that, in the threshold-aligned synthetic regime, CUTLASS logic achieves the best compact attribution-performance tradeoff: mean AUC 0.994, mean J 0.974, and only six retained feature-lag terms on average. In the top-K pane used for controlled comparison, both CUTLASS logic and EBM recover about five of the seven planted lags on average, but the full-support pane makes clear that CUTLASS does so with a deployable rule-sized representation and with a natively selected rule size k, rather than a 440-term additive model that still requires the analyst to decide how much of the active support to inspect. This direct comparator package therefore strengthens rather than weakens the RQ1 claim: there are strong transparent alternatives, but the rectification-plus-compression pipeline remains the most favorable compact method in the exact threshold-and-lag regime it was designed to address. Additionally, the CUTLASS package also yields the relevant critical ranges associated with the calculated support, which are valuable analysis artifacts in their own right.

4.6 REAL-WORLD DATA APPLICATIONS AND INTERPRETABILITY TRADEOFFS

Synthetic evidence is necessary but not sufficient. The real-world studies test whether the same pattern appears in operationally relevant longitudinal data. The HAI industrial control benchmark is particularly useful because it includes realistic multi-sensor temporal dynamics with labeled attack scenarios and known subsystem context [32]. Additional evidence from historical ionospheric radar work supports the relevance of lagged signal discrimination settings where sparse attribution can complement raw predictive performance [33].

4.6.1 Worked HAI Exemplar for the Workflow

To make the branch logic concrete, the manuscript includes one executable worked example built from the HAI industrial-control benchmark analyzed in this section and in the RQ3 chapter. The accompanying notebook `notebooks/walkthrough.ipynb`, available with the manuscript artifacts in the dissertation repository, uses the custom-built `cutlass` Python package to run the full workflow on the HAI turbine-loop case `attack_p2(a1)`: raw lag-expanded sparse baseline, train-only threshold-alignment pilot, branch selection, strict-policy anytime compression, and final deployment recommendation. The point of this exemplar is operational clarity rather than a separate scientific claim.

Table 8 compresses the worked example into one page. On this case, the raw baseline reaches test AUC 0.949 and $J = 0.574$, while the matched rectified pilot improves to AUC 0.989 and $J = 0.930$ and shifts absolute coefficient mass toward the attacked $P2$ subsystem ($0.194 \rightarrow 0.599$ share). The workflow therefore takes the "rectify" branch, promotes the rectified sparse fit to the new baseline, and then evaluates the anytime frontier under the dissertation's strict default policy.

That policy selects $k = 14$ from 39 active logical conditions ($2.79\times$ compression); the held-out deltas are $\Delta\mathrm{AUC} = -0.011$ and $\Delta J = +0.015$, with paired-bootstrap lower bounds -0.016 and -0.005, both above the dissertation's -0.02 non-inferiority margins. The resulting deployment artifact is therefore the compressed rule rather than the full rectified sparse baseline. Appendix D provides the rule-card template that should be filled out before an accepted compressed rule is handed to an operational reviewer.

4.6.2 HAI and UNICEF Case Studies

Table 9 shows that the HAI case favors transformed models on both discrimination and attribution clarity. In this setting, transformed models concentrate coefficients on turbine-related channels aligned with known attack context, whereas untransformed models spread attribution more diffusely.

One subtle but important point in the HAI case is that residual nonzero coefficients in the $P4$ block should not automatically be interpreted as attribution failure. In HAI 1.0, $P4$ is the hardware-in-the-loop simulator that explicitly couples the boiler, turbine, and water-treatment subsystems at the signal level, so a turbine-focused attack can legitimately induce secondary responses in simulator steam-side variables rather than remaining perfectly isolated within $P2$ [32]. For that reason, a transformed sparse model whose largest concentration remains in $P2$ but which retains a smaller, coherent set of $P4$ terms is more plausibly reflecting real cross-process dynamics than exposing a solver bug. In this benchmark, the appropriate attribution standard is therefore dominant concentration on the attacked subsystem with only limited, physically interpretable spillover into the coupled simulator block, not literal exclusivity of every nonzero coefficient.

Cross-dataset behavior is not uniform. In the UNICEF [46] case, untransformed models can achieve stronger raw discrimination metrics while transformed models remain substantially sparser and easier to interpret. This pattern is consistent with prior reports that rectification benefits are context dependent: strongest in threshold-and-lag aligned regimes, and more mixed when data-generating structure is less compatible with the assumed critical-range mechanism [26].

4.6.3 Cross-Domain Transfer and the Goose Bay Protocol Case Study

The practical point of this subsection is to treat the Goose Bay ionosphere data as a protocol-sensitivity case study. As such, the broader transfer package applies one standardized generic protocol across four HAI attack subsets and the UCI ionosphere radar benchmark, then compares raw and rectified lasso under the same train/test handling and fixed penalty settings. Figure 18 shows that the transfer pattern is mixed rather than uniformly positive. Rectification clearly improves both AUC and Youden's J on the HAI turbine-loop attack, improves J substantially on the combined $P1/P2$ attack, is essentially neutral on one attack subset, and loses ground on another. The ionosphere row is also negative under this standardized protocol, with $\Delta J = -0.059$. This is informative rather than contradictory: it affirms that the RQ1 claim must remain conditional on alignment between the rectification rule and the structure of the underlying domain.

The ionosphere result is well suited to show how variances in protocol can affect the outcome. Previous Goose Bay results [25] seem to show a positive outcome with respect to the rectification procedure. Figure 19 resolves that apparent contradiction by keeping the two views distinct. The published 2022 Goose Bay reference moves from $J = 0.695$ to $J = 0.854$ after rectification, a gain of 0.159. The standardized generic transfer probe instead moves from $J = 0.775$ to $J = 0.716$, a loss of 0.059. These are not the same experiment. The published result used the original study protocol, while the generic transfer probe deliberately used the same portable defaults as the other cross-domain datasets.

To determine why the Goose Bay views disagree, a repeated-split robustness audit was conducted that changes one protocol component at a time: train/test split, rectification rule, and penalty strength. Figure 20 shows that the split itself is not the main driver. Across 101 seeds, the mean rectified-minus-raw gain remains near zero when moving from the generic probe ($\Delta J = 0.006$) to a split-matched cutlass protocol ($\Delta J = 0.003$). The larger change appears when the rectifier semantics change from the default quantile rule to the positive-range min/max rule used in the earlier pipeline. Under the same $67/33$ split and $C = 0.1$, the mean gain rises to 0.066 and is positive on 85.1% of seeds. Increasing the penalty parameter to $C = 10$ under the same min/max rectifier raises the mean gain further to 0.148 and makes it positive on every tested seed. On the original Goose Bay seed 2345, the progression is $-0.033 \to 0.065 \to 0.109 \to 0.210$, which again indicates that rectifier semantics and penalty strength matter more than split matching alone.

The protocol audit then motivates a direct sensitivity analysis of the two main tuning choices left in the standardized cutlass path: quantile bounds and penalty strength. Figure 21 shows that the default $25/75$ quantile rectifier is close to neutral on Goose Bay ($\Delta J = 0.003$ on average), while wider windows improve steadily: $20/80$ raises the mean to 0.063 and $10/90$ to 0.082. Narrower windows move in the wrong direction, with $30/70$ at -0.083 and $35/65$ at -0.212. The implication is not that one fixed quantile choice is universally optimal, but that the ionosphere case is unusually sensitive to the width of the positive-range window and behaves more favorably as the rule becomes more min/max-like.

Figure 22 extends the same idea to the lasso penalty. A log-uniform sweep over 100 values from 0.01 to 100 under the Goose Bay split and min/max rectification shows that the standardized choice near $C = 0.1$ sits well below the robust high-C plateau. Near $C = 0.108$, the mean gain is only 0.077. Over the broader 10–100 range, the mean gain stabilizes between roughly 0.149 and 0.157, and almost every seed is positive. There is also a narrow low-C sweet spot around $C \approx 0.025$ where the mean gain peaks near 0.200, but the lower standard-deviation panel shows that this region is materially less stable than the broader high-C plateau. The overlaid 10-fold cross-validation curve likewise flattens in the larger-C regime. For RQ1, the implication is practical: a standardized default such as $C = 0.1$ can understate the value of rectification on Goose Bay even though the domain still supports a strong rectified solution under a nearby and more stable operating region.

Taken together, the cross-domain and Goose Bay results refine the empirical claim rather than weakening it. The standardized transfer panel shows that rectification does not dominate uniformly across every dataset when one portable protocol is imposed. The Goose Bay audit then shows why one apparent counterexample appears: the ionosphere benchmark is highly sensitive to how the positive range is defined and to how strongly the lasso penalty suppresses correlated signal fragments. That is a useful scientific boundary. It lends credence to the claim that rectification-first sparse learning is strongest in threshold-mediated regimes when the rectification semantics are aligned with the event structure, not that one untuned default will be optimal for all situations. Further, these results are consistent with the hypothesis that, in threshold-mediated regimes, aligned rectification often improves the likelihood of more favorable irrepresentable-condition behavior. Empirically, once the rectifier definition and penalty are well matched to the domain, indicated by the cross-validation curve, rectification rarely hurts and usually helps.

The practical takeaway is therefore specific. Goose Bay data shows that a portable default can obscure a strong rectified solution when the positive-range semantics and penalty are mismatched to the domain. The Goose Bay data is therefore treated as evidence for sensitivity-aware rectification, avoiding blanket application of previously successful parameters.

4.7 PRACTICAL DIAGNOSTIC FOR RECTIFICATION

The Background and RQ1 Scope section stated the operational diagnostic before the evidence: rectification should be tried when train-only event windows are stable, domain-meaningful, and associated with better support concentration, lag localization, or compactness without unacceptable held-out loss. The results in this chapter refine that diagnostic rather than replacing it. Synthetic Case 1 satisfies the diagnostic cleanly; the HAI turbine-loop case satisfies it strongly; UNICEF and the generic ionosphere probe show why the branch rules matter. In those weaker or protocol-sensitive cases, the correct operational conclusion is not that rectification is invalid, but that raw or hybrid representations may be the more defensible baseline unless the rectifier semantics and penalty regime can be aligned with the domain.

4.8 METRIC AND NOTATION GUIDE

Table 10 provides a compact reference for the main empirical metrics and notation used across the RQ chapters. Chapter-specific sections give the detailed protocol context; this table is intended as a nearby lookup point for the dense result tables and deployment-policy language that follow.

Table 4. The numbers presented are for the synthetic case showing rectified and raw comparisons from the experiments conducted for this work.

Method	Status	Rel. Time	AUC	Youden's J	F_1 Max	F_1 Exact Lag
L1	Rectified	1.0	0.997	0.946	0.966	0.571
L1	Raw	1.7	0.995	0.892	0.932	0.000
Elastic Net	Rectified	1.0	0.997	0.946	0.966	0.571
Elastic Net	Raw	1.5	0.995	0.892	0.932	0.000
Adaptive L1	Rectified	1.0	0.992	0.929	0.959	0.714
Adaptive L1	Raw	1.3	0.996	0.897	0.938	0.000
Group Lasso	Rectified	0.6	0.996	0.915	0.947	0.429
Group Lasso	Raw	0.6	0.993	0.892	0.926	0.000
Ordered Prefix	Rectified	1.2	0.993	0.920	0.953	0.000
Ordered Prefix	Raw	1.1	0.993	0.897	0.932	0.000
QP + L1	Raw	3.0	0.915	0.629	0.754	0.000
Random Forest	Rectified	2.2	0.995	0.911	0.945	0.286
Random Forest	Raw	4.5	0.999	0.964	0.987	0.143

Table 5. Repeated-resample stability and ablation summary for the synthetic Case 1 study. Values are mean $\pm$ standard deviation across three independent resamples. Pairwise stability is summarized with exact-support Jaccard over all seed pairs.

Method	AUC	Youden's J	Exact-lag F1	Pairwise Jaccard	Total Nonzero	Runtime (s)
Raw	0.986 ± 0.008	0.882 ± 0.027	0.048 ± 0.082	0.056 ± 0.096	340.667 ± 105.396	23.142 ± 0.815
Rectified	0.998 ± 0.002	0.971 ± 0.019	0.619 ± 0.082	0.452 ± 0.090	190.000 ± 117.796	12.196 ± 1.562
Rule	0.994 ± 0.003	0.974 ± 0.015	0.759 ± 0.056	0.600 ± 0.346	6.000 ± 3.464	12.201 ± 1.562

Table 6. Expanded correlated-feature baseline benchmark on the synthetic Case 1 study (normalized single-run benchmark on a fixed 70/30 split, $rseed = 1234$). Exact-lag F1 evaluates whether the selected support lands on the planted variable-lag pairs rather than merely nearby correlated surrogates.

Method	AUC	Youden's J	Exact-lag F1	Nonzero	Runtime (s)
Rectified L1	0.997	0.946	0.571	344	2.947
Raw L1	0.995	0.892	0.000	409	4.755
Raw Elastic Net	0.991	0.888	0.000	364	5.176
Raw Adaptive L1	0.982	0.835	0.000	53	2.110
Raw Group Lasso	0.990	0.888	0.000	385	1.488
Ordered Prefix	0.982	0.786	0.000	30	0.912
QP + L1	0.894	0.603	0.000	15	2.843

Table 7. Direct interpretable-model baseline benchmark on the synthetic Case 1 study. Values are mean ± standard deviation across three independent resamples. `Features used` counts active additive terms or unique lagged features directly referenced by the fitted model.

Method	AUC	Avg. Precision	Youden's *J*	Runtime (s)	Features Used
CUTLASS Sparse	0.998 ± 0.002	0.994 ± 0.006	0.971 ± 0.019	12.50 ± 1.30	190.0 ± 117.8
CUTLASS Logic	0.994 ± 0.003	0.983 ± 0.012	0.974 ± 0.015	12.50 ± 1.30	6.0 ± 3.5
RuleFit	0.974 ± 0.026	0.955 ± 0.035	0.803 ± 0.112	1.55 ± 0.03	28.7 ± 7.6
Greedy Rule List	0.785 ± 0.054	0.484 ± 0.111	0.092 ± 0.067	0.11 ± 0.03	4.7 ± 0.6
EBM	1.000 ± 0.000	1.000 ± 0.000	0.987 ± 0.004	32.96 ± 3.34	440.0 ± 0.0

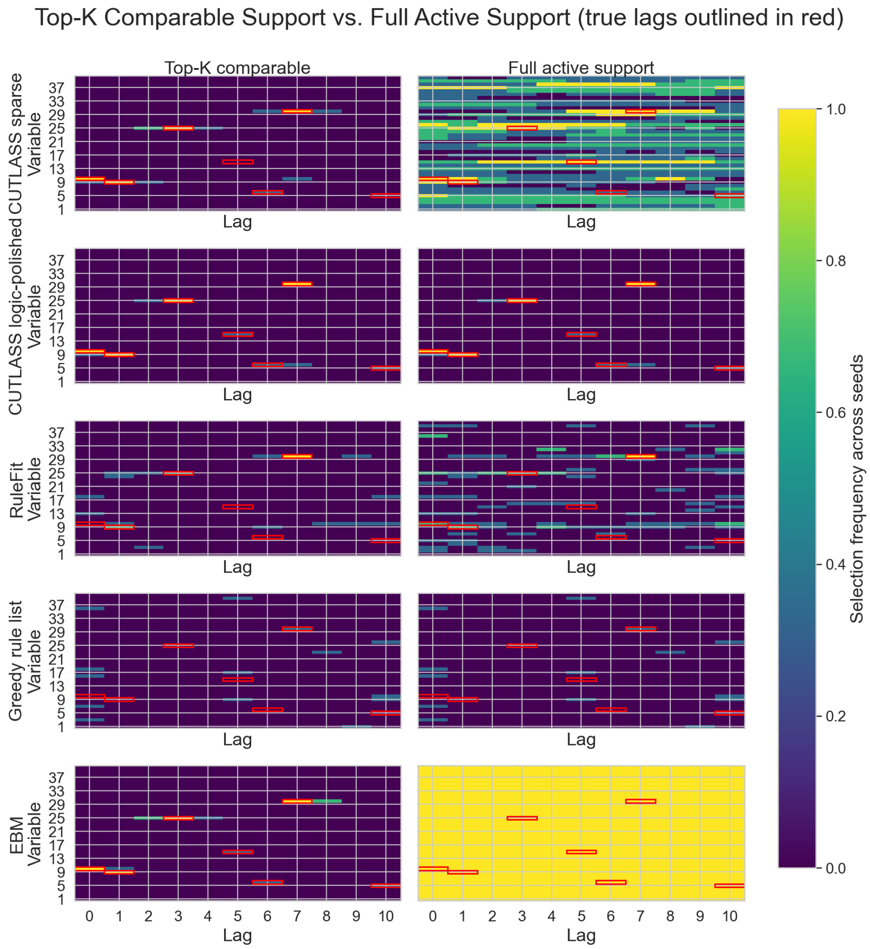


Figure 13. Dual-view exact-lag selection frequency across the direct interpretable baseline package. The left column shows the top-K comparison view, while the right column shows each model's full active support. Red boxes mark the planted variable-lag pairs. The top-K panes show that EBM and CUTLASS logic repeatedly return to the planted neighborhoods, but the full-support panes reveal the interpretive difference: EBM retains dense additive chatter across the lag-expanded design, whereas CUTLASS logic remains compact because its deployed support is already selected by the compression stage. RuleFit is faster and moderately predictive, yet its support drifts more often away from the exact planted lags, while the greedy rule list rarely recovers the planted structure.

Figure 14. Fit-complexity tradeoff for the direct interpretable baseline package on synthetic Case 1. Bubble size is proportional to runtime. EBM occupies the high-fit but high-complexity end, the greedy rule list the low-complexity but weak-fit end, RuleFit a faster but less faithful middle position, and CUTLASS logic the most favorable compact operating point in this threshold-aligned regime.

Table 8. Compact end-to-end HAI exemplar for the dissertation workflow.

Step	HAI `attack_p2(a1)` outcome	Workflow implication
Raw baseline	Test AUC 0.949 and $J = 0.574$ on the raw lag-expanded sparse fit.	Start from the conventional sparse baseline and use it as the comparator.
Train-only rectified pilot	On the same split and penalty, test AUC improves to 0.989, J improves to 0.930, and the attacked-$P2$ coefficient-mass share rises from 0.194 to 0.599.	Threshold alignment is strong enough to justify switching to rectified features.
Branch choice	Chosen branch = "rectify."	Promote the rectified sparse fit to the new baseline.
Anytime compression	The strict 2% policy selects $k = 14$ from 39 active logical conditions ($2.79\times$ compression).	The rectified baseline is compressible enough to consider a rule artifact.
Held-out adoption gate	ΔAUC $= -0.011$ and $\Delta J = +0.015$; paired-bootstrap lower bounds are -0.016 and -0.005, both above the dissertation's -0.02 margins.	The compressed candidate is deployment-eligible under the default policy.
Final deployment artifact	Deploy the compressed rule rather than the full rectified sparse baseline.	Full path: raw baseline $\rightarrow$ diagnostic $\rightarrow$ rectify $\rightarrow$ compress.

Table 9. Real-world case studies: transformed vs untransformed comparisons from the dissertation experiments.

Statistic	HAI (Transformed)	HAI (Un-Trans)	UNICEF (Transformed)	UNICEF (Un-Trans)
ACC	0.987	0.939	0.902	0.963
AUC	0.982	0.943	0.865	0.989
F1	0.987	0.942	0.894	0.958
Youden's J-Index	0.974	0.878	0.815	0.925

Table 10. Metric and notation guide for the RQ chapters.

Term	Meaning and dissertation role
AUC	Area under the ROC curve; threshold-independent discrimination used to check whether rectification or compression preserves held-out ranking quality.
Youden's J	$J = \text{TPR} + \text{TNR} - 1$ at an operating threshold; reports operating-point utility when both sensitivity and specificity matter.
Exact-lag F1	F1 against planted feature-lag ground truth; used in synthetic studies to assess recovery of true lagged triggers.
Pairwise Jaccard	Mean overlap between selected supports across seed pairs or resamples; summarizes support stability.
Total nonzero	Count of nonzero coefficients or active primitive terms before downstream compression; reports sparse-baseline size.
Active support	Selected nonzero feature-lag set; the attribution object inspected for interpretability and stability.
$k^{\star}$	Smallest training-side prefix satisfying the RQ3 compression screen; candidate rule size, not an automatic deployment decision.
m	Vote threshold in an m-of-k rule; number of retained conditions that must be active for event prediction.
Compression ratio	Baseline active-support size divided by retained size, usually $\text{nz}/k^{\star}$; simplification factor.
Deploy?	Held-out accept/reject flag for a nominated compressed rule; "No" means retain the sparse baseline.
Paired-bootstrap lower bounds	Lower confidence bounds for held-out ΔAUC and ΔJ; RQ3 deployment requires both to exceed the pre-specified non-inferiority margins.

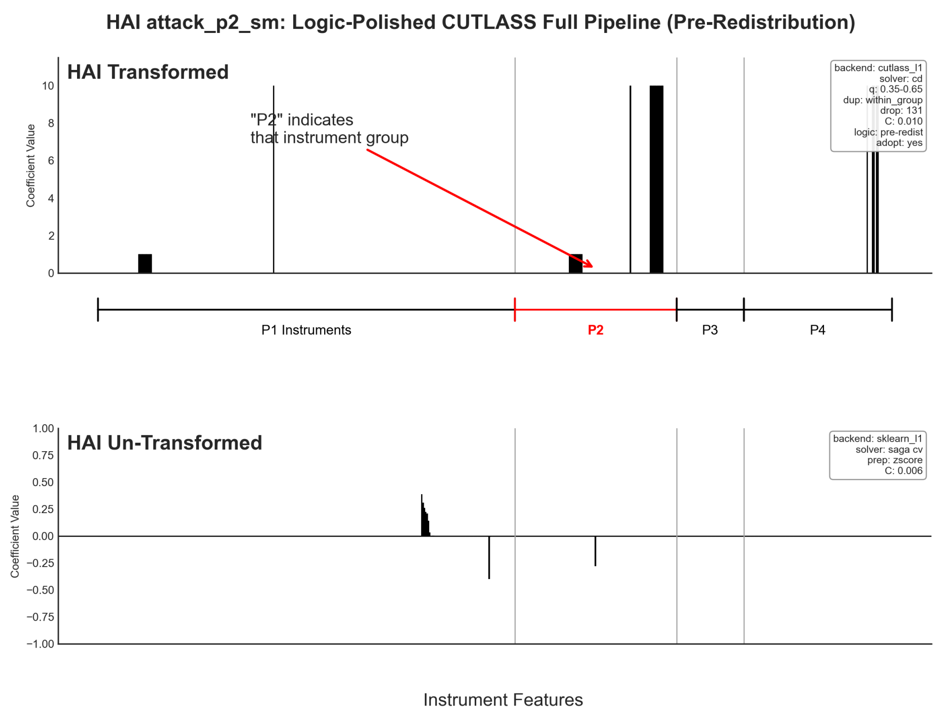


Figure 15. Real-world coefficient concentration and performance for the HAI turbine-loop case after logic polishing. Coefficients concentrate primarily in the P2 region, with only limited spillover outside the attacked subsystem. A smaller coherent P4 response is plausible because P4 is the HIL simulator block and contains signals coupled to the turbine-loop (P2) response.

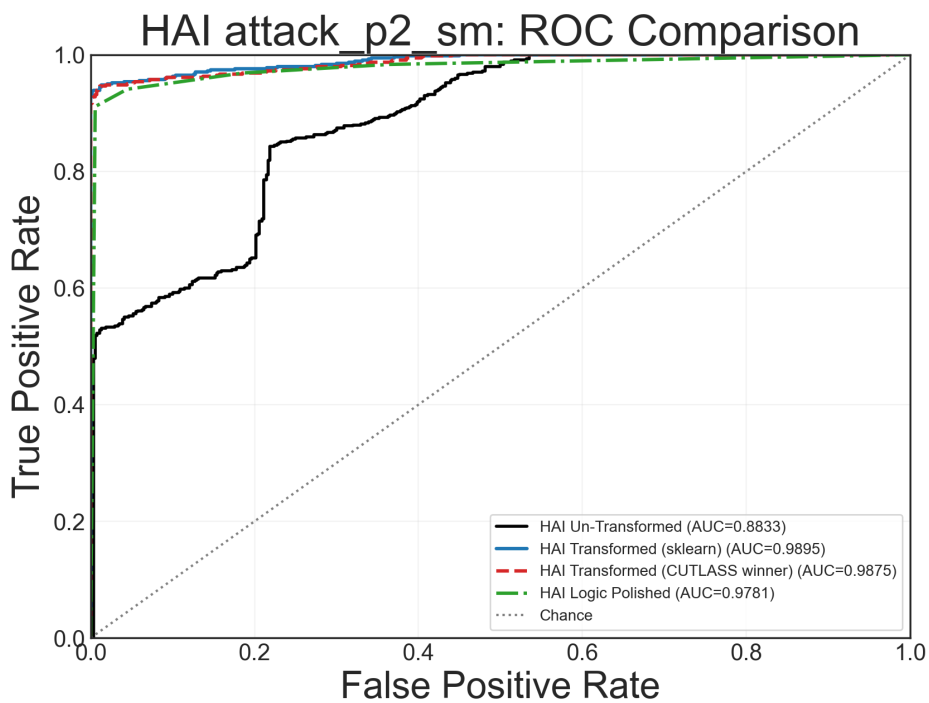


Figure 16. Receiver operating characteristic (ROC) curves for several HAI analyses. The logic-polished model lies slightly below the other transformed-data curves in AUC, but remains highly interpretable and still outperforms the model fit on untransformed data.

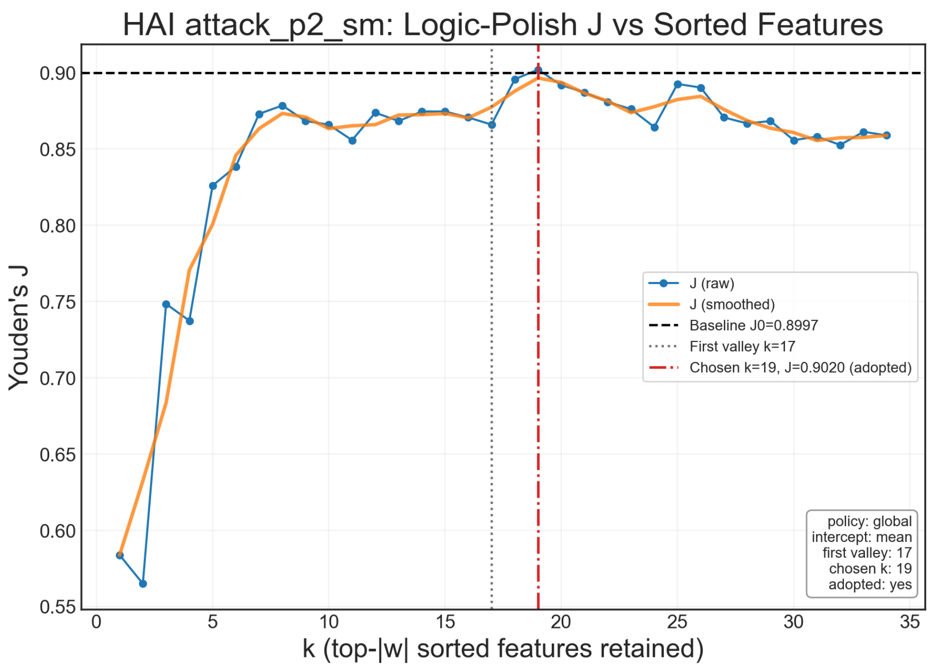


Figure 17. Youden's J is plotted against retained features sorted by coefficient weight before logic polishing. Retaining only eight features achieves nearly the same J (0.88) as the maximum value of 0.90 at 19 features. This illustrates the anytime property of the compression path: a small set of highly relevant features can produce a simple logical rule while preserving nearly all of the full model's predictive value.

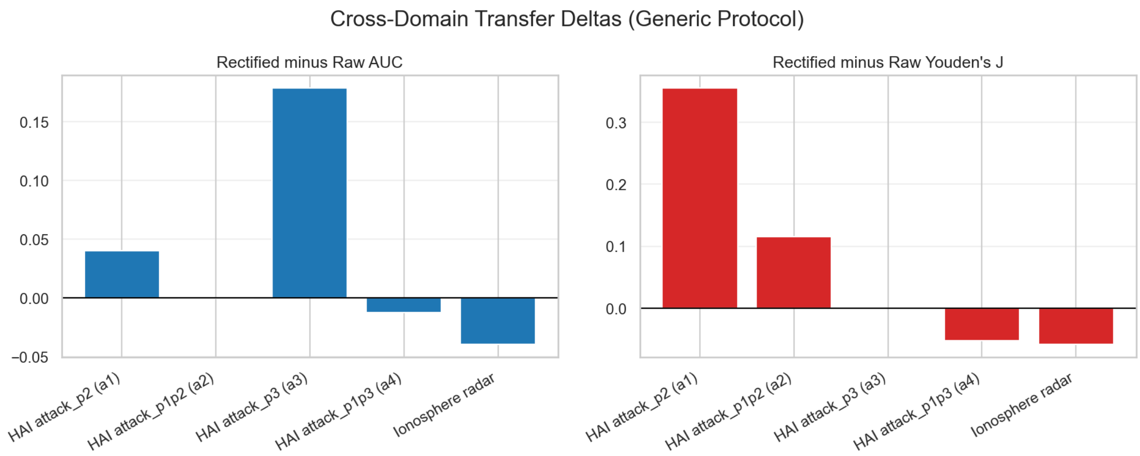


Figure 18. Cross-domain transfer deltas under one standardized generic protocol. Rectification improves some domains strongly, is neutral in others, and does not dominate uniformly across all settings.

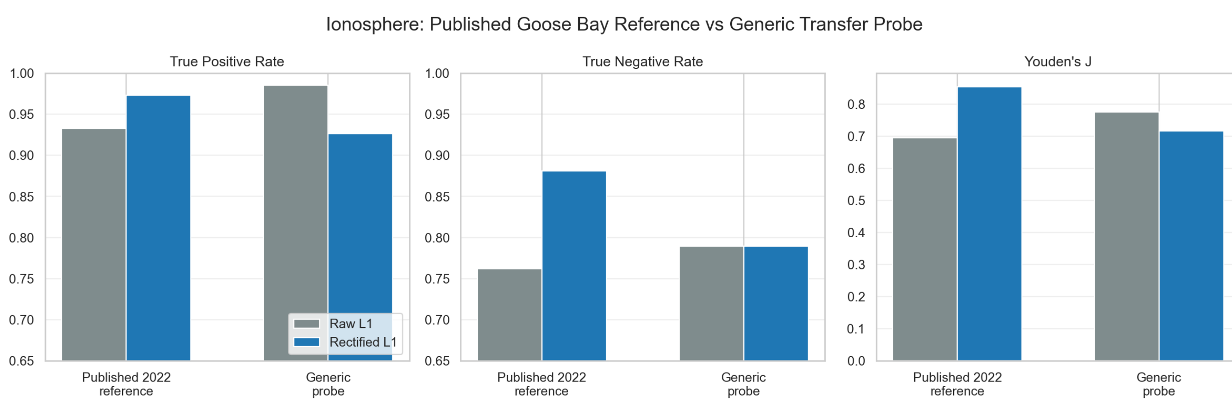


Figure 19. Ionosphere results separated into the published 2022 Goose Bay reference view and the standardized generic transfer probe. The negative generic-probe delta should therefore be read as a protocol-mismatch result, not as a replication of the original Goose Bay case study.

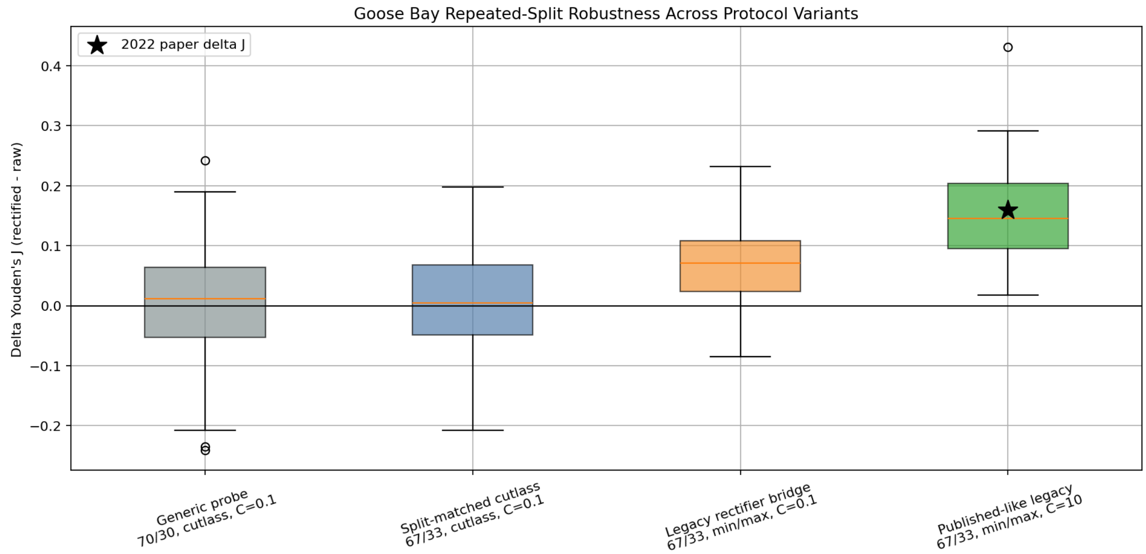


Figure 20. Repeated-split Goose Bay robustness across protocol variants. Matching the split alone does not recover the published gain; most of the movement comes from changing the rectification rule and then relaxing the penalty.

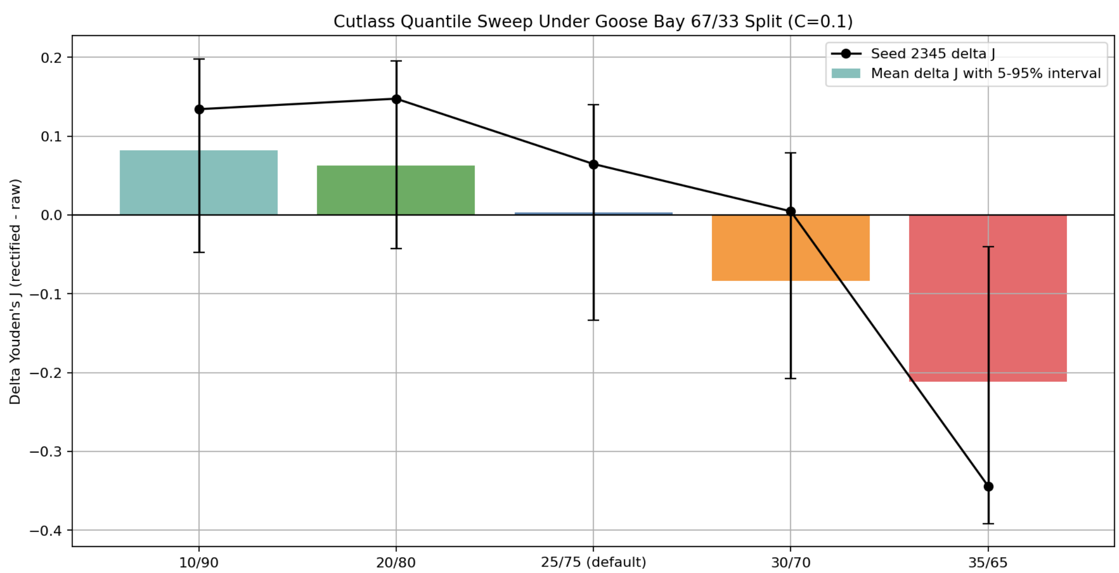


Figure 21. Sensitivity of Goose Bay performance to the quantile rectifier under the $67/33$ split with $C = 0.1$. Wider positive-range windows improve performance relative to the default $25/75$ setting, while narrower windows degrade it sharply.

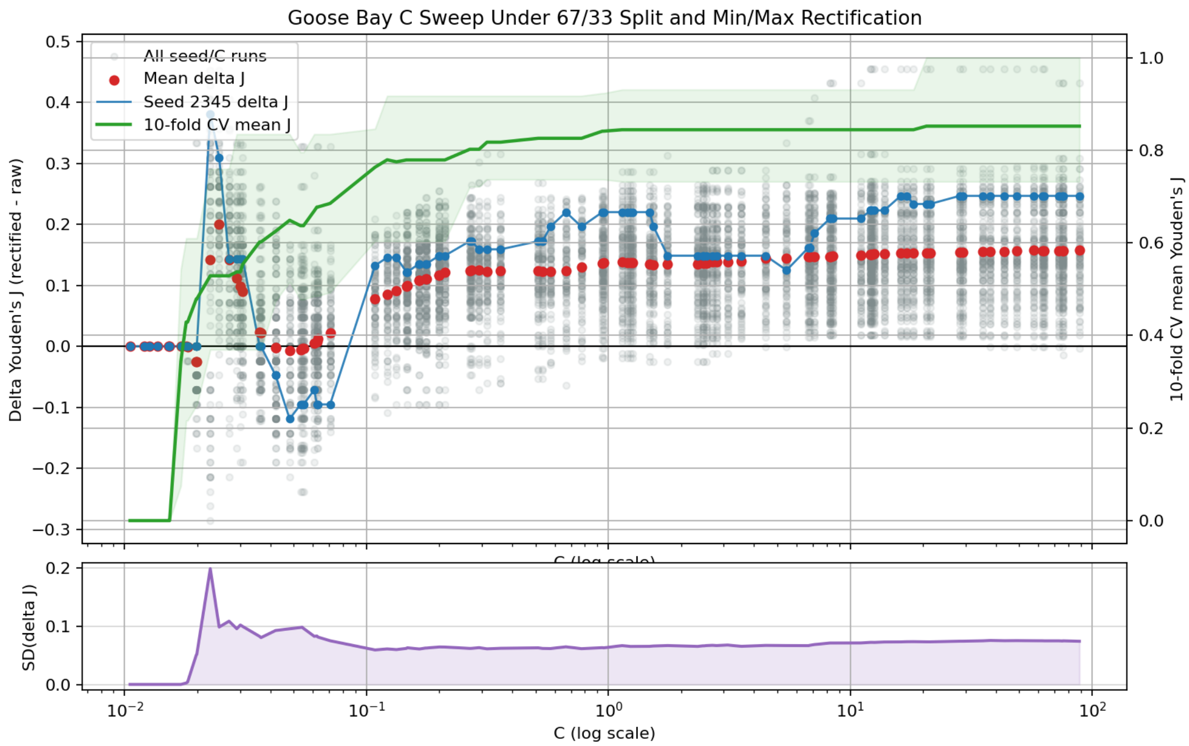


Figure 22. Goose Bay sensitivity to the lasso penalty under the 67/33 split with min/max rectification. The upper panel shows seed-level and mean ΔJ values with an overlaid 10-fold cross-validation mean-J curve; the lower panel shows the standard deviation of ΔJ at each C. The high-C regime is not uniquely optimal, but it is substantially more favorable and more stable than the standardized $C = 0.1$ setting. To be clear, all data shown and thus any conclusions drawn, are based on training data only. No tuning was performed using test data.

4.9 ANSWER TO RQ1

The answer to RQ1 is **yes, conditionally**. Critical-range rectification produces a more stable and interpretable sparse baseline in settings where events are driven by threshold-triggered lag interactions and raw lag expansion creates severe dependence. Evidence supporting this claim includes:

- improved lag-attribution concentration in controlled synthetic experiments,
- favorable or comparable discrimination with better sparsity in key real-world cases,
- and materially lower runtime than certain redundancy-oriented alternatives in tested configurations.

The repeated-resample stability and ablation results in Table 5 and Figures 7–10 reinforce this conclusion, while Table 6, Table 7, Figures 12–14, and Figures 18–22 show how the claim should be scoped. Expanded baseline coverage shows that penalty engineering alone does not recover exact lag attribution in the target synthetic regime. The direct interpretable comparator package shows that EBM can exceed the pipeline on raw fit, but only with much higher additive complexity and runtime, whereas RuleFit and greedy rule lists are faster yet materially weaker on exact-lag fidelity in the same regime. The cross-domain package shows that gains weaken or disappear when the data-generating process departs from threshold-mediated structure or when the rectification defaults are mismatched to the domain, as the UNICEF and generic Goose Bay probe illustrate. Even so, the dissertation establishes that representation-level rectification is a practical and effective intervention point for sparse longitudinal selection in the target regime.

CHAPTER 5

THEORETICAL ANALYSIS (RQ2)

RQ2 asks why rectification helps and how broadly that mechanism can be generalized. This chapter provides a tractable theoretical justification for the zero-threshold sign-binarized special case of the broader rectification step, using the irrepresentable condition (IC) as the central lens. The analysis is intentionally scoped: it focuses on zero-threshold sign binarization under joint normal feature assumptions, where correlation behavior is analytically tractable [10, 26].

The point of this chapter is not to claim a universal theorem for every thresholding scheme. Instead, it establishes a defensible mechanism: if sign binarization contracts dependence in the right way, the IC becomes easier to satisfy, and sparse support recovery becomes more likely. That mechanism aligns with both prior empirical evidence in this line of inquiry and broader observations that lasso-style selection can be unstable under strong dependence [11, 26].

5.1 METHOD

We begin from the empirical failure mode studied in RQ1: lag-expanded predictors are strongly dependent, so sparse support recovery can be unstable even when prediction is acceptable. The theoretical task is not to prove that every learned critical range helps every sparse model. It is narrower: show, in a tractable regime, that binarization can reduce the inactive-active interference that drives lasso support mistakes.

The proof therefore uses zero-threshold sign binarization under joint normal assumptions. This special case gives a closed-form arcsin relationship between raw and transformed correlations, allowing the chapter to track how representation changes affect the active Gram block, inactive-active leakage, and the IC margin. The argument proceeds from pairwise correlation contraction to matrix-level control and then to a probability statement about IC satisfaction.

This scoped strategy is deliberate. The empirical rectification step used elsewhere in the dissertation can use nonzero, one-sided, or interval-like critical ranges, but the theorem uses the zero-threshold case because it is analytically transparent. The chapter then supplements the theorem with boundary checks that test shifted one-sided thresholds, interval ranges, inactive-pool growth, strong off-support coupling, and negative-correlation repairs. In that sense, RQ2 separates three claims: theorem-level support for the positive zero-threshold case, empirical bridge evidence for nearby practical rules, and explicit out-of-scope language for regimes where theorem-level claims would overreach.

5.2 WHY RQ2 MATTERS AFTER RQ1 EVIDENCE

RQ1 showed that rectification improves support concentration and lag attribution in threshold-aligned settings. However, empirical success alone is insufficient for the central claim. Without a credible mechanism, improvements could be dismissed as dataset-specific tuning effects.

For sparse selection methods, this concern is especially important. Lasso is a strong predictive tool [8], but model-selection consistency depends on structural conditions that may fail under multicollinearity [10]. In other words, high AUC does not imply reliable feature recovery. RQ2 therefore asks whether rectification improves a known structural bottleneck rather than merely shifting surface metrics.

5.3 IC IN COMPUTER-SCIENCE TERMS

The IC can be read as an interference constraint in a sparse-recovery system. Let S be the true active-feature set and S^c the inactive set (the complement). For lasso to recover the right support, inactive features should not be too well explained by active features after accounting for active-feature geometry. Formally, one common sufficient condition is:

$$\left\| \mathbf{X}_{S^c}^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \operatorname{sign}(\beta_S) \right\|_\infty < 1. \tag{4}$$

A CS-oriented interpretation is useful:

- $\mathbf{X}_S^\top \mathbf{X}_S$ is the active-feature interaction matrix.
- $(\mathbf{X}_S^\top \mathbf{X}_S)^{-1}$ is a de-mixing operator for that interaction.
- $\mathbf{X}_{S^c}^\top \mathbf{X}_S$ measures leakage from active features into each inactive feature.
- The ℓ_∞ bound asks for worst-case leakage to stay below a hard margin.

When this leakage margin is violated, inactive features can mimic active ones and enter the model, taking the place of true support in the model, causing feature and lag misattribution due to spurious correlation. This is a structural reason for false inclusions and support instability, consistent with dependency-focused reviews [11].

5.4 TRACTABLE SETUP AND ASSUMPTIONS

To keep the analysis explicit, this chapter uses the tractable zero-threshold setting from prior work [26].

Data and transformation. Let $\mathbf{X} \in \mathbb{R}^{n\times d}$ be standardized continuous features and $\tilde{\mathbf{X}} \in \{\pm 1\}^{n\times d}$ the sign-binarized version:

$$\tilde{x}_{ij} = \operatorname{sign}(x_{ij}), \tag{5}$$

with threshold at zero. For derivations, the appendix sometimes uses an equivalent $\{0,1\}$ encoding; after centering, it preserves the same correlation-ordering arguments up to constant scaling.

In the dissertation's broader terminology, this is the theorem-scoped sign-binarized special case of the empirical rectification step.

Proof of Concept (PoC) Assumptions.

- Continuous features are jointly normal after standardization.
- The support set S is fixed in the local analysis.
- Correlation summaries are interpreted through large-sample covariance approximations.
- The goal is comparative: raw versus sign-binarized IC satisfaction probability.
- The zero-threshold is meaningful and retains the signal components relevant to the analyzed mechanism in the target regime (e.g. modeling a threshold-based phenomenon).

Joint normality and zero-threshold meaningfulness are modeling assumptions introduced for tractable analysis; they are not empirically verified prerequisites for using the full pipeline.

These assumptions are restrictive by design. They allow closed-form dependence mapping and a clean theorem statement, but they are not presented as full generality. In finite samples, especially when d is large relative to n, population-covariance approximations may deviate from realized sample behavior, so the claims here are regime-scoped and probabilistic.

5.5 WHY SIGN BINARIZATION CAN REDUCE DEPENDENCE

Under bivariate normality, sign-binarized correlation is a deterministic function of raw Pearson correlation (ρ), as shown in lemma 1:

$$\tilde{\rho} = \frac{2}{\pi}\arcsin(\rho). \tag{6}$$

For $\rho \in (-1,1)$, this mapping contracts magnitude in most of the interior, while preserving sign and endpoint behavior. Intuitively, sign mapping removes amplitude information and retains only direction, which can weaken linear coupling caused by shared scale variation. All binarized equivalent values will be annotated with a tilde (e.g. "$\tilde{\rho}$" for the binarized Pearson correlation).

From an IC viewpoint, this contraction can help in two places:

- It can reduce cross-correlation between inactive and active features.
- It can improve conditioning behavior of the active Gram block in relevant regimes.

The second point should not be read as a universal monotone guarantee for every design. It is a regime-level effect supported in the analyzed setting and tied to matrix-conditioning arguments [26, 36].

5.6 FORMAL LEMMA PATHWAY

This section states the proof pathway in theorem-style form. Appendix B provides the longer algebraic derivations; the main text gives the assumptions, conclusions, and proof logic needed to support the chapter's theorem.

Lemma 5.1 (Arcsin correlation map)**.** *Let (X,Y) be a centered, variance-one bivariate normal pair with correlation ρ, and define $\tilde{X} = \operatorname{sign}(X)$ and $\tilde{Y} = \operatorname{sign}(Y)$. Then the transformed correlation follows the classical arcsine relation [47]:*

$$\tilde{\rho} = \frac{2}{\pi}\arcsin(\rho). \tag{7}$$

Proof. Because (X,Y) is bivariate normal and centered, the classical quadrant formula gives [47]

$$P(X > 0, Y > 0) = \frac{1}{4} + \frac{\arcsin(\rho)}{2\pi}. \tag{8}$$

For sign variables, $E[\tilde{X}] = E[\tilde{Y}] = 0$ and $E[\tilde{X}\tilde{Y}] = P(\tilde{X} = \tilde{Y}) - P(\tilde{X} \neq \tilde{Y})$. By symmetry, substituting the quadrant probability gives $E[\tilde{X}\tilde{Y}] = 4P(X > 0, Y > 0) - 1 = (2/\pi)\arcsin(\rho)$. Since the sign variables have unit variance, this expectation is their correlation. Appendix B.1 gives the step-by-step derivation, including the equivalent $0/1$ encoding. □

Interpretation: Under the PoC assumptions, sign binarization induces a deterministic arcsine mapping from raw correlation to transformed correlation. In practical terms, the representation keeps polarity while suppressing magnitude information that drives linear coupling.

Lemma 5.2 (Pairwise contraction)**.** *Under the assumptions of Lemma 5.1,*

$$|\tilde{\rho}| \leq |\rho|, \tag{9}$$

with equality only at $\rho \in \{-1, 0, 1\}$ *in the closed interval.*

Proof. The map $\arcsin(\rho)$ is odd, so it is enough to consider $0 \leq \rho \leq 1$. Let $\alpha = \arcsin(\rho)$, so $\rho = \sin(\alpha)$ with $\alpha \in [0, \pi/2]$. Concavity of sine on this interval implies $\sin(\alpha) \geq 2\alpha/\pi$, the chord from $(0,0)$ to $(\pi/2, 1)$. Therefore $(2/\pi)\arcsin(\rho) \leq \rho$. Odd symmetry gives the same magnitude bound for negative ρ. Appendix B.2 gives the expanded case analysis. □

Interpretation: The transformed correlation magnitude is bounded by the raw magnitude. This bound is the key contraction property used downstream to control leakage and reduce amplification risk in de-mixing.

Lemma 5.3 (Inactive-active leakage contraction)**.** *Let $x_{S^c i}$ be an inactive standardized Gaussian feature and x_{Sj} an active standardized Gaussian feature satisfying the pairwise assumptions above. For their sign-binarized counterparts,*

$$\left|\mathrm{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{Sj})\right| \leq \left|\mathrm{Cov}(x_{S^c i}, x_{Sj})\right|. \tag{10}$$

Proof. The raw variables are standardized, so their covariance equals their correlation. The sign-binarized variables are centered and have unit variance under the symmetric Gaussian assumption, so their covariance also equals their correlation. Applying Lemma 5.2 to each inactive-active pair gives the stated covariance bound. Appendix B.3 expands this pairwise argument for the inactive-active covariance block. □

Interpretation: For each inactive-active pair, binarization weakens covariance leakage or leaves it unchanged in edge cases. In sparse-selection terms, inactive features become less able to mimic the active span.

Lemma 5.4 (Active-block inverse control)**.** *In the positive-correlation analyzed regime, assume the active block is equicorrelated with support size s and correlation $0 \leq \rho < 1$:*

$$G = (1-\rho)I_s + \rho J_s. \tag{11}$$

Let $\tilde{G} = (1-\tilde{\rho})I_s + \tilde{\rho}J_s$ with $\tilde{\rho} = (2/\pi)\arcsin(\rho)$.

Then,

$$\|\tilde{G}^{-1}\|_\infty \leq \|G^{-1}\|_\infty. \tag{12}$$

Proof. For an equicorrelation matrix, Sherman-Morrison gives

$$G^{-1} = \frac{1}{1-\rho} I_s - \frac{\rho}{(1-\rho)(1+(s-1)\rho)} J_s, \tag{13}$$

and the same expression holds for $\tilde{G}^{-1}$ after replacing ρ with $\tilde{\rho}$. In the positive-correlation regime, the resulting row-sum norm is monotone increasing in ρ on $[0,1)$. Lemma 5.2 gives $0 \leq \tilde{\rho} \leq \rho$, so the transformed inverse row-sum norm is no larger than the raw one. Appendix B.4 contains the expanded matrix calculation. $\square$

Interpretation: The transformed active block requires no greater inverse amplification than the raw block in the target regime. Combined with Lemma 5.3, this yields a smaller IC interference term and therefore a higher chance of satisfying the IC margin.

5.7 THEOREM AND PROOF

Theorem 5.5 (Sufficient IC-bound probability under sign binarization)**.** *Let*

$$\Theta(\mathbf{X}) = \left\| \mathbf{X}_{S^c}^{\top} \mathbf{X}_S (\mathbf{X}_S^{\top} \mathbf{X}_S)^{-1} \operatorname{sign}(\beta_S) \right\|_{\infty}, \tag{14}$$

and define $\Theta(\tilde{\mathbf{X}})$ analogously for the sign-binarized design. For each $i \in S^c$, let

$$B_i = \|\mathbf{c}_i\| \, \|\mathbf{G}^{-1}\| \, \|\operatorname{sign}(\beta_S)\|, \tag{15}$$

$$\tilde{B}_i = \|\tilde{\mathbf{c}}_i\| \, \|\tilde{\mathbf{G}}^{-1}\| \, \|\operatorname{sign}(\beta_S)\|. \tag{16}$$

and set $B_\star = \max_{i \in S^c} B_i$ and $\tilde{B}_\star = \max_{i \in S^c} \tilde{B}_i$. Assume the PoC conditions, the positive equicorrelation regime of Lemma 5.4, and the large-sample covariance approximation used in this chapter. Then sign binarization weakly improves the probability that this sufficient IC bound certifies the margin:

$$P\big(B_\star < 1\big) \leq P\big(\tilde{B}_\star < 1\big). \tag{17}$$

Proof. For each inactive feature index $i \in S^c$, define its IC interference score

$$\theta_i = \left| \mathbf{c}_i \mathbf{G}^{-1} \operatorname{sign}(\beta_S) \right|, \tag{18}$$

where $\mathbf{c}_i$ is its covariance vector with the active features and $\mathbf{G}$ is the active Gram matrix. In the large-sample regime, sample Gram and covariance blocks concentrate around their population counterparts, so raw and binarized designs can be compared through (ρ_{iS}, ρ_{SS}) and $(\tilde{\rho}_{iS}, \tilde{\rho}_{SS})$.

Lemma 5.3 contracts each inactive-active covariance term, giving a no-larger transformed leakage vector in the analyzed norm. Lemma 5.4 gives no greater active-block inverse amplification in the positive/equicorrelated regime. Therefore, by submultiplicativity,

$$\theta_i^{\text{bin}} \leq \|\tilde{\mathbf{c}}_i\| \, \|\tilde{\mathbf{G}}^{-1}\| \, \|\operatorname{sign}(\beta_S)\| \leq \|\mathbf{c}_i\| \, \|\mathbf{G}^{-1}\| \, \|\operatorname{sign}(\beta_S)\|, \tag{19}$$

using the same compatible norm on both sides. Thus $\tilde{B}_i \leq B_i$ for each inactive feature in the theorem-scoped regime. Taking the maximum over $i \in S^c$ preserves the weak inequality, giving $\tilde{B}_\star \leq B_\star$. Therefore the event $\{B_\star < 1\}$ is contained in $\{\tilde{B}_\star < 1\}$ up to the finite-sample concentration error, which gives the probability inequality in (17). Because $\Theta(\mathbf{X}) \leq B_\star$ and $\Theta(\tilde{\mathbf{X}}) \leq \tilde{B}_\star$, $B_\star < 1$ and $\tilde{B}_\star < 1$ are sufficient certificates for the raw and transformed IC margins, respectively. □

The theorem states a scoped probabilistic improvement claim for a sufficient IC bound, not deterministic per-instance dominance of the exact IC score. Strict improvement is expected only in stylized regimes; the empirical boundary checks below identify where the theorem's direction survives and where theorem-level claims should stop.

5.8 PRACTICAL INTERPRETATION

The theorem can be read as a data-representation effect on sparse search geometry.

- **Before transformation:** lag-expanded features create dense dependency graphs. Multiple inactive nodes have high edge weight to active nodes, increasing support ambiguity.

- **After transformation:** edge weights are attenuated in the target regime, reducing ambiguity and lowering the number of near-tie candidates along the regularization path.

This interpretation connects theory to observed outcomes in earlier chapters: fewer false positives, clearer lag attribution, and higher support stability in threshold-and-lag aligned datasets [25, 26].

Importantly, the argument does *not* require modifying lasso internals. The solver, objective, and tuning workflow remain standard; the intervention is in representation. That separation is operationally useful because it preserves mature optimization tooling while changing the statistical geometry presented to the solver.

5.9 A SMALL INTERFERENCE EXAMPLE

Consider a simplified setting with one active feature x_1 and one inactive feature x_2 that is strongly correlated with x_1. If the true signal is carried only by x_1, lasso can still assign weight to x_2 when x_2 is highly predictable from x_1. In IC language, this is a high-leakage case: x_2 sits close to the active span, so the inactive-feature margin shrinks.

Now apply sign binarization. In the PoC regime, the effective correlation between the two channels is contracted by the arcsin mapping. The inactive channel remains related to the active channel, but less linearly confounded in the space used by the sparse solver. This can move the system from a near-violation regime toward a satisfied-margin regime.

For an advanced CS reader, this is analogous to reducing feature aliasing in a compressed representation. The model family is unchanged, but representational overlap among candidate predictors is reduced enough that greedy path updates and KKT-driven screening are less likely to elevate spurious coordinates early in the path.

5.10 HOW THIS DIFFERS FROM PENALTY-ONLY FIXES

A natural question is why this chapter emphasizes representation rather than only stronger penalties. Existing work already proposes many penalty-level fixes for dependence, including elastic net, adaptive penalties, and grouped or ordered structures [12–14, 16]. These methods are important and remain part of the baseline set.

The RQ2 claim is narrower: even with the same downstream sparse learner, changing the representation can improve the structural conditions that drive support recovery. Penalty redesign and representation redesign are not mutually exclusive; they attack different layers of the pipeline.

From a systems perspective:

- Penalty-level methods change optimization geometry in coefficient space.
- Rectification changes statistical geometry in feature space before optimization.

This distinction matters for engineering reuse. If the mechanism is mainly representational, the approach can be plugged into mature sparse-learning stacks without custom solvers, while still improving support behavior in targeted dependence regimes.

5.11 WHAT THIS THEOREM DOES NOT CLAIM

To avoid overreach, the following non-claims are explicit:

- It does not prove universal improvement for arbitrary nonzero thresholds.

- It does not guarantee gains for every dataset or every correlation sign pattern.
- It does not replace empirical validation; it explains a mechanism under a tractable regime.

These caveats are consistent with prior reports that transformed and untransformed tradeoffs can vary by regime, with some datasets favoring raw discrimination while transformed models improve interpretability and attribution fidelity [11, 26].

5.12 EMPIRICAL BOUNDARY CHECKS BEYOND THE THEOREM

The closed-form theorem is intentionally conservative. The dissertation therefore supplements it with an explicit empirical boundary-condition package that tests how much of the IC-oriented mechanism survives once its simplifying assumptions are perturbed. These studies do not claim a second theorem. Their purpose is narrower and more useful: identify where the positive-correlation proof pathway is strongly supported, where it remains directionally informative but weaker, and where the chapter should stop making theorem-level claims.

5.12.1 Construction of the Empirical Boundary Package

All studies presented use controlled synthetic covariance families rather than one fixed observed dataset. That choice is deliberate. RQ2 is about mechanism, so direct control is needed over the dependence geometry that drives IC behavior. The baseline construction fixes an active support of size $s = 4$ (except in the case of the heat map in figure 24, in which case s is expanded to 25 in order to remain consistent with the HAI dataset that is plotted over it) and begins with one inactive feature. The active block is equicorrelated with parameter ρ_{active}, and the inactive-active coupling is controlled by a multiplier c, which induces

$$\rho_{\text{cross}} = \operatorname{sign}(\rho_{\text{active}})\, c \sqrt{\frac{1+(s-1)\rho_{\text{active}}}{s}}. \tag{20}$$

This scaling keeps the covariance construction positive definite while making it possible to increase off-support leakage strength systematically.

For each $(\rho_{\text{active}}, c)$ configuration, the comparison draws multivariate Gaussian samples, constructs both the raw design and the zero-threshold sign-binarized design, estimates their sample correlation matrices, and evaluates the empirical IC proxy

$$\theta_i = \left|\mathbf{c}_i G^{-1}\mathbf{1}\right|, \qquad \Theta = \max_{i\in S^c} \theta_i, \tag{21}$$

where G is the active-block sample correlation matrix, $\mathbf{c}_i$ is the correlation vector between inactive feature i and the active block, and $\mathbf{1}$ is a fixed positive sign vector used as a simplified stand-in for $\operatorname{sign}(\beta_S)$. A pseudoinverse is used for numerical robustness near poorly conditioned settings. The baseline sweep uses $n = 6000$ samples and five random seeds; the harder cross-scale maps use $n = 5000$ and three seeds per grid point; the inactive-count stress test uses $n = 5000$ and five seeds while increasing the inactive pool from 1 to 10 at fixed $s = 4$ relevant (active) features; and the threshold-bridge and complement studies use 250000 draws so that threshold-placement effects are not dominated by Monte Carlo noise.

These calculations still inherit simplifying assumptions and should be read accordingly:

- Joint normality is retained so the zero-threshold sign case remains connected to the arcsin proof pathway.
- The active support is treated as known. The package studies IC geometry, not full support discovery from unknown truth.

- The baseline covariance toy begins with one explicit inactive feature for interpretability, then relaxes that simplification in the inactive-count sweep.
- The proxy is correlation-based and finite-sample, so it is an empirical approximation to the theorem rather than a symbolic extension of it.

5.12.2 Baseline Sweep Across the Positive and Negative Regimes

Figure 23 is the first empirical check on the theorem's logic. The upper-left panel asks whether the most basic proof-of-concept signature is present: does the transformed active block follow the arcsin contraction? The answer is yes on the positive side. At $\rho_{\text{active}} = 0.60$, the raw mean active correlation is 0.600 and the rectified mean active correlation is 0.415, which sits close to the theoretical arcsin value of 0.410. This matters because the theorem is not merely about lower correlation in the abstract; it is about lower correlation in the specific form predicted by sign binarization under joint normality.

The upper-right panel then converts that pairwise contraction into the quantity that matters for support recovery: the inactive-active leakage proxy Θ. In the moderate positive regime where the baseline slice is most informative, rectification moves the system from a raw near-failure state into a safer IC region. For example, at $\rho_{\text{active}} = 0.10$, mean raw Θ is 1.060, while mean rectified Θ falls to 0.753. The lower-left panel shows why this is more than cosmetic: the empirical safe frequency $P(\Theta < 1)$ changes from 0 to 1 at the same point. The lower-right panel provides the active-block counterpart. At $\rho_{\text{active}} = 0.60$, the inverse norm falls from 3.633 in the raw design to 2.383 after rectification, indicating that the transformed active block is easier to invert and less likely to amplify nuisance correlations.

The negative side of Figure 23 is equally important because it defines the chapter's boundary language. At $\rho_{\text{active}} = -0.30$, raw Θ is 3.815 and rectified Θ falls to 0.562, so the empirical proxy can still improve numerically. However, the red pole marker at $(-1/(s-1) = -1/3)$ discontinuity signals that this region is not necessarily a clean extension of the positive-correlation theorem. While an interesting effect, the main theorem focus is on the segment of the plot with ρ greater than zero.

One limitation of Figure 23 is intentional: it uses a moderate fixed coupling level $c = 0.60$. Under that particular slice, rectified Θ remains below 1 across the sweep. That is useful as a baseline, but it is too forgiving to establish a genuine failure boundary. The next figures make the coupling strength explicit.

5.12.3 Cross-Scale Maps: Where the Rectified Mechanism Fails

Figure 24 addresses the most important objection to the baseline sweep: perhaps the original coupling level was simply too mild. The heatmaps vary both ρ_{active} and c (the cross-scale factor) and therefore expose the actual rectified failure boundary. The top row shows mean Θ itself. In these panels, the white contour is the operational threshold $\Theta = 1$. Everything below that contour is IC-favorable in the proxy sense, and everything above it is not. The visual comparison is therefore direct: the method with the larger dark region below the white contour retains IC favorability over more of the parameter space.

The raw and rectified maps differ in exactly the way the theorem suggests. At $\rho_{\text{active}} = 0.00$, raw Θ first crosses 1 at $c = 0.50$, whereas rectified Θ does not cross until $c = 0.80$. At $\rho_{\text{active}} = 0.20$, the crossings shift from 0.65 to 0.85. At $\rho_{\text{active}} = 0.60$, they move from 0.85 to 0.95. The meaning of those numbers is concrete: rectification allows a stronger inactive-active coupling before the IC proxy breaks. The lower row restates the same fact as a frequency statement rather than a contour statement. The rectified safe region is not flat at 1.0 anymore; it shrinks as coupling grows. That is exactly the right result for a scoped mechanism claim, because it shows boundary displacement rather than universal dominance.

The negative side remains useful but must be described carefully. At $\rho_{\text{active}} = -0.20$, raw Θ crosses near $c = 0.45$, while rectified Θ waits until about 0.80. This still suggests that the transformed representation can be empirically less fragile, but Figure 24 should not be cited as extending the theorem into the negative regime. It should be cited as evidence that the empirical boundary is asymmetric and that negative-side behavior must be framed as stress-test evidence rather than theorem-level support.

Modifying the heatmap so that it includes an overlay of the HAI data in figure 25, we can see that during binarization, relevant features become more event-aligned, which makes them more similar to each other, while non-relevant features become less event-aligned with their relevant counterparts. This has the effect of pushing the Active-block rho to the right, and increasing the margin to the 1.0 theta contour. Hence, the selected support lives in a more favorable region of that boundary map.

5.12.4 Inactive-Pool Scaling: the Maximum Over Inactive Features Matters

Figure 26 isolates a separate effect that the one-inactive toy model cannot show by itself: Θ is a maximum over inactive features. Even if each nuisance feature is only moderately aligned with the active block, increasing the number of inactive features creates more chances for one of them to become the worst offending leakage direction. This figure therefore keeps the active support fixed at $s = 4$, fixes the coupling at $c = 0.80$, and varies the inactive pool, increasing from 1 to 10.

The left panel shows the max effect directly. At $\rho_{\text{active}} = 0.20$, mean raw Θ rises from 1.188 with one inactive feature to 1.230 with ten, while mean rectified Θ rises from 0.920 to 0.950. The deterioration is not enormous because the active geometry is held constant, but it is systematic: more nuisance features make the worst-case leakage direction worse. The same pattern appears at $\rho_{\text{active}} = 0.00$, where raw Θ rises from 1.389 to 1.511 and rectified Θ rises from 0.937 to 1.007.

The right panel is the operational view. For $\rho_{\text{active}} = 0.00$, the raw design is never safe in this sweep, while the rectified design stays at $P(\Theta < 1) = 1.0$ through eight inactive features before dropping to 0.8 at nine and 0.4 at ten. For $\rho_{\text{active}} = 0.60$, both raw and rectified designs remain empirically safe across the sweep, but the left panel still shows that raw Θ is consistently higher. The lesson is not just that rectification helps. The lesson is that dimensionality on the inactive side still matters even after rectification, because the IC proxy is governed by the worst nuisance competitor rather than by an average competitor.

5.12.5 Threshold Bridge: From Exact Sign Theory to Practical Critical Ranges

Figure 27 connects the theorem to how rectification is actually used in practice. The exact theorem lives at the zero-threshold sign case, but deployed critical ranges are often shifted and may collapse one boundary to an observed extreme, making the operational rule behave more like a one-sided threshold than like a perfectly symmetric interval. The left panel therefore asks a focused question: if the threshold moves away from zero, how much of the correlation-contraction mechanism survives?

The answer is that the bridge is strongest for one-sided rules and weaker for bounded intervals. In the underlying Gaussian pair with raw correlation 0.599, the zero-threshold sign case produces indicator correlation 0.408. At a moderately shifted one-sided threshold of $t = 0.75$, the indicator correlation is still only 0.379, which remains well below the raw correlation and therefore preserves the same directional story. The curve peaks near the exact sign case and tapers as the threshold moves deeper into either tail because prevalence drops and the indicator becomes increasingly extreme-event specific. That is why the strongest practical generalization is not “all thresholds behave like the theorem.” It is “moderately shifted one-sided thresholds preserve the theorem’s direction best.”

The right panel explains why interval rules should be written up more cautiously. The shifted interval $[0.25, 1.25]$ yields indicator correlation 0.186, while the tail interval $[1.0, 2.0]$ yields 0.237. Both remain below the raw Pearson correlation, so the contractive direction survives empirically, but the exact arcsin mapping is gone and prevalence becomes a major driver of the result. In other words, once the rectifier becomes a bounded interval rather than a sign rule, placement and occupancy of the interval matter almost as much as the original covariance geometry.

5.12.6 Negative Correlation and the Complement-Feature Repair

Negative correlation is the regime where this work most clearly distinguishes theorem scope from engineering repair. In the complement study, the original sign pair remains negatively associated at -0.411, which leaves it outside the chapter’s positive-correlation proof path. Complementing one of the features flips the same relationship to $+0.411$. This does not extend the theorem. Instead, it re-expresses the problematic relation in a form that the rectified representation can use more naturally. That is why complement features should be described as a practical repair strategy rather than as a hidden corollary of the zero-threshold proof.

5.13 EMPIRICAL IMPLICATIONS OF RQ2

Table 11. Summary of the empirical boundary package. “Strong” means the theorem’s direction is well supported empirically; “weak” means the direction survives but the exact theorem does not; “fail” means that making theorem-level claims is not supported; “recovered” means a practical representation repair is available.

Setting	Status	Key empirical evidence
Positive zero-threshold sign case ($\rho = 0.60$)	strong	$\Theta : 0.719 \to 0.600$; active correlation: $0.600 \to 0.415$.
Shifted one-sided cutoff ($t = 0.75$)	strong	Indicator correlation: $0.599 \to 0.379$.
Shifted interval $[0.25, 1.25]$	weak	Indicator correlation: $0.599 \to 0.186$.
Tail interval $[1.0, 2.0]$	weak	Indicator correlation: $0.599 \to 0.237$.
Negative-side IC sweep near the pole ($\rho = -0.30$)	fail	$\Theta : 3.815 \to 0.562$; empirical only, not theorem-scoped.
Negative pair with complement feature	recovered	Sign correlation: $-0.411 \to +0.411$.

The theorem’s implications are now presented as a structured empirically supported story rather than merely as a placeholder proof-of-concept claim.

1. Within the theorem-scoped positive-correlation regime, the data show the same mechanism at each level of resolution: pairwise dependence contracts, inactive-active leakage drops, the active block is easier to invert, and the empirical safe frequency $P(\Theta < 1)$ rises (Figures 23 and 24).

2. Once off-support coupling or inactive-pool size grows, rectification remains helpful but does

not guarantee safety. The failure boundary moves outward, but rectification may not remove failure altogether (Figures 24 and 26).

3. Once the threshold rule departs from the exact sign case, one-sided rules remain the strongest empirical bridge, interval rules become weaker and more prevalence-sensitive, and negative-correlation settings require either explicit out-of-scope language or a complement-feature repair (Figure 27 and Table 11).

These empirical patterns convert RQ2 from an abstract theorem into a concrete explanation for the behavior observed in RQ1 and RQ3. RQ1 reported better support concentration, lower false-positive lag selection, and more stable sparse attribution in threshold-aligned settings. The empirical studies explain why those gains should be expected in the favorable regime and, equally importantly, where they should not be overgeneralized.

5.14 PRACTICAL SIGNIFICANCE

For RQ2, the core result is a principled explanation for why a simple preprocessing step can improve sparse support recovery in a difficult longitudinal regime. The significance is twofold:

- **Scientific:** it links empirical improvements to established model-selection theory through IC-oriented analysis [10].
- **Engineering:** it supports a modular pipeline design where preprocessing changes correlation geometry while downstream sparse solvers remain standard and efficient.

In summary, the theorem does not end the theoretical story, but it establishes a coherent bridge from sign binarization to IC favorability. That bridge is sufficient to support the mechanistic claim while also identifying the boundary conditions that matter for later extensions.

5.15 ANSWER TO RQ2

The answer to RQ2 is **yes, in a scoped form**. Rectification can improve sparse support recovery because, in the analyzed regime, binarization changes the correlation geometry that drives inactive-active interference. In the zero-threshold, jointly normal, positive-correlation setting, sign binarization yields the arcsine correlation map, contracts pairwise dependence, reduces the inactive-active leakage term, and weakly improves the probability that a sufficient IC bound certifies the support-recovery margin.

The mechanism should not be read as a universal guarantee for every critical-range rule. The empirical boundary checks show that the theorem's direction is strongest for the exact sign case and remains most persuasive for moderately shifted one-sided thresholds. Interval rules are more prevalence-sensitive, large inactive pools can still break the IC proxy, and negative-correlation settings require explicit caution or a complement-feature repair rather than a theorem-level extension. Thus RQ2 provides a defensible explanation for why rectification works in the favorable threshold-aligned regimes studied in RQ1 and used by RQ3, while also defining the conditions under which the claim should be limited.

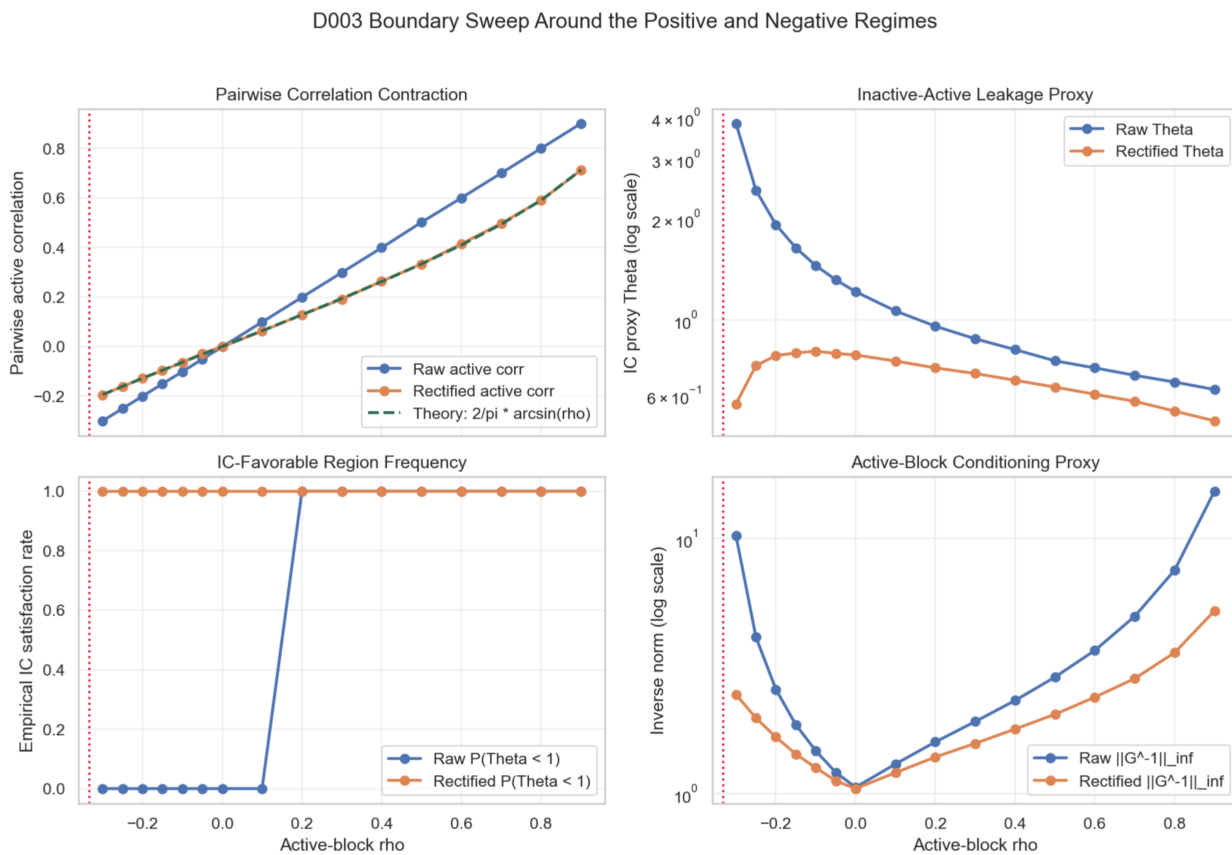


Figure 23. Baseline empirical IC boundary sweep over the active-block correlation. Upper-left: pairwise active correlation contracts toward the arcsin prediction after sign binarization. Upper-right: the inactive-active leakage proxy Θ drops after rectification. Lower-left: the empirical safe frequency $P(\Theta < 1)$ rises in the positive-correlation regime. Lower-right: the active-block inverse norm falls after rectification. The crimson vertical line marks the negative-side pole, beyond which the theorem-scoped interpretation no longer applies cleanly.

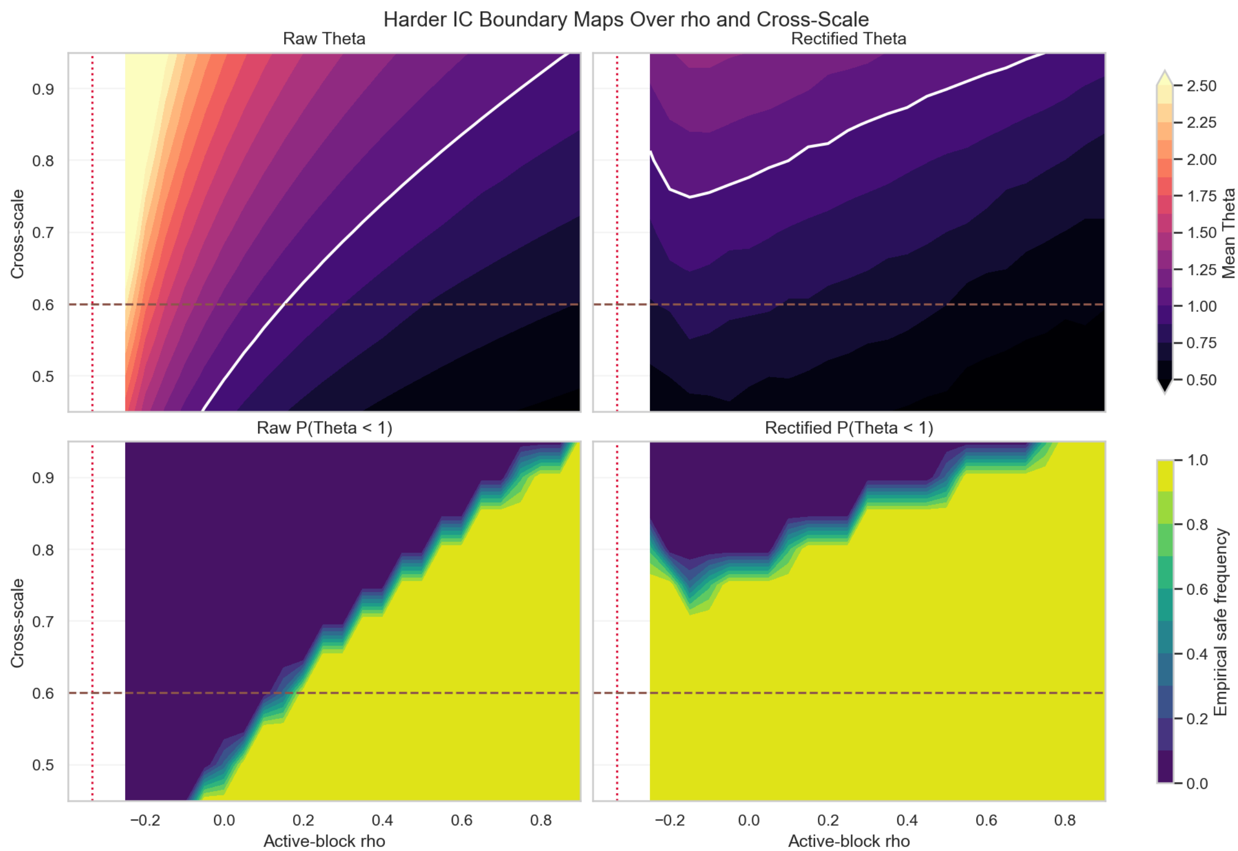


Figure 24. Harder IC boundary maps over active-block correlation and inactive-active coupling. The top row shows mean Θ and the white contour marks the operative failure boundary $\Theta = 1$. The bottom row reports the empirical safe frequency $P(\Theta < 1)$. Rectification pushes the failure boundary outward but does not eliminate failure.

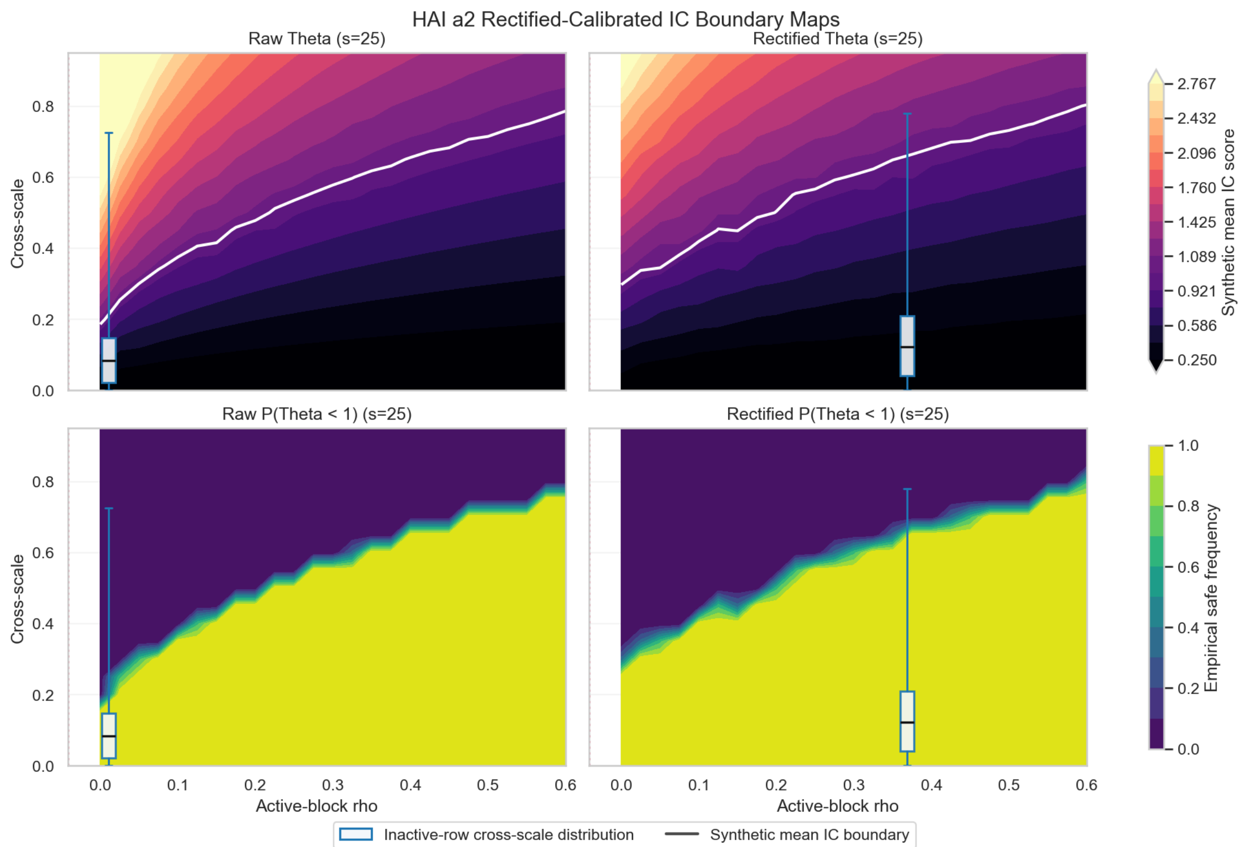


Figure 25. Harder IC boundary maps over active-block correlation and inactive-active coupling expanded to s=25 and with the HAI data included. As with figure 24, the top row shows mean Θ and the white contour marks the operative failure boundary $\Theta = 1$. The bottom row reports the empirical safe frequency $P(\Theta < 1)$. The box-plot overlay shows the real-world HAI dataset, and illustrates the effect of the process of binarization.

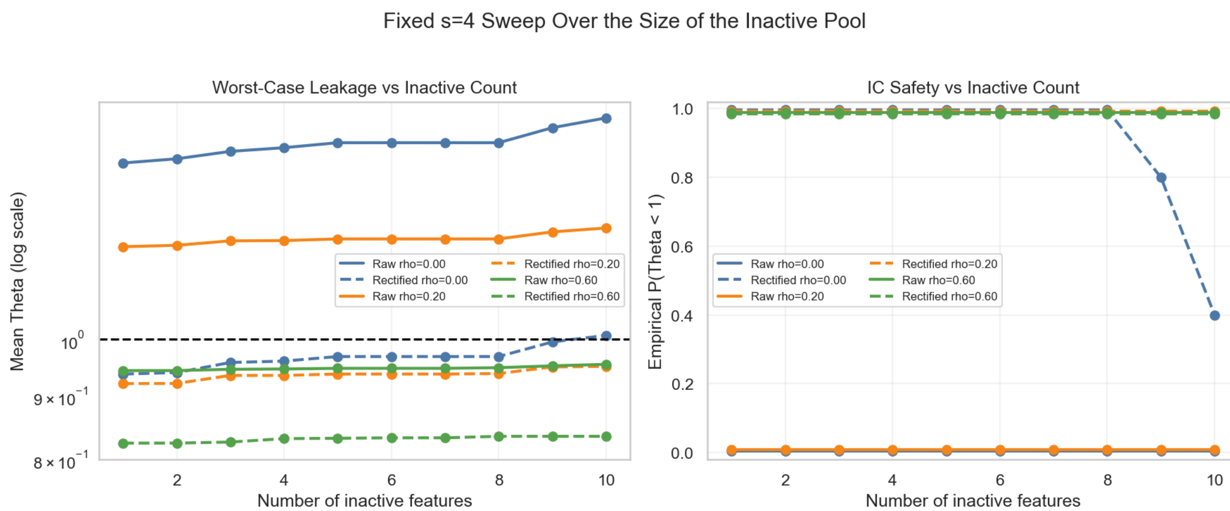


Figure 26. Fixed-$s = 4$ sweep over the size of the inactive pool. The left panel shows how the max-based leakage proxy Θ grows as more nuisance features are added. The right panel shows the corresponding empirical safe frequency $P(\Theta < 1)$. Small vertical offsets at exact 0 and 1 in the right panel are display-only and are used solely to separate coincident curves visually.

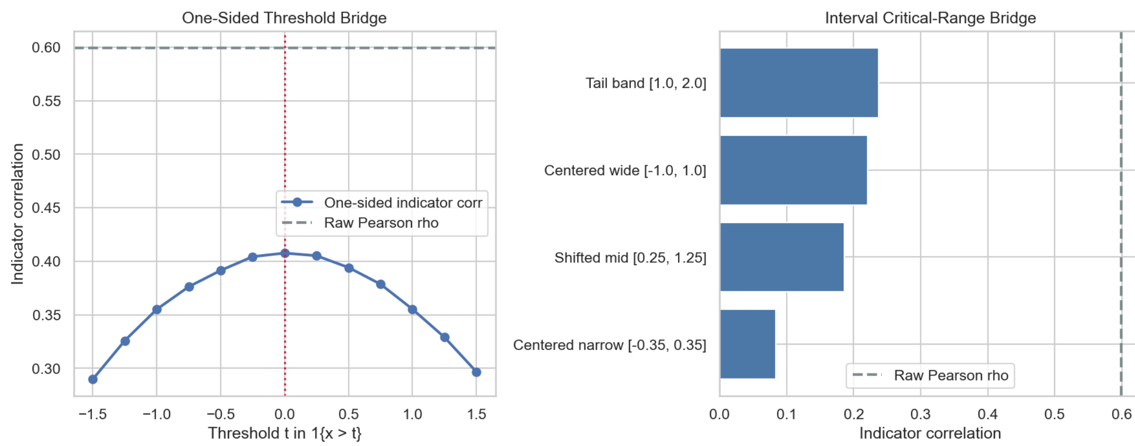


Figure 27. Threshold bridge from exact zero-threshold sign binarization to practical critical ranges. The left panel varies a one-sided threshold and reports the resulting indicator correlation. The right panel compares interval-based critical ranges. The zero-threshold sign case is the exact theoretical anchor; shifted one-sided rules remain directionally contractive; interval rules are weaker and more prevalence-sensitive.

CHAPTER 6

ANYTIME RULE COMPRESSION (RQ3)

Research Question 3 asks whether a sparse, rectified linear model can be compressed into a compact rule form while preserving practical discrimination quality. In this work, the target is not just sparsity in coefficient space, but operational interpretability: a model that can be read, explained, and validated as an explicit decision rule under realistic constraints. The anytime rule compression framework addresses this requirement by converting a fitted sparse model into an ordered sequence of candidate logic rules and allowing compression to stop as soon as performance is good enough for the application context [26, 27].

6.1 WHY RQ3 MATTERS AFTER RQ1 AND RQ2

RQ1 and RQ2 establish two foundations: first, binarization plus sparse learning can recover relevant structure more reliably in threshold-driven settings; second, the theoretical conditions for support recovery can improve when feature interactions are reframed through the binarized representation. RQ3 addresses the remaining translational gap: whether those improvements can be turned into human-auditable, low-complexity rules without giving back the predictive benefit [26, 27].

This is important for longitudinal and event-focused domains where users need a concrete trigger logic rather than a dense weight vector. In healthcare monitoring, safety surveillance, and maintenance settings, end users often need to know which few conditions jointly trigger escalation, and they need that logic in a form that is stable under repeated deployment [1, 2, 32]. In interpretability terms, this chapter focuses on the transition from post hoc explanations to inherently interpretable rule structure [20, 21].

6.2 MOTIVATION AND APPLICATION CONTEXT

The practical motivation for anytime rule compression is strongest in settings where decisions must be explained and audited quickly:

1. Clinical and public-health monitoring: produce concise trigger logic for intervention review and cross-site validation in longitudinal streams [1, 2].

2. Safety and reliability operations: support transparent condition-based escalation when false negatives are costly and operators need explicit criteria.

3. Policy-facing analytics: convert technical models into rules stakeholders can inspect, challenge, and recalibrate without retraining a complex model stack.

Across these settings, the benefit is not only readability. Compact rules also reduce implementation burden, simplify drift monitoring, and support deterministic replay during audits. The anytime mechanism then provides a tunable point on the complexity-performance frontier for each deployment constraint [21, 27].

6.3 METHOD

We treat sparse fitting as an intermediate representation rather than as the final deployment artifact. Even when rectification improves support concentration, a weighted linear score with dozens of nonzero lag terms can still be too large for direct human use. The compression stage therefore starts from the active support recovered upstream and searches for a smaller explicit rule that preserves practical operating performance.

This stage is part of the dissertation's method rather than an unrelated post-processing convenience. The sparse model creates a focused candidate set; logic polishing converts that candidate set into a form that humans can audit, challenge, implement, and monitor. The approach preserves the front-end efficiency of convex sparse learning while recovering a model form closer to how domain experts reason about triggers and deployment criteria.

Given a rectified design matrix and an L1-regularized logistic model, the baseline predictor can be written as

$$\hat{p}(y_i = 1 \mid \mathbf{x}_i) = \sigma\left(\beta_0 + \sum_{j=1}^{p} \beta_j \tilde{x}_{ij}\right), \tag{22}$$

where $\tilde{x}_{ij}$ is the binarized/rectified feature and β_j is sparse due to the L1 penalty [8, 28, 29]. Let the active set be $\mathscr{A} = \{j : \beta_j \neq 0\}$. RQ3 starts from this fitted model and asks whether we can replace the full weighted sum with a smaller rule that preserves operating performance.

The compression idea is to map the active coefficients to a shared magnitude (logic polishing), keep their signs, and evaluate truncated models along an ordered path. Ordering is by absolute effect size from the trained sparse model so that early steps prioritize the most influential terms [27].

6.3.1 Anytime Compression Objective

The objective is to optimize a practical trade-off:

$$\max_{\text{rule complexity } C} J(C) \quad \text{subject to} \quad C \leq C_{\max}, \tag{23}$$

where J is Youden's statistic ($J = \text{TPR} + \text{TNR} - 1$), and complexity is represented by the number of retained conditions k (and optionally the vote threshold m in an m-of-k rule) [27]. Rather than commit to a single fixed complexity in advance, the anytime procedure scans candidate compressions along an ordered frontier. In the generic anytime framework, one may stop at the first prefix whose achieved J falls within an application-defined tolerance of the baseline. In this dissertation, however, that generic tolerance idea is used only to nominate a candidate prefix from the training frontier; actual deployment is decided later by held-out non-inferiority gates.

This aligns with a deployment reality: in some environments, a 1–2% relative loss may be acceptable if complexity falls by an order of magnitude; in others, compression is only accepted at near-zero loss. The framework supports either case by explicitly parameterizing the tolerance [27].

6.3.2 Algorithmic Pipeline

The compression process can be summarized in four stages [27].

1. Train Baseline Sparse Rectified Model

Fit L1-regularized logistic regression on the rectified matrix and compute baseline operating metrics (AUC, J, sensitivity, specificity) at the chosen operating threshold. This baseline defines the reference quality floor for compression [8, 27–29].

2. Logic Polishing and Ordered Prefix Construction

Restrict to active features $\mathscr{A}$ and sort them by $|\beta_j|$ descending. For prefix length k, keep the top-k signed indicators and assign common magnitude K (or equivalently use signed votes). This yields candidate rule families of increasing complexity, each nested in the next [27].

3. Vote Threshold Optimization

For each prefix, evaluate an m-of-k decision threshold. Intuitively, m controls how many supportive conditions are required before predicting the positive class. Sweeping m gives a local complexity-performance frontier at fixed k; sweeping k gives the global anytime frontier [27].

4. Dissertation Strict Policy: Candidate Prefix Then Deployment Decision

The anytime frontier supplies the candidate family, but the dissertation's implemented real-data policy is stricter than the generic relative-tolerance rule alone:

1. **Training-side candidate selection.** Choose the smallest prefix $k^\star$ whose training frontier contains a rule satisfying $J_{\text{train}}(k^\star) \geq 0.98\, J_{\text{train,base}}$. This nominates the smallest passing prefix and its associated vote threshold on training data.

2. **Held-out non-inferiority check.** Evaluate that candidate rule on held-out test data with paired bootstrap and require lower bounds $\Delta\text{AUC} > -0.02$ and $\Delta J > -0.02$.

3. **Deployment decision.** Deploy the compressed rule only if both held-out gates pass. Otherwise retain the upstream sparse baseline. Reported candidate values of $k^\star$ for rejected cases indicate compression headroom, not an automatically adopted artifact.

This keeps the anytime frontier as a search procedure while making the final deployment rule explicit and conservative. Accepted compressed rules should then be documented with the rule-card template in Appendix D, which records the selected feature-lag conditions, vote rule, validation deltas, non-inferiority margins, intended use, and monitoring or retraining triggers.

6.4 WHY THE METHOD IS "ANYTIME"

The method is "anytime" in the algorithmic sense: each additional step (larger k or alternative m) refines quality, but every intermediate rule is already a valid candidate model on the frontier. Under a permissive application policy, the current incumbent could be deployed immediately; under the dissertation's strict policy, it still must clear the held-out gates listed above. If more budget is available, scanning continues to improve operating characteristics. This property is practical for iterative model governance and constrained environments [27].

6.5 EMPIRICAL RESULTS

Across the experiments, compression reduces model complexity by large factors only when the deployment rule is made explicit. The real-data validation package therefore reports both the training-side candidate prefix $k^\star$ nominated by the strict policy above and the final held-out deployment decision. This separates compression headroom from actual deployment eligibility and avoids claiming success merely because a short rule exists somewhere on the frontier.

Figure 28 shows that the frontier shape is strongly domain-specific. In HAI $a1$, $a2$, $a3$, and $a4$, held-out J plateaus well before the full baseline support, so the training-side strict screen nominates candidate prefixes $k^\star = 14$ from 39 active terms (2.8$\times$ compression), $k^\star = 8$ from 25 (3.1$\times$), $k^\star = 7$ from 23 (3.3$\times$), and $k^\star = 8$ from 34 (4.3$\times$), respectively. In the case of the ionosphere probe, the strict training screen does not justify simplification beyond the baseline support, so the candidate remains $k^\star = 10$. Table 12 then determines whether those candidates are actually deployment-eligible. The $a3$ panel is the main cautionary case: the baseline test AUC remains 0.730 even though baseline test $J = -0.077$, so the large J improvement at $k^\star = 7$ reflects threshold mismatch in the deployed operating point rather than new ranking signal created by polishing.

Table 12 makes the deployment result explicit. In the controlled synthetic threshold-and-lag

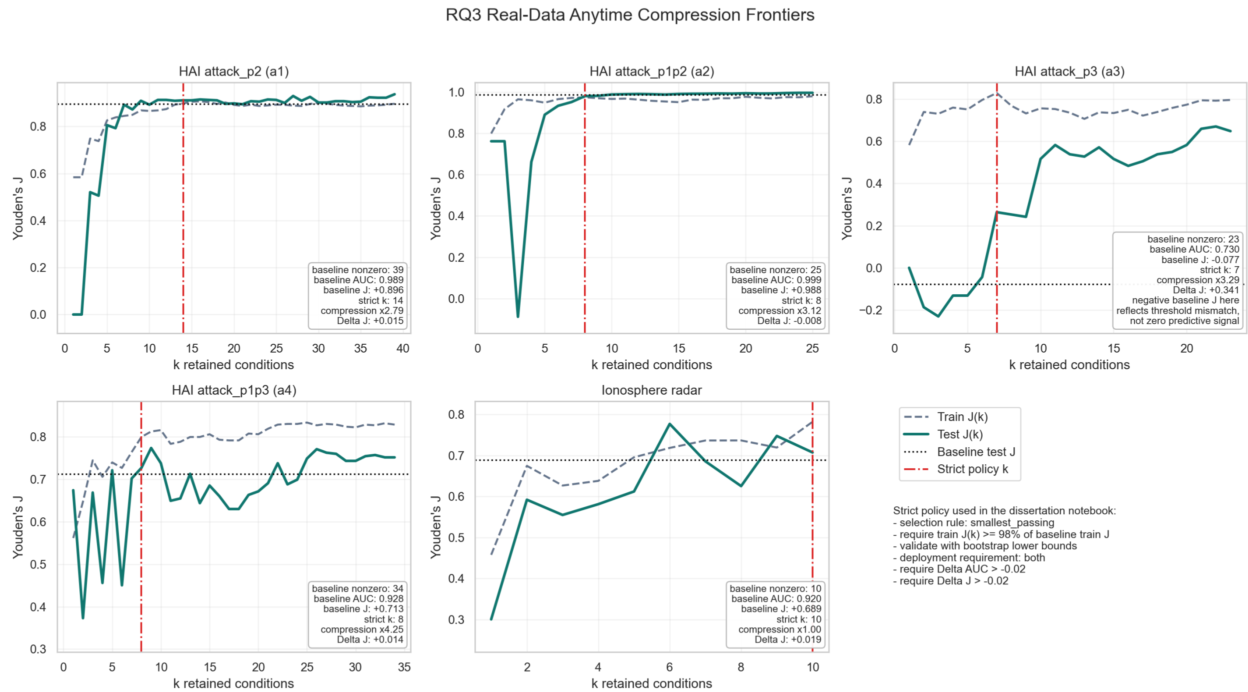


Figure 28. Real-data anytime compression frontiers for the four HAI attack subsets and the ionosphere radar benchmark. Each panel shows training and held-out $J(k)$, the held-out baseline J, and the smallest candidate prefix nominated by the strict $\varepsilon = 0.02$ training screen. Final held-out deployment decisions are reported in Table 12.

regime studied earlier in this chapter, the anytime frontier shows that compression itself works cleanly: once the sparse rectified baseline has isolated the planted event logic, the rule path can recover substantially smaller models without meaningful degradation. The real-data setting is more complicated. Under the strict 2% policy with the relaxed AUC non-inferiority margin, two of the five held-out studies satisfy both gates for their nominated candidate prefixes: HAI $a1$, where the deployed compressed rule shrinks from 39 active terms to 14 ($2.79\times$ compression) with only a modest AUC decrease and a slight J increase, and HAI $a2$, where the deployed compressed

Table 12. Strict real-data compression policy summary using the smallest-passing training-side candidate prefix under $\varepsilon = 0.02$. Deployment eligibility requires paired-bootstrap lower bounds above the prespecified non-inferiority margins $\Delta\text{AUC} > -0.02$ and $\Delta J > -0.02$ on held-out test data.

Domain	Baseline nz	Candidate $k^\star$	Compression	ΔAUC [95% CI]	ΔJ [95% CI]	Deploy?
HAI attack_p2 ($a1$)	39	14	2.79×	−0.011 [−0.016, −0.006]	+0.015 [−0.005, +0.036]	yes
HAI attack_p1p2 ($a2$)	25	8	3.12×	−0.001 [−0.002, −0.001]	−0.008 [−0.012, −0.004]	yes
HAI attack_p3 ($a3$)	23	7	3.29×	−0.003 [−0.054, +0.050]	+0.341 [+0.223, +0.457]	no
HAI attack_p1p3 ($a4$)	34	8	4.25×	+0.011 [+0.002, +0.023]	+0.014 [−0.026, +0.055]	no
Ionosphere radar	10	10	1.00×	+0.010 [−0.014, +0.038]	+0.019 [−0.123, +0.163]	no

rule shrinks from 25 terms to 8 (3.12× compression) while keeping both lower bounds above the prespecified margins. HAI $a3$ retains clear compression headroom but its nominated candidate still fails the AUC lower-bound gate, HAI $a4$ reaches an even smaller candidate $k^\star = 8$ with positive point estimates for both ΔAUC and ΔJ yet misses the bootstrap J lower-bound requirement, and the ionosphere probe clears the relaxed AUC margin but still fails the J gate because the smaller-sample test estimate is too unstable. This mixed real-data outcome is expected when threshold-triggered effects coexist with more nearly additive relationships and with measurement uncertainty. The real-data answer is therefore not that every sparse model should be shortened, but that the anytime frontier provides an auditable basis for either "compress" or "do not compress" decisions under an explicit deployment policy. The specific acceptance criteria used here are intentionally conservative and application-dependent rather than universal. Under other operating priorities, especially when compression gains are large and the held-out shortfall is small, some near-miss cases could reasonably be judged acceptable.

Under that strict policy, the decision summary is simple: deploy the compressed rule for HAI $a1$ and $a2$; keep the upstream sparse baseline for HAI $a3$ and $a4$ because at least one held-out gate fails for their nominated candidate prefixes; and keep the full sparse baseline for the ionosphere probe because the frontier does not justify simplification. This is the operative accept/reject output of the chapter.

The broader tolerance sweep $\varepsilon \in \{0.01, 0.02, 0.05\}$ clarifies why this work now centers the 2% rule in the manuscript. HAI $a1$ changes only gradually across the grid ($k = 15 \to 14 \to 9$), and under the relaxed AUC margin the $k = 14$ candidate now clears both held-out gates. HAI $a2$ likewise remains nearly lossless at $\varepsilon = 0.02$ with $k = 8$. By contrast, HAI $a4$ moves from $k = 9$ to $k = 8$ and keeps favorable point estimates even though the stricter held-out J lower-bound check is no longer met at the smaller candidate. The $\varepsilon = 0.05$ candidate for HAI $a2$ still collapses to $k = 3$ and loses 0.234 AUC and 1.076 points of J on test, and the ionosphere probe likewise resists compression until the loosest policy. This is exactly the role of the anytime frontier: it supports simplification when the curve is flat and rejects it when the curve turns sharply downward [27].

Figure 29 illustrates the same central behavior on a synthetic case: J is tracked as a function of retained rule size k, creating an explicit curve from highly compressed to minimally compressed models. The answer to RQ3 is based on this entire curve, not on a single arbitrarily chosen point.

6.6 STATISTICAL FRAMING OF "NO MATERIAL DEGRADATION"

RQ3 uses "no material degradation" in a practical, decision-oriented sense rather than a literal requirement of zero difference. Classical null-hypothesis difference tests are not sufficient to show retained utility, because failure to reject a difference is not evidence that a simplified rule

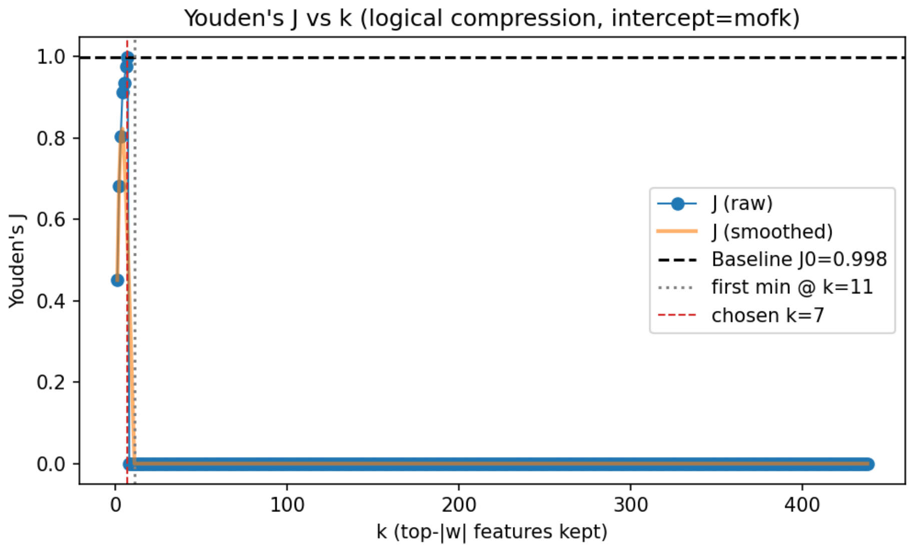


Figure 29. J versus retained rule size k for anytime compression. The curve gives a direct complexity-performance trade-off for selecting a deployable m-of-k rule. After the algorithm reaches $J = 1.0$, the plotted sequence is zero-padded because larger prefixes are not evaluated; those trailing zeros are therefore not measured performance values. See Figure 17 for a real-world example without this early-stop truncation.

remains acceptable. Prior work in this line used an equivalence framing (TOST-style logic) with a predeclared margin (for example, ± 0.01 AUC) to justify practical non-loss claims [27, 34, 48]. The held-out validation package in this dissertation uses a prespecified one-sided non-inferiority screen instead: paired-bootstrap lower bounds on ΔAUC and ΔJ must remain above the declared loss margins. That choice is methodologically aligned with the chapter objective: demonstrate that simplified rules remain fit for purpose under explicit situational tolerances.

6.7 RELATIONSHIP TO ADJACENT INTERPRETABLE METHODS

Rule-list and scoring-system approaches provide strong interpretability baselines, but they often optimize directly in combinatorial rule space with different scalability profiles [20, 22]. The anytime compression strategy differs by leveraging a sparse convex fit first, then performing structured compression in a constrained post-processing stage. This separation is useful when the rectified feature space is large and sparse optimization is already part of the workflow.

The approach is also related to Logical Analysis of Data (LAD), particularly in its use of thresholded indicators and logical structure, but it targets a different optimization path and deployment interface [23, 24]. In the context of this analysis, the criteria presented can be viewed as a bridge between sparse statistical learning and logic-level decision rules tailored to threshold-driven events [25, 26].

6.8 LIMITATIONS AND SCOPE

The results are strongest for the target regime: sparse, threshold-mediated phenomena represented through rectified features. This is a scoped framework, not a universal guarantee that every sparse model admits large compression at negligible loss. Performance depends on the shape of the learned coefficient spectrum, class balance, and operating-point priorities. These boundaries are consistent with the chapter's scope and explain why compression is best treated as a policy-controlled operation rather than an unconditional simplification step.

6.9 ANSWER TO RQ3

The answer to RQ3 is **yes**, but in two distinct senses. In controlled threshold-mediated settings with explicit planted event logic, the compression phase works cleanly: the anytime path can recover substantially smaller m-of-k rules from the sparse rectified baseline without meaningful loss.

In real-world data, however, threshold-triggered structure can be mixed with more straightforward additive relationships and with measurement uncertainty, so it is not always clear that further simplification is desirable. Under the strict 2% policy with $\Delta\text{AUC} > -0.02$, HAI $a1$ and $a2$ support deployed rules that are $2.79\times$ and $3.12\times$ smaller, respectively, while HAI $a3$ and $a4$ remain exactly the cases where the correct conservative decision is "do not compress further" and the ionosphere probe offers no justification for simplification. The key contribution is therefore not only that logic polishing can work, but that the anytime formulation supplies an auditable baseline for accepting or rejecting compression under explicit operational tolerances [26, 27].

CHAPTER 7

FUTURE WORK

This dissertation establishes a rectification-first pipeline for interpretable longitudinal modeling and shows how sparse fitting and anytime compression can be integrated into a single decision-support workflow. The most useful future work is therefore not to revisit the core pipeline, but to broaden its theoretical coverage, expand its empirical boundary, and deepen its deployment evidence.

7.1 EXTENDING THE THEORETICAL SCOPE

The RQ2 theorem is intentionally centered on zero-threshold sign binarization under joint normality. A natural next step is to extend the analysis to learned nonzero critical ranges, finite-sample concentration bounds, and adverse covariance geometries. The most valuable outcome would be a family of regime-specific guarantees that clarifies when rectification helps because of dependence contraction and when other mechanisms dominate [10, 26].

7.2 BROADER EMPIRICAL VALIDATION

The dissertation's strongest gains appear in threshold-mediated settings. Future work should therefore test the same pipeline on additional clinical, industrial, and socio-environmental longitudinal benchmarks with standardized stability reporting and prospective evaluation splits [1, 2, 19, 32]. Those studies would sharpen the applicability boundary by separating regimes where rectification improves both discrimination and attribution from regimes where its main benefit is interpretability.

7.3 RICHER RULE FAMILIES AND HYBRID MODELS

Anytime compression currently targets compact m-of-K style rules. Future work should explore richer rule structures, including hierarchical rules, interaction-specific clauses, group-aware compression, and hybrid models that retain both threshold indicators and selected continuous magnitude terms. Such extensions could preserve the current pipeline's transparency while reducing performance loss in settings where post-threshold magnitude still carries predictive information [22–24].

7.4 PROSPECTIVE DEPLOYMENT AND HUMAN EVALUATION

The framework is designed for auditable decision support, but deployment studies remain an important next step. Future work should evaluate how analysts, operators, or clinicians use the extracted rules in practice, whether compression improves review speed and consistency, and how rule updates should be governed under concept drift. Human-centered validation is especially important in high-stakes environments where explanation quality matters alongside discrimination [20, 21].

7.5 TOOLING AND REPRODUCIBILITY

The core reproducibility path already exists: the `cutlass` package is public on PyPI, the dissertation repository already exposes notebook- and script-based study regeneration, and cached outputs are retained for auditability. Another useful direction is therefore not first release of the artifacts, but hardening that public release into a more polished artifact bundle with dataset manifests, versioned splits, and ideally one-command regeneration of every figure and table. That packaging work would make the framework easier to audit, benchmark, and extend in later work, and would lower the cost of reproducing both positive and negative findings.

7.6 CLOSING PERSPECTIVE

Taken together, these directions move this work from a strong pipeline result toward a more general methodology for interpretable longitudinal decision support. The core framework is now established; the next phase is to broaden its guarantees, domains, and deployment evidence.

CHAPTER 8

CONCLUSION

This dissertation examined longitudinal feature learning under three simultaneous constraints: dependence induced by lag expansion, the need for reliable feature-lag attribution, and the requirement for compact, human-auditable model outputs. The central claim is that a representation-first pipeline, followed by sparse fitting and anytime rule compression, provides a practical middle ground between unstable raw sparse selection and computationally heavy direct rule search [25–27].

8.1 RESEARCH PROGRAM SUMMARY

The work was organized around three research questions:

- **RQ1:** whether critical-range rectification improves sparse attribution reliability in longitudinal settings;
- **RQ2:** why rectification helps under a defensible theoretical mechanism and where that mechanism is scoped;
- **RQ3:** whether sparse rectified models can be compressed into compact logical rules without unacceptable operational loss.

Together, these questions move from empirical behavior (RQ1), to theoretical plausibility (RQ2), to deployment usability (RQ3). This sequencing is intentional: interpretability claims are weak if support recovery is unstable, and practical rule compression is less meaningful without a credible upstream selection mechanism [10, 11, 20, 21].

8.2 CONCLUSIONS BY RESEARCH QUESTION

8.2.1 RQ1 Conclusion: Rectification Can Improve Attribution Reliability in Target Regimes

The accumulated evidence supports a conditional positive conclusion for RQ1. In threshold-and-lag aligned settings, critical-range rectification tends to improve support concentration, lag localization, and sparse-model usability under multicollinearity stress [25, 26]. This finding is consistent with known dependence-related limits of penalty-only sparse selection, where prediction can remain acceptable while support identification degrades [10, 11].

The dissertation does not claim universal dominance over all baseline families. Direct interpretable comparators clarify this boundary: EBM can achieve even stronger raw fit, but with substantially higher additive complexity and runtime, while RuleFit and greedy rule lists are faster yet materially weaker on exact-lag fidelity in the target synthetic regime. The resulting conclusion is therefore not that rectification wins every comparison, but that representation-level intervention provides the strongest compact attribution-performance tradeoff when events are naturally threshold-mediated.

8.2.2 RQ2 Conclusion: A Scoped IC-Based Mechanism is Theoretically Defensible

For RQ2, this work provides a scoped theoretical result: in a tractable regime (zero-threshold sign binarization with explicit assumptions), dependence contraction can improve the likelihood of satisfying IC-related sparse-recovery conditions [10, 26]. The theoretical contribution is therefore mechanistic rather than universal.

This distinction is central to rigor. The chapter argues for a plausible structural pathway from binarization to improved sparse-support behavior, while explicitly separating theorem-scoped claims from broader engineering intuition. The resulting conclusion is that rectification is not just a heuristic preprocessing trick; it has a coherent connection to established sparse-selection theory under defined conditions.

8.2.3 RQ3 Conclusion: Anytime Compression Makes Interpretability Operational

For RQ3, this work reaches a two-part conclusion. In controlled threshold-mediated settings with explicit event logic, sparse rectified models can be compressed cleanly into substantially smaller rule representations without meaningful practical loss, showing that the compression phase itself works in the regime it targets [26, 27]. In real-world data, however, threshold-triggered effects can coexist with smoother additive relationships and with measurement uncertainty, so the more important contribution is the explicit acceptance policy that determines when simplification should be adopted and when the upstream sparse model should be retained [20, 21].

The key advance is the anytime framing itself: rule complexity can be reduced incrementally with explicit stop/adoption criteria, making the interpretability-performance tradeoff controllable rather than ad hoc. Prespecified non-inferiority framing further supports practical non-loss claims when simplification is the objective and provides a baseline for accepting or rejecting compression in mixed real-world regimes [34, 48].

8.3 PRIMARY DISSERTATION CONTRIBUTIONS

The dissertation contributes the following integrated advances:

1. **A unified rectification-first longitudinal pipeline** that preserves lag structure while map-

ping signals into critical-range indicators with clear operational semantics [25, 26].

2. **A scoped theoretical bridge to sparse-recovery conditions** showing how representation-level dependence contraction can increase IC favorability in a tractable regime [10, 26].

3. **An anytime rule-compression stage for interpretable deployment** that converts sparse coefficients into compact m-of-k-style logic with tunable performance-complexity tradeoffs [27].

4. **A dissertation-level empirical validation package** that adds repeated-resample stability and ablation, expanded sparse and direct interpretable baselines, cross-domain transfer and protocol audits, empirical boundary-condition stress tests, and held-out policy-based compression validation.

5. **A reproducible implementation pathway** that exposes the reusable modeling core through the public `cutlass` package and preserves notebook-backed study regeneration in the dissertation repository.

Beyond the prior papers in this research line, the dissertation-level delta lies in the repeated-resample stability and ablation package, the direct interpretable-model baselines, the cross-domain plus Goose Bay audit, the empirical boundary-condition package, and the strict held-out compression validation.

From a computer-science perspective, the dissertation's distinct contribution is the combination of representation-first feature redesign, reuse of mature sparse solvers, and policy-controlled interpretable compression in one auditable longitudinal modeling pipeline.

8.4 POSITIONING RELATIVE TO PRIOR WORK

This work extends, rather than replaces, prior sparse and interpretable modeling literature. Relative to classical L1 and elastic-net families [8, 12, 13, 28, 29], the main difference is intervention point: representation is modified before sparse optimization. Relative to grouped and ordered penalties [14–16, 35, 37], this work emphasizes threshold logic and lag-attribution clarity in a longitudinal context. Relative to direct interpretable additive, rule-ensemble, and rule-list approaches [22, 43, 44], it uses sparse convex fitting as a scalable front end and performs structured rule simplification afterward.

Table 3 in Chapter 3 compresses this cross-family comparison into one view, making the dissertation's intervention point, output form, and deployment-policy distinction explicit rather than only descriptive.

Accordingly, the work is best interpreted as a bridge architecture between sparse statistical learning and rule-level interpretability, not as a claim that one method family should replace all others.

8.5 SCOPE BOUNDARIES AND LIMITATIONS

Several limitations remain explicit:

- Benefits are strongest in regimes with threshold-mediated event structure and may be weaker when that structure is absent.
- The current theory is intentionally scoped and does not provide universal guarantees across arbitrary threshold schemes or dependence geometries.
- Compression quality depends on operating-point policy, class balance, and the shape of the learned sparse support.

- Interpretability gains can trade against raw discrimination in some datasets.

These boundaries are consistent with broader evidence that no sparse-selection or interpretable-modeling method is uniformly optimal across all dependence structures and objectives [11, 21].

8.6 PRACTICAL IMPLICATIONS

This work's practical implication is straightforward: when longitudinal decisions require both predictive utility and audit-ready rationale, it can be more effective to restructure features around event-relevant critical ranges first, then apply mature sparse optimization, and finally compress to explicit rule logic. From a computer-science perspective, the primary technical users are ML engineers building sensor-stream classifiers, ICS security analysts, and clinical-informatics researchers who need lag-aware models with explicit deployment rules rather than only high raw fit. This design is particularly relevant in domains such as ICS monitoring and clinical trajectory analysis, where lag effects, threshold triggers, and traceable decision criteria are operationally important [1, 2, 32].

8.7 FINAL STATEMENT

This work concludes that rectification-first sparse longitudinal learning with anytime rule compression is a credible and useful framework for interpretable event modeling under dependence. Its value lies in integration: empirical support behavior (RQ1), scoped mechanistic theory (RQ2), and operational rule simplification (RQ3) are addressed as a single pipeline rather than isolated techniques [25–27]. Within its stated scope, this provides a defensible contribution to interpretable machine learning for longitudinal data.

REFERENCES

[1] C. Hiploylee, P. Dufort, H. Davis, R. Wennberg, M. Tartaglia, D. Mikulis, and C. Tator, "Longitudinal study of postconcussion syndrome: Not everyone recovers," *Journal of Neurotrauma*, vol. 34, pp. 1511–1523, 2017.

[2] C. Wong, A. Caspi, B. Williams, I. Craig, R. Houts, A. Ambler, T. Moffitt, and J. Mill, "A longitudinal study of epigenetic variation in twins," *Epigenetics*, vol. 5, pp. 516–526, 2010.

[3] J. Sorvali, J. Kaseva, and P. Peltonen-Sainio, "Farmer views on climate change—a longitudinal study of threats, opportunities and action," *Climatic Change*, vol. 164, pp. 1–19, 2021.

[4] D. Hemraj, K. Solis, and N. e. a. Petersen, "Towards ecosystem-based techniques for tipping point detection," *Biological Reviews*, vol. 100, no. 2, pp. 892–919, 2025.

[5] A. S. Mata and S. C. Ferreira, "Multiple transitions of the susceptible-infected-susceptible epidemic model on complex networks," *Physical Review E*, vol. 91, no. 1, p. 012 816, 2015. DOI: 10.1103/PhysRevE.91.012816

[6] C. Peng and T. C. Yang, "Event-triggered communication and h-infinity control co-design for networked control systems," *Automatica*, vol. 49, no. 5, pp. 1326–1332, 2013. DOI: 10.1016/j.automatica.2013.01.038

[7] K. Ters and J. Urban, "Estimating unknown arbitrage costs: Evidence from a three-regime threshold vector error correction model," Bank for International Settlements, BIS Working Papers 689, 2018, Working paper version used in this dissertation. [Online]. Available: https://www.bis.org/publ/work689.htm

[8] R. Tibshirani, "Regression shrinkage and selection via the lasso," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 58, pp. 267–288, 1996.

[9] A. Y. Ng, "Feature selection, l1 vs. l2 regularization, and rotational invariance," in *Proc. ICML*, 2004.

[10] P. Zhao and B. Yu, "On model selection consistency of lasso," *Journal of Machine Learning Research*, 2006.

[11] L. Freijeiro-González, M. Febrero-Bande, and W. González-Manteiga, "A critical review of lasso and its derivatives for variable selection under dependence among covariates," *International Statistical Review*, vol. 90, pp. 118–145, 2022.

[12] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *Journal of the Royal Statistical Society: Series B*, 2005.

[13] H. Zou, "The adaptive lasso and its oracle properties," *Journal of the American Statistical Association*, 2006.

[14] M. Yuan and Y. Lin, "Model selection and estimation in regression with grouped variables," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 68, pp. 49–67, 2006.

[15] L. Meier, S. VanDeGaer, and P. Bühlmann, "The group lasso for logistic regression," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 70, pp. 53–71, 2006.

[16] R. Tibshirani and X. Suo, "An ordered lasso and sparse time-lagged regression," *Technometrics*, 2016.

[17] H. Zhu, W. Xu, Z. Zhang, Y. Xu, and J. Fan, “A variable selection approach for highly correlated predictors in high-dimensional genomic data,” *Bioinformatics*, 2021.

[18] A. Katrutsa and V. Strijov, “Comprehensive study of feature selection methods to solve multicollinearity problem according to evaluation criteria,” *Expert Systems with Applications*, vol. 76, pp. 1–11, 2017.

[19] L. Yang and T. Wu, “Model-based clustering of high-dimensional longitudinal data via regularization,” *Biometrics*, vol. 79, pp. 761–774, 2023.

[20] C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Machine Intelligence*, 2019.

[21] W. J. Murdoch, C. Singh, K. Kumbier, R. Abbasi-Asl, and B. Yu, “Definitions, methods, and applications in interpretable machine learning,” *Proceedings of the National Academy of Sciences*, 2019.

[22] E. Angelino, N. Larus-Stone, D. Alabi, M. Seltzer, and C. Rudin, “Learning certifiably optimal rule lists for categorical data,” *Journal of Machine Learning Research*, 2018.

[23] E. Boros, P. L. Hammer, T. Ibaraki, and A. Kogan, “Logical analysis of numerical data,” *Mathematical Programming*, 1997.

[24] E. Boros, P. L. Hammer, T. Ibaraki, A. Kogan, E. Mayoraz, and I. Muchnik, “An implementation of logical analysis of data,” *IEEE Transactions on Knowledge and Data Engineering*, 2000.

[25] J. Orender, M. Zubair, and J. Sun, "Lasso logic engine: Harnessing the logic parsing capabilities of the lasso algorithm for longitudinal feature learning," in *2022 IEEE International Conference on Big Data (Big Data)*, IEEE, 2022, pp. 309–318.

[26] J. Orender, J. Sun, and M. Zubair, "Efficient longitudinal feature selection via binarized transformation: Theory and case studies," in *IEEE Big Data*, 2025.

[27] J. Orender, J. Sun, and M. Zubair, "Anytime rule compression and rectified logistic modeling for longitudinal signals," in *1st International Conference on Big Data Analytics and Applications (BDAA)*, 2025.

[28] J. Friedman, R. Tibshirani, and T. Hastie, "Regularization paths for generalized linear models via coordinate descent," *Journal of Statistical Software*, vol. 33, pp. 1–22, 2010. DOI: `10.18637/jss.v033.i01`

[29] J. Tay, B. Narasimhan, and T. Hastie, "Elastic net regularization paths for all generalized linear models," *Journal of Statistical Software*, vol. 106, pp. 1–31, 2023. DOI: `10.18637/jss.v106.i01`

[30] J. Orender, *cutlass: Rectified L1 logistic regression with critical-range encoding*, Python package on PyPI, version 0.1.0, 2026.

[31] J. Orender, *Dissertation artifact repository for interpretable sparse modeling of longitudinal signals*, Public GitHub repository with notebooks, regeneration scripts, cached outputs, and manuscript figures, 2026.

[32] H. Shin, W. Lee, J. Yun, and H. Kim, "Hai 1.0: Hil-based augmented ics security dataset," in *13th USENIX Workshop on Cyber Security Experimentation and Test*, Santa Clara, CA, USA: CSET, 2020.

[33] V. Sigillito, S. Wing, L. Hutton, and K. Baker, "Classification of radar returns from the ionosphere using neural networks," *Johns Hopkins APL Technical Digest*, vol. 10, pp. 262–266, 1989.

[34] D. J. Schuirmann, "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability," *Journal of Pharmacokinetics and Biopharmaceutics*, 1987.

[35] Y. Yang and H. Zou, "A fast unified algorithm for solving group-lasso penalize learning problems," *Statistics and Computing*, vol. 25, pp. 1129–1141, 2015.

[36] R. A. Horn and C. R. Johnson, *Matrix Analysis*. Cambridge University Press, 2012.

[37] B. Turlach, W. Venables, and S. Wright, "Simultaneous variable selection," *Technometrics*, vol. 47, pp. 349–363, 2005.

[38] M. Rejchel and M. Bogdan, "Rank-based lasso: Efficient methods for high-dimensional robust model selection," *Journal of Machine Learning Research*, 2020.

[39] Y. Kim, J. Kim, and Y. Kim, "Blockwise sparse regression," *Statistica Sinica*, pp. 375–390, 2006.

[40] M. Roozbeh, S. Babaie-Kafaki, and A. Sadigh, "A heuristic approach to combat multicollinearity in least trimmed squares regression analysis," *Applied Mathematical Modelling*, vol. 57, pp. 105–120, 2018.

[41] L. Ladha and T. Deepa, "Feature selection methods and algorithms," *International journal on computer science and engineering*, vol. 3, pp. 1787–1797, 2011.

[42] M. KP and P. Thiyagarajan, “Feature selection using efficient fusion of fisher score and greedy searching for alzheimer’s classification,” *Journal of King Saud University-Computer and Information Sciences*, vol. 34, pp. 4993–5006, 2022.

[43] Y. Lou, R. Caruana, and J. Gehrke, “Intelligible models for classification and regression,” in *Proceedings of the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2012, pp. 150–158. DOI: 10.1145/2339530.2339556

[44] J. Friedman and B. Popescu, “Predictive learning via rule ensembles,” *The Annals of Applied Statistics*, vol. 2, no. 3, pp. 916–954, 2008. DOI: 10.1214/07-AOAS148

[45] T. Inoue, H. Qiu, R. Ueji, and Y. Kimura, “Ductile-to-brittle transition and brittle fracture stress of ultrafine-grained low-carbon steel,” *Materials*, vol. 14, p. 1634, 2021.

[46] UNICEF, *UNICEF Data Warehouse*, https://data.unicef.org/dv_index/, 2023.

[47] W. F. Sheppard, “On the application of the theory of error to cases of normal distribution and normal correlation,” *Philosophical Transactions of the Royal Society of London. Series A*, vol. 192, pp. 101–167, 1899. DOI: 10.1098/rsta.1899.0003

[48] Student, “The probable error of a mean,” *Biometrika*, 1908.

APPENDIX A

SYNTHETIC DATA DEVELOPMENT

The synthetic studies use a common generator for bounded, time-varying sensor streams with delayed threshold-triggered events. Each data set begins as continuous channels. A small subset is designated as relevant, labels are generated from lag-shifted relevant channels, and the observed unshifted channels are flattened into fixed-width feature-lag rows for model fitting. When pseudo-random draws are required, the generator uses the Mersenne–Twister algorithm.

1. **Base oscillatory components.** Generate a large collection of sine waves using one of five randomly selected frequencies per cycle. The generated object is a bounded time-varying component, and its purpose is to provide sensor-like variation without an unbounded trend. This supports the threshold-and-lag experiments by giving later steps realistic temporal structure.

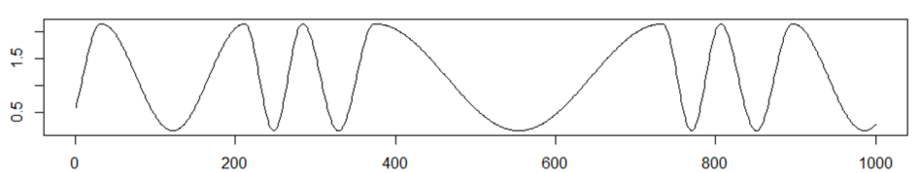


Figure 30. Example base sine-wave component used to construct bounded time-varying synthetic channels.

2. **Composite sensor channels.** Superpose multiple base waves with varying frequencies and amplitudes to form each continuous channel. The generated object is a more complex sensor

stream, and its purpose is to avoid a single easily visible periodic pattern. This keeps the synthetic task focused on feature, threshold, and lag recovery rather than on recognizing one simple waveform.

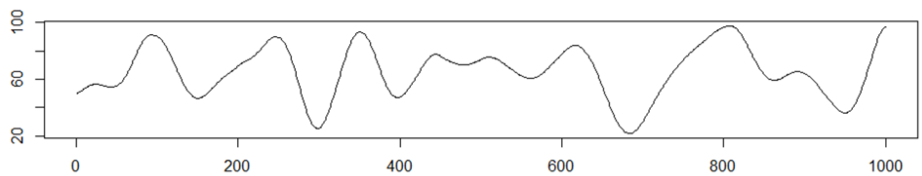


Figure 31. Example composite channel formed by superposing base waves with varying frequencies and amplitudes.

3. **Magnitude scaling.** Assign channel ranges using pseudorandom draws from a normal distribution centered at zero with standard deviation 0.05, scaled by 100. The generated object is a set of channels with heterogeneous magnitudes, and its purpose is to prevent the method from depending on one fixed amplitude scale. This supports the range-agnostic interpretation of critical-range discovery.

4. **Lagged event mechanism.** Shift the designated relevant channels by predetermined offsets before label generation. The generated object is a delayed copy of each relevant channel, and its purpose is to represent processes in which a sensor change precedes the observed event by an unknown delay. The shifted channels determine when the event occurs, while the unshifted channels remain the measurements supplied to the model.

5. **Relevant-feature selection and label construction.** Assemble a pool of candidate channels

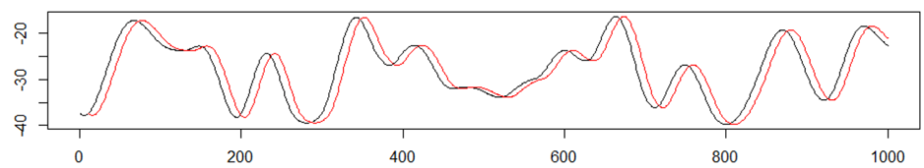


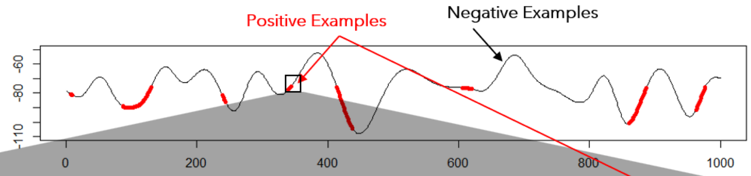
Figure 32. Example delayed channel used to generate lagged threshold-triggered event labels while preserving unshifted measurements for model fitting.

and designate selected channels as relevant features. At each time step, assign the label `True` only when all relevant shifted channels satisfy their threshold criteria; otherwise assign `False`. The generated object is a binary event label with known ground-truth feature and lag structure. Figure 1 illustrates the threshold-and-lag interpretation used by this construction.

6. **Lag flattening.** Convert the longitudinal channels into supervised learning rows by concatenating the current measurements with the previous ten time steps for each channel; this window length is adjustable in the script. The generated object is a fixed-width feature matrix whose columns encode feature-lag pairs. Consecutive chronological rows have overlapping histories, but each row exposes the model to only the flattened measurements and the corresponding event label.

7. **Row randomization and split.** Randomize the flattened rows before modeling so that chronological adjacency does not determine train/test membership. The generated object is the final supervised data set, split into training and testing partitions with 70% of rows used for training and 30% used for testing.

The resulting data set contains bounded continuous sensor-like measurements, known relevant



	V10TM10	V10TM9	V10TM8	V10TM7	V10TM6	V10TM5	V10TM4	V10TM3	V10TM2	V10TM1	V10TM0	INDC
330	-80.94031	-80.79972	-80.61806	-80.39601	-80.13446	-79.83451	-79.49742	-79.12467	-78.71790	-78.27890	-77.80965	TRUE
331	-80.79972	-80.61806	-80.39601	-80.13446	-79.83451	-79.49742	-79.12467	-78.71790	-78.27890	-77.80965	-77.31225	TRUE
332	-80.61806	-80.39601	-80.13446	-79.83451	-79.49742	-79.12467	-78.71790	-78.27890	-77.80965	-77.31225	-76.78894	TRUE
333	-80.39601	-80.13446	-79.83451	-79.49742	-79.12467	-78.71790	-78.27890	-77.80965	-77.31225	-76.78894	-76.24209	TRUE
334	-80.13446	-79.83451	-79.49742	-79.12467	-78.71790	-78.27890	-77.80965	-77.31225	-76.78894	-76.24209	-75.67417	TRUE
335	-79.83451	-79.49742	-79.12467	-78.71790	-78.27890	-77.80965	-77.31225	-76.78894	-76.24209	-75.67417	-75.08774	FALSE
336	-79.49742	-79.12467	-78.71790	-78.27890	-77.80965	-77.31225	-76.78894	-76.24209	-75.67417	-75.08774	-74.47798	FALSE
337	-79.12467	-78.71790	-78.27890	-77.80965	-77.31225	-76.78894	-76.24209	-75.67417	-75.08774	-74.47798	-73.84015	FALSE
338	-78.71790	-78.27890	-77.80965	-77.31225	-76.78894	-76.24209	-75.67417	-75.08774	-74.47798	-73.84015	-73.17711	FALSE
339	-78.27890	-77.80965	-77.31225	-76.78894	-76.24209	-75.67417	-75.08774	-74.47798	-73.84015	-73.17711	-72.49184	FALSE
340	-77.80965	-77.31225	-76.78894	-76.24209	-75.67417	-75.08774	-74.47798	-73.84015	-73.17711	-72.49184	-71.78736	FALSE

Figure 33. Lag-flattening layout that converts longitudinal measurements into fixed feature-lag columns for supervised learning.

features, background candidate features, binary labels generated by a logical AND over lagged threshold conditions, and a fixed feature-lag representation for model fitting. This construction preserves ground-truth feature and lag information while making the raw curves sufficiently complex that simple temporal order or amplitude scale does not determine the result.

APPENDIX B

LEMMAS

B.1 LEMMA 1

Lemma B.1 (Arcsin correlation map)**.** *For jointly standard normal random variables thresholded at zero, the relationship between the pre-binarization correlation ρ and the post-binarization correlation $\tilde{\rho}$ is the classical arcsine relation [47]:*

$$\tilde{\rho} = \frac{2}{\pi} \arcsin(\rho) \tag{24}$$

The following derivation is included for the reader's convenience and is written in notation consistent with this dissertation. It follows the same median-classification argument underlying Sheppard's classical result [47].

Assumptions.

Let X and Y be jointly standard normal random variables with:

- **Means:** $E[X] = E[Y] = 0$ (centered)
- **Variances:** $\text{Var}(X) = \text{Var}(Y) = 1$ (standardized)
- **Correlation:** $\text{Corr}(X,Y) = \rho$, where $-1 \leq \rho \leq 1$

The variables are binarized using threshold zero:

$$\tilde{X} = \begin{cases} 1 & \text{if } X > 0 \\ \\ 0 & \text{if } X \leq 0 \end{cases} \quad \text{and} \quad \tilde{Y} = \begin{cases} 1 & \text{if } Y > 0 \\ \\ 0 & \text{if } Y \leq 0 \end{cases} \tag{25}$$

This $0/1$ encoding is affine-equivalent to the $\{\pm 1\}$ sign encoding used in the RQ2 chapter; after centering, the correlation-ordering arguments are unchanged up to constant scaling.

Proof. Since X and Y are symmetric about zero:

- Mean of $\tilde{X} : E[\tilde{X}] = P(X > 0) = 0.5$
- Variance of $\tilde{X} : \text{Var}(\tilde{X}) = E[\tilde{X}^2] - (E[\tilde{X}])^2 = 0.5 - (0.5)^2 = 0.25$
- Similarly for $\tilde{Y} : E[\tilde{Y}] = 0.5, \quad \text{Var}(\tilde{Y}) = 0.25$

The correlation $\tilde{\rho}$ between $\tilde{X}$ and $\tilde{Y}$ is:

$$\tilde{\rho} = \frac{E[\tilde{X}\tilde{Y}] - E[\tilde{X}]E[\tilde{Y}]}{\sqrt{\text{Var}(\tilde{X})\text{Var}(\tilde{Y})}} = 4\left(P(\tilde{X} = 1, \tilde{Y} = 1) - 0.25\right) \tag{26}$$

Since X and Y are assumed to be jointly normally distributed, $P(\tilde{X} = 1, \tilde{Y} = 1)$ can be expressed using the **bivariate standard normal cumulative distribution function (CDF)**.

$$P(\tilde{X} = 1, \tilde{Y} = 1) = P(X > 0, Y > 0) \tag{27}$$

For standard normal variables X and Y with correlation ρ, the bivariate normal quadrant identity holds [47]:

$$P(X > 0, Y > 0) = \frac{1}{4} + \frac{\arcsin(\rho)}{2\pi} \tag{28}$$

Explanation:

- The joint probability $P(X > 0, Y > 0)$ corresponds to the *orthant probability* in the positive quadrant.
- The term $\arcsin(\rho)$ arises from integrating the bivariate normal density over this quadrant.

Substituting this quadrant probability into the binarized correlation expression gives:

$$\tilde{\rho} = 4\left(\frac{1}{4} + \frac{\arcsin(\rho)}{2\pi} - 0.25\right) = \frac{2}{\pi}\arcsin(\rho) \tag{29}$$

□

Closing. Thus, zero-threshold binarization maps the original Gaussian correlation to the arcsine-transformed correlation in Equation 24.

B.2 LEMMA 2

Lemma B.2 (Correlation contraction)**.** *Under the assumptions of Lemma B.1, zero-threshold binarization does not increase the absolute pairwise correlation:*

$$|\tilde{\rho}| \leq |\rho| \tag{30}$$

Assumptions.

Assume $-1 \leq \rho \leq 1$ and use the arcsine map from Lemma B.1. The needed properties of the arcsine function are:

- The **arcsine function** $\arcsin(x)$ is defined for $x \in [-1, 1]$.
- The range of $\arcsin(x)$ is $[-\frac{\pi}{2}, \frac{\pi}{2}]$.
- The function $\arcsin(x)$ is **increasing** on $[-1, 1]$.

Proof. It is sufficient to consider ρ in one of three intervals:

1. $\rho = 0$
2. $0 < \rho < 1$
3. $-1 < \rho < 0$

When $\rho = 0$, $\arcsin(0) = 0$, so $\tilde{\rho} = 0$ and the desired inequality holds. The ratio is indeterminate at zero, but its limiting value is $2/\pi$ because $\arcsin(\rho) \approx \rho$ as $\rho \to 0$.

For $0 < \rho < 1$, both ρ and $\arcsin(\rho)$ are positive. Define the ratio function:

$$f(\rho) = \frac{2}{\pi\rho} \arcsin(\rho) \tag{31}$$

For $0 < \rho < 1$, $\arcsin(\rho) < \frac{\pi}{2}\rho$. This inequality holds because the function $\arcsin(\rho)$ is strictly convex on $(0,1)$, and its graph lies below the straight line $y = \frac{\pi}{2}\rho$ [26]. Therefore, $f(\rho) < 1$ for all $0 < \rho < 1$.

For $-1 < \rho < 0$, both ρ and $\arcsin(\rho)$ are negative, so $\tilde{\rho}/\rho$ is positive. In this interval, $\arcsin(\rho) > \frac{\pi}{2}\rho$; dividing by the negative value ρ reverses the inequality and again gives $f(\rho) < 1$. At the endpoints, $\arcsin(-1) = -\frac{\pi}{2}$ and $\arcsin(1) = \frac{\pi}{2}$, so the ratio is equal to one only when $\rho = \pm 1$.

$$|\tilde{\rho}| \leq |\rho| \tag{32}$$

$\square$

Closing. Binarization therefore reduces or preserves pairwise correlation magnitude and provides the pairwise contraction result used by the matrix-level RQ2 argument.

B.3 LEMMA 3

Lemma B.3 (Inactive-active covariance contraction)**.** *For any irrelevant feature $x_{S^c i} \in X_{S^c}$ and relevant feature $x_{Sj} \in X_S$, zero-threshold binarization satisfies*

$$\left|\mathrm{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{Sj})\right| \leq \left|\mathrm{Cov}(x_{S^c i}, x_{Sj})\right| \tag{33}$$

Assumptions.

1. **Joint Normality:** The continuous features $x_{S^c i}$ and x_{Sj} are jointly normally distributed with zero means and unit variances:

$$x_{S^c i}, x_{Sj} \sim \mathcal{N}(0,1) \tag{34}$$

and correlation $\rho_{ij} = \mathrm{Corr}(x_{S^c i}, x_{Sj})$.

2. **Binarization at Zero Threshold** - The binarized variables are defined as:

$$\tilde{X}_k = \begin{cases} 1 & \text{if } X_k > 0 \\ 0 & \text{if } X_k \leq 0 \end{cases}, \quad k = i, j \tag{35}$$

Proof. The covariance between $x_{S^c i}$ and x_{Sj} is:

$$\mathrm{Cov}(x_{S^c i}, x_{Sj}) = \rho_{ij} \cdot \sigma_{x_{S^c i}} \sigma_{x_{Sj}} = \rho_{ij} \cdot (1)(1) = \rho_{ij} \tag{36}$$

The covariance between $\tilde{x}_{S^c i}$ and $\tilde{x}_{Sj}$ is:

$$\mathrm{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{Sj}) = \mathbb{E}[\tilde{x}_{S^c i} \tilde{x}_{Sj}] - \mathbb{E}[\tilde{x}_{S^c i}]\mathbb{E}[\tilde{x}_{Sj}] \tag{37}$$

Since $x_{S^c i}$ and x_{Sj} are symmetric about zero:

$$\mathbb{E}[\tilde{x}_{S^c i}] = P(x_{S^c i} > 0) = \frac{1}{2}, \quad \mathbb{E}[\tilde{x}_{Sj}] = \frac{1}{2} \tag{38}$$

Compute the joint probability:

$$\mathbb{E}[\tilde{x}_{S^c i}\tilde{x}_{Sj}] = P(x_{S^c i} > 0, x_{Sj} > 0) \tag{39}$$

For jointly standard normal variables, this joint probability is given by the same bivariate normal quadrant identity [47]:

$$P(x_{S^c i} > 0, x_{Sj} > 0) = \frac{1}{4} + \frac{\arcsin(\rho_{ij})}{2\pi} \tag{40}$$

Therefore, the covariance is:

$$\mathrm{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{Sj}) = \left(\frac{1}{4} + \frac{\arcsin(\rho_{ij})}{2\pi}\right) - \left(\frac{1}{2}\cdot\frac{1}{2}\right) = \frac{\arcsin(\rho_{ij})}{2\pi} \tag{41}$$

We need to show that:

$$\left|\frac{\arcsin(\rho_{ij})}{2\pi}\right| \leq |\rho_{ij}| \tag{42}$$

When $\rho_{ij} \geq 0$, the function $f(\rho) = \frac{\arcsin(\rho)}{2\pi}$ is increasing on $[0,1]$. Since $\arcsin(\rho) \leq \frac{\pi}{2}$, we have:

$$0 \leq \frac{\arcsin(\rho_{ij})}{2\pi} \leq \frac{1}{4} \tag{43}$$

Also, $\rho_{ij} \leq 1$, so:

$$0 \leq \frac{\arcsin(\rho_{ij})}{2\pi} \leq \rho_{ij} \tag{44}$$

because $\arcsin(\rho_{ij}) \leq \frac{\pi}{2}\rho_{ij}$ in $[0,1]$.

When $\rho_{ij} \leq 0$, the function $f(\rho) = \frac{\arcsin(\rho)}{2\pi}$ is increasing on $[-1,0]$. Since $\arcsin(\rho_{ij}) \geq -\frac{\pi}{2}$, we have:

$$-\frac{1}{4} \leq \frac{\arcsin(\rho_{ij})}{2\pi} \leq 0 \tag{45}$$

Also, $\rho_{ij} \geq -1$, so:

$$\rho_{ij} \leq \frac{\arcsin(\rho_{ij})}{2\pi} \leq 0 \tag{46}$$

because $\arcsin(\rho_{ij}) \geq \frac{\pi}{2}\rho_{ij}$ in $[-1, 0]$.

Therefore, in both cases:

$$\left|\mathrm{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{S j})\right| = \left|\frac{\arcsin(\rho_{ij})}{2\pi}\right| \leq |\rho_{ij}| = \left|\mathrm{Cov}(x_{S^c i}, x_{S j})\right| \tag{47}$$

$\square$

Closing. The magnitude of the covariance between $\tilde{x}_{S^c i}$ and $\tilde{x}_{S j}$ is less than or equal to the magnitude of the covariance between $x_{S^c i}$ and $x_{S j}$. Therefore, binarization reduces the covariances between irrelevant and relevant features.

B.4 LEMMA 4

Lemma B.4 (Positive-correlation inverse norm bound)**.** *When* $0 < \tilde{\rho} < \rho < 1$*,*

$$\|\tilde{G}^{-1}\|_\infty \leq \|G^{-1}\|_\infty \tag{48}$$

Assumptions.

- $n > 0$: Positive scalar (e.g., sample size).
- $s \geq 2$: Dimension of the square matrices ($s \times s$).

- $\mathbf{I}_s$: Identity matrix of size s.
- $\mathbf{J}_s$: All-ones matrix of size s.
- $0 < \tilde{\rho} < \rho < 1$: Constants with $\tilde{\rho}$ always less than ρ.

The original Gram matrix is:

$$G = n\left((1-\rho)\mathbf{I}_s + \rho\mathbf{J}_s\right) \tag{49}$$

The binarized Gram matrix is:

$$\tilde{G} = n\left((1-\tilde{\rho})\mathbf{I}_s + \tilde{\rho}\mathbf{J}_s\right) \tag{50}$$

Proof. First, rewrite G as

$$G = n(1-\rho)\left(\mathbf{I}_s + \frac{\rho}{1-\rho}\mathbf{J}_s\right) \tag{51}$$

Let $\alpha = \dfrac{\rho}{1-\rho}$.

Compute $(\mathbf{I}_s + \alpha\mathbf{J}_s)^{-1}$:

- Observation: $\mathbf{J}_s$ is rank-1.
- Sherman-Morrison-Woodbury formula for the inverse:

$$(\mathbf{A} + \mathbf{U}\mathbf{C}\mathbf{V}^T)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}^T\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}^T\mathbf{A}^{-1} \tag{52}$$

- $\mathbf{A} = \mathbf{I}_s \Rightarrow \mathbf{A}^{-1} = \mathbf{I}_s$
- $\mathbf{U} = \mathbf{V} = \mathbf{1}_s$
- $\mathbf{C} = \alpha$
- $\mathbf{C}^{-1} = \frac{1}{\alpha}$ (the inverse of a scalar being its reciprocal)

- $\mathbf{V}^T\mathbf{A}^{-1}\mathbf{U} = \mathbf{1}_s^T\mathbf{1}_s = s$

For $\mathbf{I}_s + \alpha\mathbf{J}_s$:

$$(\mathbf{I}_s + \alpha\mathbf{J}_s)^{-1} = \mathbf{I}_s - \frac{\alpha}{1+\alpha s}\mathbf{J}_s \tag{53}$$

Therefore,

$$G^{-1} = \frac{1}{n(1-\rho)}\left(\mathbf{I}_s - \frac{\rho}{1+\rho(s-1)}\mathbf{J}_s\right) \tag{54}$$

Similarly, for $\tilde{G}$:

$$\tilde{G}^{-1} = \frac{1}{n(1-\tilde{\rho})}\left(\mathbf{I}_s - \frac{\tilde{\rho}}{1+\tilde{\rho}(s-1)}\mathbf{J}_s\right) \tag{55}$$

The Infinity norm for matrices is defined as:

$$\|A\|_\infty = \max_{1\le i\le s}\sum_{j=1}^{s}|a_{ij}| \tag{56}$$

The entries of G^{-1} are:

- Diagonal Entries: $(G^{-1})_{ii} = \frac{1}{n(1-\rho)}(1-\gamma)$

- Off-Diagonal Entries: $(G^{-1})_{ij} = -\frac{\gamma}{n(1-\rho)} \quad$ for $i \neq j$

Where: $\gamma = \frac{\rho}{1+\rho(s-1)}$

Row Sum:

$$\text{RowSum}_{G^{-1}} = \frac{1}{n(1-\rho)}\left[(1-\gamma)+(s-1)\gamma\right] \tag{57}$$

This row sum simplifies to:

$$\begin{aligned}\text{RowSum}_{G^{-1}} &= \frac{1}{n(1-\rho)}\left[1-\gamma+\gamma(s-1)\right] \\ &= \frac{1}{n(1-\rho)}\left[1+\gamma(s-2)\right]\end{aligned} \tag{58}$$

Where: $\tilde{\gamma} = \frac{\tilde{\rho}}{1+\tilde{\rho}(s-1)}$

Similarly:

$$\text{RowSum}_{\tilde{G}^{-1}} = \frac{1}{n(1-\tilde{\rho})}\left[1+\tilde{\gamma}(s-2)\right] \tag{59}$$

Define $f(x) = \frac{x}{1+x(s-1)}$. Because $f(x)$ is increasing on $\left(\frac{-1}{s-1}, 1\right)$, $x_1 < x_2$ implies $f(x_1) < f(x_2)$. It follows that:

- $f(x)$ is *increasing* wrt x on $\left(\frac{-1}{s-1}, 1\right)$.
- $|\tilde{\rho}| \leq |\rho|$ (Lemma 2), and both are bounded by the interval $(-1, 1)$
- Since $\tilde{\rho} \geq \rho$ when $\tilde{\rho}$ and ρ are less than zero (meaning that ρ is *more* negative), the monotonicity of f(x) implies that $|f(\tilde{\rho})| \leq |f(\rho)|$
- Therefore, we have $|\tilde{\gamma}| = |f(\tilde{\rho})| \leq |f(\rho)| = |\gamma|$, with a discontinuity at $x = \frac{-1}{s-1}$.
- Furthermore, $\tilde{\rho}, \rho$, and hence γ, $\tilde{\gamma}$ will all be of the same sign in the interval $\left[\frac{-1}{s-1}, 1\right)$ for $\tilde{\rho}$ and ρ, and γ will be positive otherwise.

Since $|\gamma| \geq |\tilde{\gamma}|$ and $s-2 \geq 0$ (because $s \geq 2$),

$$\gamma(s-2) \geq \tilde{\gamma}(s-2) \text{ when } \gamma > 0 \tag{60}$$

Therefore,

$$1+\gamma(s-2) \geq 1+\tilde{\gamma}(s-2) \text{ when } \gamma > 0 \tag{61}$$

Since $|\tilde{\rho}| \leq |\rho|$,

- $1-\tilde{\rho} \geq 1-\rho \Rightarrow \frac{1}{n(1-\tilde{\rho})} \leq \frac{1}{n(1-\rho)}$ when $\tilde{\rho},\rho > 0$
- $1-\tilde{\rho} \leq 1-\rho \Rightarrow \frac{1}{n(1-\tilde{\rho})} \geq \frac{1}{n(1-\rho)}$ when $\tilde{\rho},\rho < 0$

The Infinity norms when $\tilde{\rho},\rho$ and $\gamma > 0$ are:

$$\|G^{-1}\|_\infty = \frac{1}{n(1-\rho)}\left[1+\gamma(s-2)\right] \tag{62}$$

$$\|\tilde{G}^{-1}\|_\infty = \frac{1}{n(1-\tilde{\rho})}\left[1+\tilde{\gamma}(s-2)\right] \tag{63}$$

Since both terms examined in the expression for G^{-1} are greater than the corresponding terms in the expression for $\tilde{G}^{-1}$ when $\tilde{\rho},\rho$ and γ are greater than zero,

$$\|G^{-1}\|_\infty \geq \|\tilde{G}^{-1}\|_\infty \tag{64}$$

Furthermore, when $s=2$ and $\tilde{\rho},\rho$ and γ are less than zero it can be shown that the reverse is guaranteed to be true. When $s>2$ the answer is dependent upon where $\tilde{\rho}$ or ρ fall with respect to the interval $\left[\frac{-1}{s-1},1\right)$. When ρ is below the lower bound but $\tilde{\rho}$ is above, the relative norm values will be ambiguous. By constraining ρ and $\tilde{\rho}$ to be greater than zero, the inequality is unambiguous.

As a result, in the positive-correlation PoC regime, the derived inequality supports the IC-oriented matrix statement

$$\|\tilde{G}^{-1}\|_\infty \leq \|G^{-1}\|_\infty. \tag{65}$$

□

Closing.

Combined with the inactive-active leakage contraction, this supports the appendix-level weak probability claim in RQ2: sign binarization makes IC satisfaction at least as likely under the analyzed zero-threshold, jointly normal, positive-correlation conditions.

The negative-correlation cases above should be read as boundary conditions rather than as an extension of the theorem. Their behavior depends on where ρ and $\tilde{\rho}$ fall relative to the discontinuity at $\frac{-1}{s-1}$, so they require empirical stress checks. Adding complement features is an engineering mitigation explored in those boundary checks: it can re-express some negative associations as positive ones, but it is not a theorem-level guarantee for arbitrary negative-correlation designs.

APPENDIX C

REPRODUCIBILITY MAP

This appendix provides the direct manuscript-to-artifact map referenced in the introduction and RQ chapters. Each row names the notebook-backed study package, its paired generator, the cached run directory preserved under `notebooks/runs_new/`, the principal manuscript artifact(s) supported by that package, and the chapter that consumes the result.

From the repository root, the minimal regeneration pattern is:

```
python scripts/_generate_<study>_notebook.py
jupyter nbconvert --to notebook --execute --inplace \
  --ExecutePreprocessor.kernel_name=cutlass \
  notebooks/<study>.ipynb
```

where `<study>` is replaced by one of the notebook stems listed below. If the HAI parquet inputs are absent for the HAI-backed studies, run the preprocessing helper first:

```
python notebooks/build_hai_cutlass_data.py
```

Table 13. Direct regeneration map for the introduction and RQ1 study packages.

Study package, generator, cached outputs	Principal manuscript artifacts	Chapter(s)
notebooks/stability_ablation.ipynb scripts/_generate_stability_ablation_notebook.py notebooks/runs_new/stability_ablation/	Table 5; Figures 7, 8, 9, and 10.	RQ1 chapter.
notebooks/cross_domain.ipynb scripts/_generate_cross_domain_notebook.py notebooks/runs_new/cross_domain/	Table 6; Figures 18 and 19.	RQ1 chapter.
notebooks/goose_bay_robustness.ipynb scripts/_generate_goose_bay_robustness_notebook.py notebooks/runs_new/goose_bay_robustness/	Figures 20, 21, and 22.	RQ1 chapter.
notebooks/interpretable_baselines.ipynb scripts/_generate_interpretable_baselines_notebook.py notebooks/runs_new/interpretable_baselines/	Table 7; Figures 12, 13, and 14.	RQ1 chapter.
notebooks/walkthrough.ipynb scripts/_generate_walkthrough_notebook.py notebooks/runs_new/walkthrough/	Worked HAI exemplar subsection in the RQ1 chapter and Table 8.	RQ1 chapter.

Table 14. Direct regeneration map for the RQ2 and RQ3 study packages.

Study package, generator, cached outputs	Principal manuscript artifacts	Chapter(s)

`notebooks/boundary_conditions.ipynb`
`scripts/_generate_boundary_conditions_`
`notebook.py`
`notebooks/runs_new/boundary_conditions/`
	Table 11; Figures 23, 24, 26, and 27.	RQ2 chapter.

`notebooks/compression_validation.ipynb`
`scripts/_generate_compression_`
`validation_notebook.py`
`notebooks/runs_new/compression_`
`validation/`
	Table 12 and Figure 28.	RQ3 chapter.

APPENDIX D

RULE-CARD TEMPLATE FOR ACCEPTED COMPRESSED RULES

This appendix defines the handoff template for a compressed rule that passes the dissertation's held-out adoption gate. The template is not another statistical test. It is the operational record that makes an accepted m-of-k rule reviewable after model fitting: what rule was selected, what evidence justified acceptance, where it may be used, and when it must be revisited.

For the worked HAI exemplar in the RQ1 chapter, the selected-rule and deployment-summary artifacts are written under `notebooks/runs_new/walkthrough/`. A completed rule card would combine those files with the validation policy in the RQ3 chapter and the intended-use notes from the domain review before the compressed rule is treated as deployable.

Table 15. Rule-card template for accepted compressed rules: identity, rule logic, and intended use.

Field	Required content
Rule identifier	Dataset or subsystem, outcome label, model version, training window, held-out evaluation split, author or owner, reviewer, and date.
Intended use	Operational decision supported by the rule, target population or process, expected user, and explicit out-of-scope uses.
Upstream model	Raw or rectified sparse baseline identifier, active-support size before compression, penalty-selection rule, and training-only threshold-estimation policy.
Selected feature-lag conditions	One row per retained condition: variable name, lag or time window, direction or sign, critical-range lower bound, critical-range upper bound, and whether the condition is one-sided or interval-like.
Vote rule	Retained prefix k, vote threshold m, tie handling, predicted class when the vote passes, and how negative-signed conditions or complements are represented.
Baseline metrics	Held-out AUC, Youden's J, sensitivity, specificity, operating threshold, and any domain metric used to justify the upstream sparse baseline.
Rule metrics	Held-out AUC, Youden's J, sensitivity, specificity, primitive-condition count, and compression ratio relative to the upstream active support.

Table 16. Rule-card template for accepted compressed rules: validation, risks, and monitoring.

Field	Required content
Validation deltas	ΔAUC, ΔJ, paired-bootstrap confidence intervals or lower bounds, and the exact non-inferiority margins used for acceptance.
Acceptance decision	Whether both held-out gates passed, whether the deployed artifact is the compressed rule or the upstream sparse baseline, and who approved the decision.
Known failure modes	Weak threshold structure, unstable threshold windows, meaningful post-threshold magnitude effects, concept drift, sensor changes, missing channels, or a failed compression gate in adjacent studies.
Monitoring trigger	Data-quality, drift, performance, or domain-event conditions that require review before continued use. Examples include missing retained variables, sustained AUC or J loss, shifted event prevalence, or process maintenance that changes sensor semantics.
Retraining trigger	Time interval, new-sample threshold, performance-loss threshold, sensor replacement, process redesign, or incident-review condition that requires rebuilding both the sparse baseline and compressed rule.
Artifact links	Paths or identifiers for the selected-rule table, deployment-summary table, notebook, generator script, cached output directory, and manuscript table or figure supported by the rule.

APPENDIX E

ACRONYM DEFINITIONS

This appendix defines acronyms and recurring shorthand used in the dissertation. Some entries, such as package names or benchmark labels, are included because they function as acronyms in the manuscript even when they are proper names rather than fully expanded initialisms.

ACC

Accuracy; the fraction of predictions classified correctly.

AP

Average precision; a summary of precision-recall behavior used as a discrimination metric in the direct interpretable-model comparison.

AUC

Area under the receiver operating characteristic curve; a threshold-independent measure of ranking discrimination.

B.S.

Bachelor of Science.

BDAA

Big Data Analytics and Applications; used in the vita as part of the conference name "International Conference on Big Data Analytics & Applications."

CDF

Cumulative distribution function; used in the appendix derivation of the bivariate normal quadrant identity.

CI

Confidence interval; used in held-out compression summaries to report uncertainty around performance differences.

CR

Critical range; an event-associated value interval estimated from training observations.

CS

Computer science; used when describing a computer-science-oriented interpretation of the theoretical argument.

CUTLASS

The dissertation's public Python package and method label for critical-range rectified L1 logistic modeling with optional logic compression. It is used as a package/model name rather than as a separately expanded acronym.

EBM

Explainable Boosting Machine; an additive model used as an interpretable comparator in the RQ1 direct baseline study.

F1

The harmonic mean of precision and recall; used for exact-lag recovery and classification summaries.

HAI

HIL-based Augmented ICS; the industrial-control security benchmark used in the real-world studies.

HIL

Hardware-in-the-loop; a simulation-and-control setup that couples physical or realistic process components with simulated dynamics.

IC

Irrepresentable condition; a lasso support-recovery condition used as the central theoretical lens in RQ2.

ICS

Industrial control system; the operational domain represented by the HAI benchmark and discussed in the dissertation's practical implications.

IEEE

Institute of Electrical and Electronics Engineers; used in the vita for conference presentations.

KKT

Karush-Kuhn-Tucker; first-order optimality conditions used when discussing sparse-solver screening behavior.

L1

The ell-one norm or penalty; used for sparse logistic regression and lasso-style feature selection.

L2

The ell-two norm or penalty; used when contrasting L1 sparse penalties with rotationally invariant or ridge-like alternatives.

LAD

Logical Analysis of Data; a rule-learning tradition used as an interpretable-model comparator.

LASSO

Least Absolute Shrinkage and Selection Operator; the L1-penalized sparse modeling method used as the dissertation's main sparse-learning baseline.

M.B.A.

Master of Business Administration.

M.S.

Master of Science.

ML

Machine learning; used when identifying the dissertation's technical audience and application context.

NumPy

Numerical Python; the Python array-computing library named in the runtime comparison discussion.

nz

Nonzero; shorthand used in compression-ratio notation and table headings for the number of nonzero active terms.

P1–P4

HAI process-area identifiers. In this dissertation, P2 denotes the turbine-loop subsystem in the HAI discussion, and P4 denotes the HIL simulator block that can legitimately couple to the turbine-loop response.

PoC

Proof of concept; used for the scoped assumptions and boundary of the RQ2 theoretical analysis.

PyPI

Python Package Index; the public package index where the `cutlass` implementation is released.

QP

Quadratic programming; used for a dependence-aware sparse-selection comparator.

ROC

Receiver operating characteristic; the curve underlying the AUC metric.

RQ

Research question; used for the dissertation's organizing questions and chapter labels.

RQ1

Research Question 1; asks whether critical-range rectification improves sparse longitudinal selection and feature-lag attribution.

RQ2

Research Question 2; asks why rectification can help and how far that mechanism can be generalized.

RQ3

Research Question 3; asks whether sparse rectified models can be compressed into compact logical rules without unacceptable loss in operating performance.

TNR

True negative rate; specificity, used in Youden's $J = \mathrm{TPR} + \mathrm{TNR} - 1$.

TOST

Two one-sided tests; an equivalence-testing framework discussed in the RQ3 non-inferiority framing.

TPR

True positive rate; sensitivity or recall, used in Youden's $J = \mathrm{TPR} + \mathrm{TNR} - 1$.

U.S.

United States.

UCI

University of California, Irvine; used for the UCI ionosphere benchmark.

UNICEF

United Nations Children's Fund; the cross-domain real-world data source used as a boundary-condition case.

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CONFERENCE PRESENTATIONS

1. 2022 IEEE International Conference on Big Data, "LASSO Logic Engine: harnessing the logic parsing capabilities of the LASSO algorithm for longitudinal feature learning", 17-20 December 2022
2. 2025 IEEE International Conference on Big Data, "Efficient Longitudinal Feature Selection via Binarized Transformation: Theory and Case Studies", 08-11 December 2025
3. 2025 International Conference on Big Data Analytics & Applications (BDAA 2025), "Any-time Rule Compression and Rectified Logistic Modeling for Longitudinal Signals", 25-27 November 2025